simplydata

simple words for data science

The manual: lessons, every word, the ideas behind them, and solutions

```
import simplydata as sdata

table = sdata.table(sdata.example("marks"))
table.compare_averages("marks", between="class")
chart = table.chart()
chart.boxes(x="class", y="marks")
chart.save("marks.png")
```

Version 0.1.0.dev0

Open source (MIT licence): https://github.com/bioforkgithub/simplydata

Made from the code by `docs/make_manual.py`; every example in it was run to make its chart.

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PART I

Getting started

Welcome

simplydata gives you one small vocabulary of plain English words for working with data: reading a table, cleaning it, summing it up, testing a difference and drawing a chart. Underneath, the work is done by the libraries that scientists use every day: pandas and Polars for tables, NumPy for numbers, SciPy for statistics, and matplotlib and seaborn for charts. You learn the words; simplydata picks the right call in the right library, with the right settings.

```
import simplydata as sdata

table = sdata.table(sdata.example("marks"))   # or your own: sdata.table("marks.xlsx")
table.keep_rows("marks", above=50)
table.average_of("marks", for_each="class")   # says the answer in a sentence
table.save("good_marks.xlsx")

chart = table.chart()                         # or sdata.chart(table)
chart.bars(x="class", y="marks", error="standard error")
chart.title_top("Good marks in each class")
chart.save("marks.png")
```

The program reads like a list of instructions, one step on each line. A table, a chart and a list of numbers are separate objects, as in pandas, NumPy and matplotlib: one cleaned table can feed many charts, and several charts can share one page.

Who this manual is for

Everyone who wants to learn data science with Python: school pupils, students, teachers and researchers. You do not need to know Python first. Every word is simple enough for a child to read, and every result names the technical term (“Welch’s t-test”, “linear regression”) so that you can write it up in a report or a paper.

How this manual is built

Part II: Lessons

Fifteen short lessons that teach the whole of simplydata in a day. Each lesson shows code you can run, says what every line does, shows the real chart and output it makes, and ends with tasks to try yourself.

Part III: Dictionary of words

Every word, with what it does, its options, an example you can run and the library it uses underneath.

Part IV: Data science ideas

The ideas behind the numbers in plain words: average and median, standard deviation and standard error, confidence intervals, p-values, effect sizes, and a guide to which statistical test to use when.

Part V: Solutions

A worked solution for every “Try it yourself” task.

Everything in this manual is made from simplydata's own code by `docs/make_manual.py`: the examples were run to make the charts and the printed output you see, so they are exactly what you will get.

Install and first steps

Install

```
pip install simplydata           # coming soon
pip install simplydata[fast]     # adds Polars, the fast engine for big data
```

Where to type the code

Any place that runs Python works. In **Jupyter**, **VS Code** or **Spyder**, a chart appears by itself when you write `chart` on the last line of a cell. In a script, `chart.show()` opens it in a window, and `chart.save("name.png")` always saves it as a file.

simplydata teaches itself

```
import simplydata as sdata
sdata.lesson()                      # lists the lessons
sdata.lesson(1)                     # the first lesson
sdata.lesson(1, solutions=True)     # the first lesson, with worked solutions to its tasks
sdata.words()                       # every word, with a short example
sdata.help("bars")                  # one word: what it does, its options and an example
sdata.example()                     # the made-up tables you can practise with
```

Your own data

`sdata.table("my_file.xlsx")` and `sdata.chart("my_file.xlsx")` read Excel, CSV, TSV, Parquet and JSON files. They also take a pandas or Polars table you already have, or a small table typed by hand: `sdata.table({"name": ["Asha", "Ben"], "height": [144, 152]})`. Your file is never changed: only `table.save(...)` writes a file.

When something goes wrong

simplydata explains its errors in plain words. Misspell a column and it says which columns there are and which one you probably meant. Ask for a word that does not exist and `sdata.help` suggests the nearest one. A missing word can be requested on GitHub, at https://github.com/bioforkgithub/simplydata/issues.

How the words are built

A few rules make the words easy to guess. If you know one word of a pattern, you can usually guess its siblings.

Pattern	Examples
verb + thing changes a table or numbers	`keep_rows`, `remove_columns`, `fill_empty_cells`, `add_column`, `keep_numbers`
chart types are nouns on a chart	`chart.bars(...)`, `chart.histogram(...)`, `chart.dots(...)`
thing + where places something on a chart	`title_top`, `label_y`, `line_at_y`, `legend_bottom`, `shade_between_x`
measure + _of sums up one column; **relation + _between** links two; **compare_ + measure** tests	`average_of`, `median_of`, `correlation_between`, `compare_averages`
show_ / hide_ + thing looks without changing	`show_first_rows`, `show_grid`, `hide_legend`
as_ + library hands the data on; **save** writes a file	`as_pandas`, `as_polars`, `as_numpy`; `table.save("clean.xlsx")`

- **One option name, one meaning, everywhere:** `for_each=` (groups), `colour=` (one colour, or a colour per group), `between=[low, high]`, `from_column=`, `with_value=`, `biggest_first=`, `per_row=`, `label=`, `fast=`.
- **Measure names are the same wherever they appear:** `"average"`, `"median"`, `"total"`, `"count"`, `"smallest"`, `"biggest"`, `"standard deviation"`, `"standard error"`, `"confidence interval"`.
- **Axes are x and y.** Anything that belongs to one axis ends in `_x` or `_y`: `label_y`, `range_x`, `log_scale_y`.
- **The word names the idea**, and the printed answer names the technical term, so you can follow (and write) the Methods section of a paper.
- **Order does not matter** on a chart: after the chart type, write the other steps in any order. Both `colour` and `color`, `grey` and `gray` work.
- **Every change can be taken back**: tables and numbers remember each step, so `undo()`, `steps()` and `start_again()` always work.

The practice tables

All the examples use small made-up tables that come with simplydata, so they run anywhere without a file. Every number in them is invented. Get one with `sdata.example("marks")`, and list them all with `sdata.example()`.

marks

Test marks of 18 made-up children in three classes. `sdata.example("marks")`

Column	What it holds
`name`	the child's first name
`class`	A, B or C
`subject`	Maths, Science or English
`hours`	hours spent studying for the test
`marks`	marks in the test, out of 100
`height`	the child's height in cm

weather

Weather in three made-up cities, month by month. `sdata.example("weather")`

Column	What it holds
`month`	Jan, Feb ... Dec
`month_number`	1 to 12
`city`	Sunnyvale, Rainbridge or Frostby
`temperature`	average temperature in °C
`rain`	rain that month in mm
`wind_direction`	where the wind mostly came from, in degrees (0 to 360)
`wind_speed`	average wind speed in km/h

fruit

Fruit sold by two made-up shops in one week. One row is one sale. `sdata.example("fruit")`

Column	What it holds
`day`	Mon to Fri
`shop`	Corner shop or Market stall
`fruit`	apple, banana, cherry, grape, orange or pear
`count`	how many pieces of fruit were sold
`price`	price of one piece (made-up money)

growth

Population of four made-up countries, every two years from 2000 to 2024. `sdata.example("growth")`

Column	What it holds
`year`	2000, 2002 ... 2024
`country`	Applia, Bananaland, Cherria or Dateland
`population`	people, in millions
`forest`	how much of the land is forest, in %

island

A map of a made-up island with two hills, on a 15 x 15 grid. `sdata.example("island")`

Column	What it holds
`east`	how far east, 0 to 14 (km)
`north`	how far north, 0 to 14 (km)
`height`	height of the ground in m (below 0 is under the sea)

PART II

Lessons

LESSON 1

Your first chart

In this lesson you will learn

- how to get a table to practise with
- how to start a chart and draw bars
- how to save your chart as a picture

Example

A simplydata program is made of small steps, one step on each line, and each step says what it does.

```
import simplydata as sdata
sdata.example()
marks = sdata.example("marks")
print(marks)
chart = sdata.chart(marks)
chart.bars(x="name", y="marks")
chart.save("lesson1.png")
```

What each line does

1. Load simplydata. From now on we call it by its short name, `"sdata"`.
2. List the made-up tables you can practise with, and their columns.
3. Get the marks table: test marks of made-up children. We call it marks.
4. Look at the table. Each row is one child in one test. The names at the top (name, class, subject ...) are the columns.
5. Start a new, empty chart that uses the marks table. We call it chart, and every next line starts with that name.
6. Draw bars: one bar for each name (along x, the bottom), as tall as the marks (up y, the side). Each child did three tests, so the bar shows the average: add the three marks up and divide by 3.
7. Save the chart as a picture called lesson1.png, in the folder you are working in. Open it to see your chart!

What it prints

```
Example tables you can use (all made up, for practice):

  sdata.example("marks")
      Test marks of 18 made-up children in three classes.
      Columns: name, class, subject, hours, marks, height

  sdata.example("weather")
      Weather in three made-up cities, month by month.
      Columns: month, month_number, city, temperature, rain, wind_direction, wind_speed

  sdata.example("fruit")
```

```
      Fruit sold by two made-up shops in one week. One row is one sale.
      Columns: day, shop, fruit, count, price

  sdata.example("growth")
      Population of four made-up countries, every two years from 2000 to 2024.
      Columns: year, country, population, forest

  sdata.example("island")
      A map of a made-up island with two hills, on a 15 x 15 grid.
      Columns: east, north, height

Use one like this:  table = sdata.example("marks")  then  chart = sdata.chart(table)
      name class  subject  hours  marks  height
0     Asha     A     Maths    6.3     61     144
1     Asha     A  Science    3.8     66     144
2     Asha     A  English    7.6     77     144
3      Ben     A     Maths    6.4     73     152
4      Ben     A  Science    9.2     98     152
5      Ben     A  English    6.2     71     152
6     Chen     A     Maths    7.9     81     151
7     Chen     A  Science    3.5     59     151
8     Chen     A  English    2.5     46     151
9     Dina     A     Maths    6.5     74     155
10    Dina     A  Science    4.5     79     155
11    Dina     A  English    2.3     57     155
12    Emil     A     Maths    3.2     47     141
13    Emil     A  Science    8.4     86     141

... (40 more lines)
```

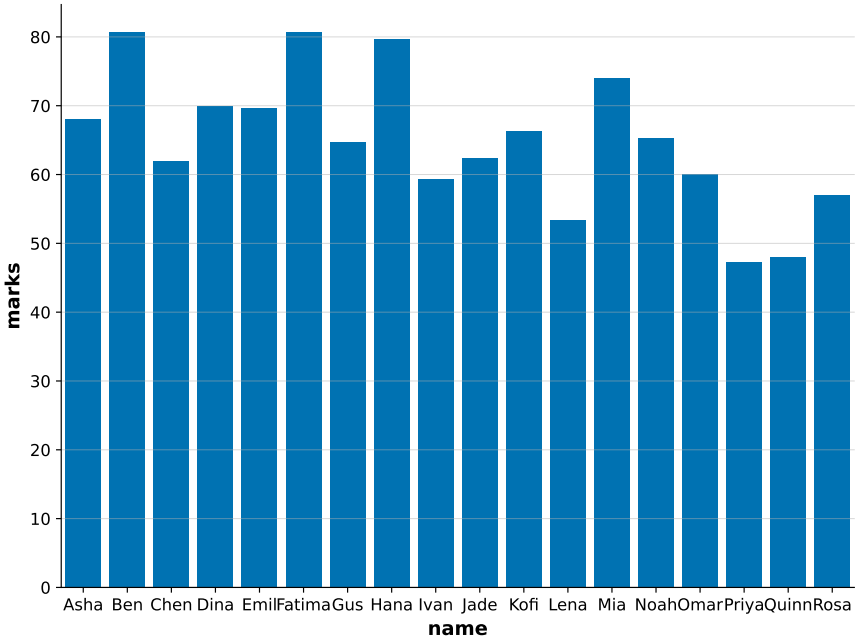


Figure 1.1: Lesson 1, example: `lesson1.png`.

Good to know

- Chart types are nouns on a chart: `chart.bars`, `chart.lines`, `chart.dots` ... The chart is the thing, and bars is what it shows.
- For your own data, write chart = `sdata.chart("my_marks.xlsx")`. Excel, CSV, TSV,

Parquet and JSON files all work.

- `chart.save("lesson1.pdf")` saves a PDF instead: the end of the name decides the type.
- In Jupyter, VS Code or Spyder, write chart on the last line of a cell and the chart appears by itself. In a script, `chart.show()` opens it in a window.
- Are the names under the bars bumping into each other? Lesson 2 shows how to turn them.

Try it yourself

1. Change `x="name"` to `x="subject"`. What does each bar show now?
2. Save the chart as `"my_first_chart.png"` and also as `"my_first_chart.pdf"`.
3. Make a chart of the `"fruit"` table: bars with `x="fruit"` and `y="count"`.

Worked solutions: page 156, or type `sdata.lesson(1, solutions=True)`.

Words in this lesson: `sdata.example()`, `sdata.chart()`, `chart.bars()`, `chart.save()`, `chart.show()`.

LESSON 2

Titles and labels

In this lesson you will learn

- how to put a title on top
- how to name the x-axis and the y-axis
- how to add a small caption and change the size of the text

Example

A chart is easier to read when it says what it shows. First draw, then add the words.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="name", y="marks")
chart.title_top("Our class marks")
chart.label_x("Student")
chart.label_y("Average mark")
chart.rotate_labels_x(45)
chart.caption_bottom("Made-up data from sdata.example")
chart.font_size(14)
chart.save("lesson2.png")
```

What each line does

1 Load simplydata.
2 Get the marks table.
3 Start a chart.
4 Draw one bar for each child.
5 Write a title on top of the chart.
6 Write a name under the x-axis (the bottom).
7 Write a name beside the y-axis (the side).
8 Turn the names under the bars by 45 degrees, so long names do not bump into each other.
9 Write a small line under the whole chart.
10 Make all the text a bit bigger (normal is 12).
11 Save it.

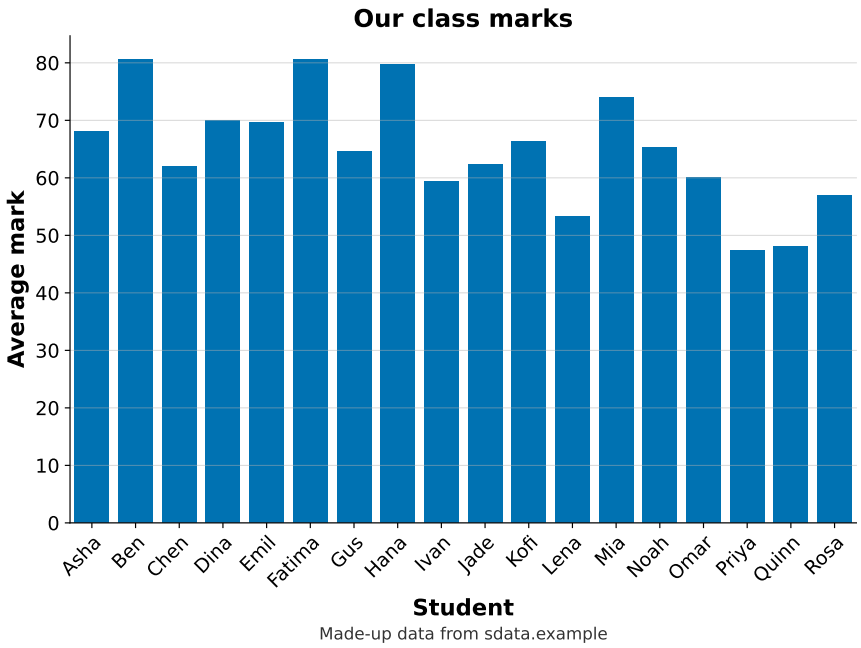


Figure 2.1: Lesson 2, example: `lesson2.png`.

Good to know

- After the chart type, the order of words such as `title_top` and `label_y` does not matter: the chart is drawn when you save or show it.
- Words that belong to one axis end in _x or _y: `label_x`, `label_y`, `rotate_labels_x`.

Try it yourself

1. Give the chart your own title, and label the y-axis `"Marks out of 100"`.
2. Try `chart.font_size(20)`. Then try 9. Which is easier to read?
3. Take away the line `chart.rotate_labels_x(45)`. What happens to the names?

Worked solutions: page 158, or type `sdata.lesson(2, solutions=True)`.

Words in this lesson: `chart.title_top()`, `chart.label_x()`, `chart.label_y()`, `chart.rotate_labels_x()`, `chart.caption_bottom()`, `chart.font_size()`.

LESSON 3

Colours, groups and legends

In this lesson you will learn

- how to give each group its own colour
- how to choose your own colours
- how to move the legend (the box that says which colour is which)
- how to change the whole look with a style

Example 1

`colour=` inside a chart type splits the data into groups.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="subject", y="marks", colour="class")
chart.colours("orange", "teal", "purple")
chart.legend_bottom()
chart.title_top("Marks in each subject")
chart.save("lesson3_groups.png")
```

What each line does

1. Load simplydata.
2. Get the marks table.
3. Start a chart.
4. For each subject, draw one bar per class, each class in its own colour.
5. Choose the colours yourself, in order: class A orange, B teal, C purple.
6. Put the legend under the chart.
7. Add a title.
8. Save it.

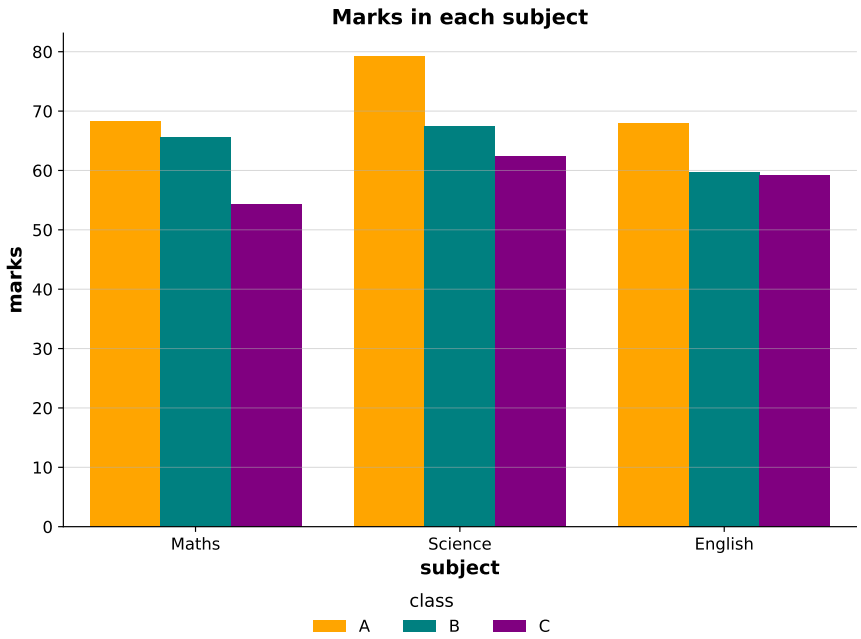


Figure 3.1: Lesson 3, example 1: `lesson3_groups.png`.

Example 2

Without groups, one colour is used for everything. You can choose it.

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month", y="rain")
chart.colour("blue")
chart.style("school")
chart.save("lesson3_look.png")
```

What each line does

1. Load simplydata.
2. Get the weather table: three made-up cities.
3. Start a chart.
4. Draw a line of rain through the months. With three cities in each month, the line shows their average.
5. Make the line blue.
6. Change the whole look. Try `"paper"`, `"school"`, `"presentation"` and `"dark"`.
7. Save it.



Figure 3.2: Lesson 3, example 2: `lesson3_look.png`.

Good to know

- `colour="class"` (a column) gives each group a colour. `colour="red"` (a colour) makes that step red. color works too.
- The legend can go on any side: `legend_top`, `legend_bottom`, `legend_left`, `legend_right`. `hide_legend` takes it away and `show_legend` brings it back.

Try it yourself

1. In the weather chart, add `colour="city"` to `chart.lines(...)`. How many lines do you get?
2. Move the legend to the top with `chart.legend_top()`.
3. Try `chart.style("dark")` and choose three colours that look good on it.

Worked solutions: page 161, or type `sdata.lesson(3, solutions=True)`.

Words in this lesson: `chart.colours()`, `chart.colour()`, `chart.legend_bottom()`, `chart.legend_top()`, `chart.hide_legend()`, `chart.style()`, `chart.lines()`.

LESSON 4

Different chart types

In this lesson you will learn

- lines, dots, histograms, boxes and pies
- which chart suits which question
- how to put two chart types on top of each other

Example 1

Lines show how something changes, for example over time.

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month", y="temperature", colour="city", dots=True)
chart.save("lesson4_lines.png")
```

What each line does

1 Load simplydata.
2 Get the weather table.
3 Start a chart.
4 One line per city. `dots=True` also marks each month with a dot.
5 Save it.

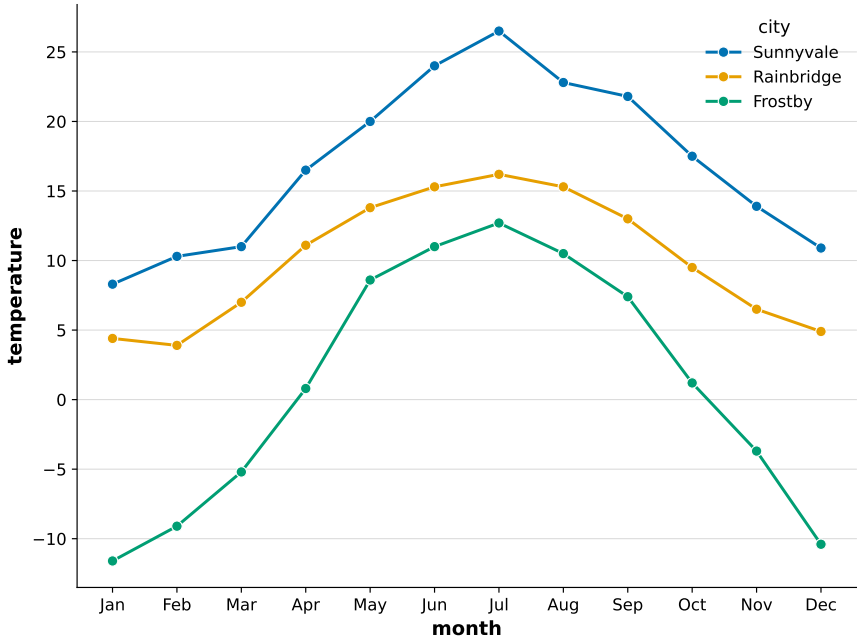


Figure 4.1: Lesson 4, example 1: `lesson4_lines.png`.

Example 2

Dots show if two things go together. This is also called a scatter chart.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks", colour="class")
chart.save("lesson4_dots.png")
```

What each line does

1. Load simplydata.
2. Get the marks table.
3. Start a chart.
4. One dot for each test: how long the child studied, and the mark they got.
5. Save it. Do more hours give more marks?

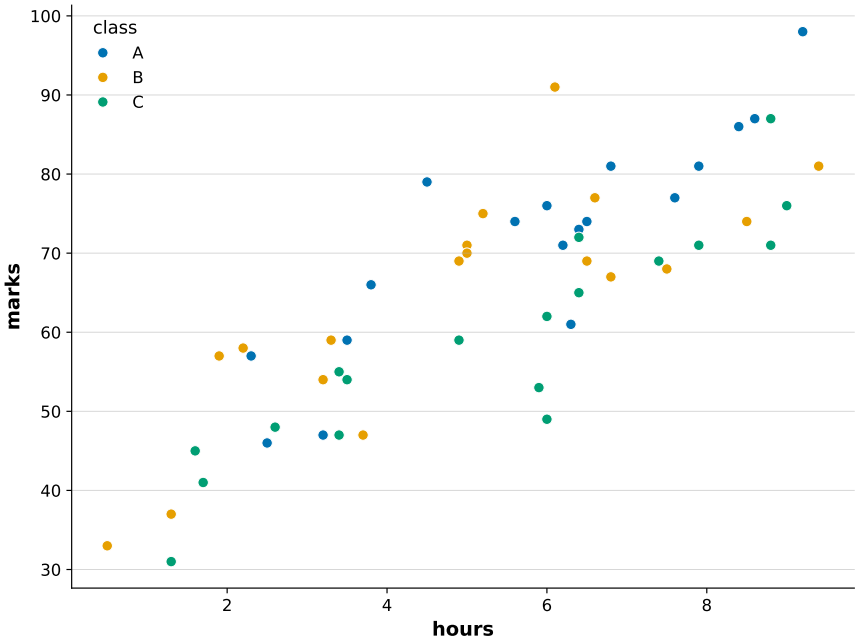


Figure 4.2: Lesson 4, example 2: `lesson4_dots.png`.

Example 3

A histogram counts how many values fall in each range, like 40 to 50, 50 to 60 ... Its shape is called the distribution of the values.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.histogram(x="marks", bins=8)
chart.save("lesson4_histogram.png")
```

What each line does

1 Load simplydata.
2 Get the marks table.
3 Start a chart.
4 Cut the marks into 8 ranges (bins) and draw a bar for how many marks are in each.
5 Save it.

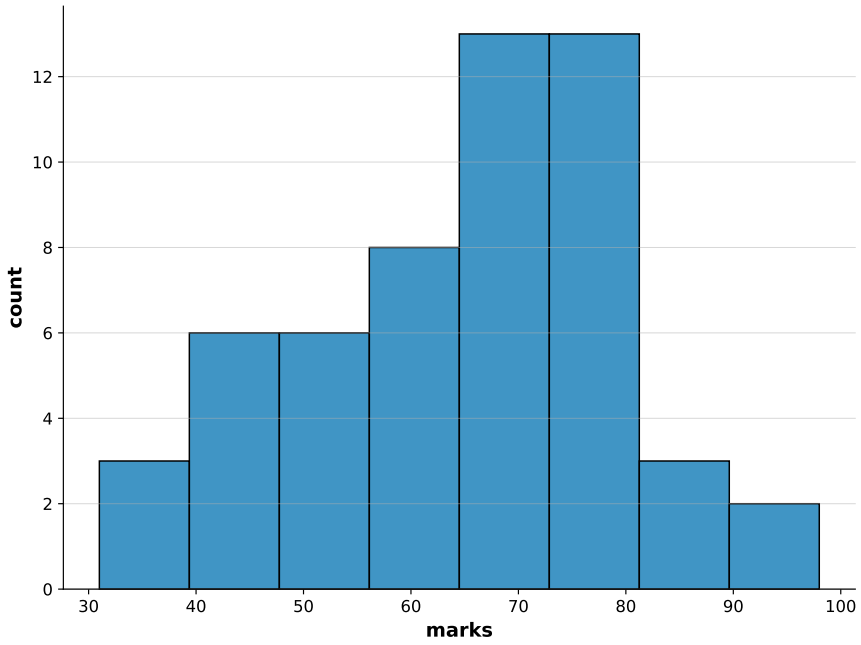


Figure 4.3: Lesson 4, example 3: `lesson4_histogram.png`.

Example 4

Boxes show how spread out the values are. Two chart types on one chart are drawn one on top of the other.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.boxes(x="class", y="marks")
chart.all_dots(x="class", y="marks")
chart.save("lesson4_boxes.png")
```

What each line does

1 Load simplydata.
2 Get the marks table.
3 Start a chart.
4 One box per class. The line in the box is the median: the middle mark when they are put in order. The box holds the middle half of the marks (the interquartile range).
5 Also show every single mark as a dot, on top of the boxes.
6 Save it.

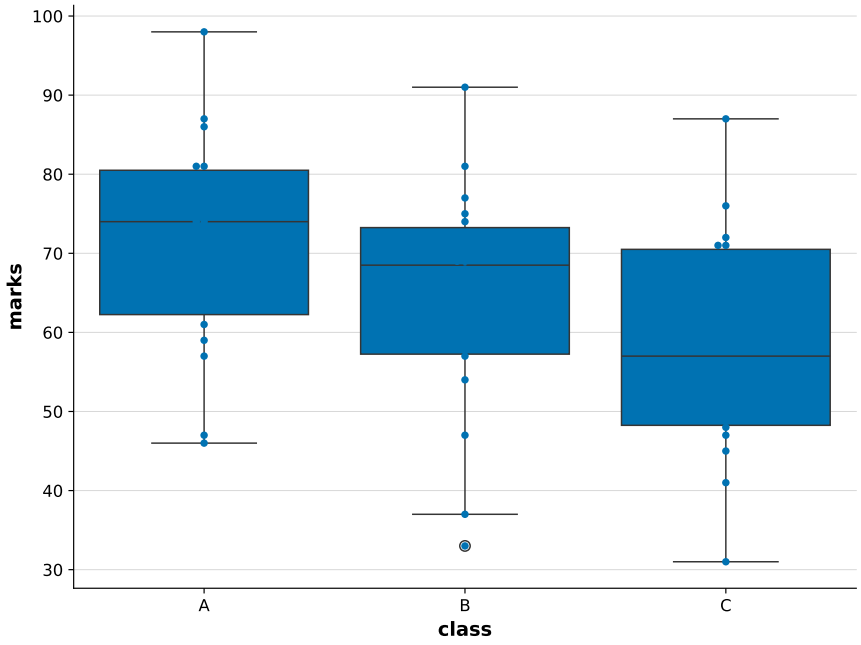


Figure 4.4: Lesson 4, example 4: `lesson4_boxes.png`.

Example 5

A pie shows how a whole is shared out.

```
import simplydata as sdata
fruit = sdata.example("fruit")
chart = sdata.chart(fruit)
chart.pie(labels="fruit", values="count")
chart.save("lesson4_pie.png")
```

What each line does

1. Load simplydata.
2. Get the fruit table: what two shops sold in a week.
3. Start a chart.
4. One slice per fruit, as big as the number sold.
5. Save it.

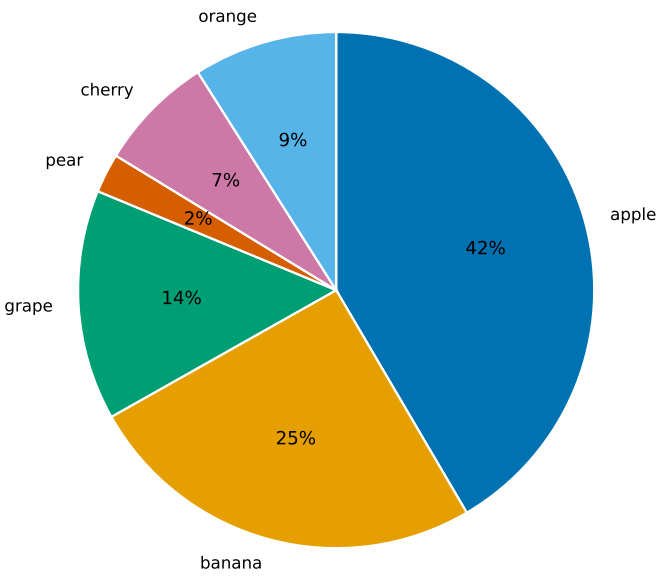


Figure 4.5: Lesson 4, example 5: `lesson4_pie.png`.

Good to know

- Chart types on the same chart are layers of one chart. Lesson 7 shows how to make several charts and put them on one page.
- There are more chart types: area, violins, heatmap, polar, contour, surface and `dots_3d`. Type `sdata.words()` to see them all and `sdata.help("violins")` to learn one.

Try it yourself

1. Draw `chart.violins(x="class", y="marks")` instead of boxes. What is different?
2. Draw a histogram of the weather `"temperature"` column with `bins=6`.
3. Make a chart of `sdata.example("growth")` and draw `chart.area(x="year", y="population", colour="country")`.

Worked solutions: page 164, or type `sdata.lesson(4, solutions=True)`.

Words in this lesson: `chart.lines()`, `chart.dots()`, `chart.histogram()`, `chart.boxes()`, `chart.all_dots()`, `chart.pie()`, `chart.violins()`, `chart.area()`.

LESSON 5

Statistics: averages, error bars and fit lines

In this lesson you will learn

- what a bar shows when a group has many values: the average or the median
- how to add error bars, and the difference between standard deviation and standard error
- how to draw the straight line that fits some dots best

Example 1

The average (also called the mean) is all the values added up, divided by how many there are. Error bars show how far the values spread around it.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="class", y="marks", show="average", error="standard deviation")
chart.numbers_on_bars()
chart.save("lesson5_average.png")
```

What each line does

1 Load simplydata.
2 Get the marks table.
3 Start a chart.
4 One bar per class, as tall as the average mark. The error bars show the standard deviation: how far marks usually are from the average. A line is printed that says which error bars were drawn.
5 Write the average on each bar.
6 Save it.

What it prints

```
Error bars: the standard deviation of the values (how spread out they are around the average).
```

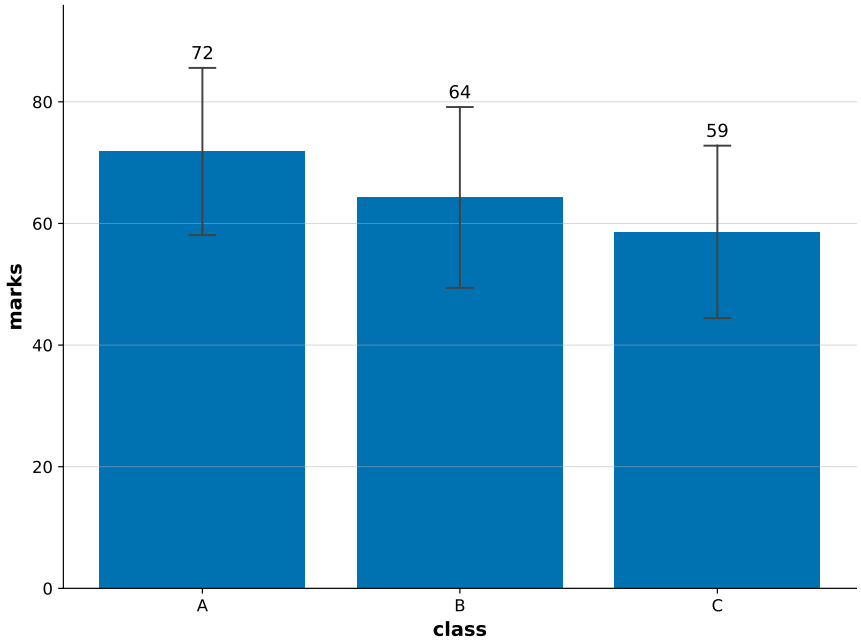


Figure 5.1: Lesson 5, example 1: `lesson5_average.png`.

Example 2

A fit line is the straight line that goes as close as possible to all the dots.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.fit_line(confidence_band=True, show_equation=True)
chart.save("lesson5_fit.png")
```

What each line does

1. Load simplydata.
2. Get the marks table.
3. Start a chart.
4. One dot per test: hours of study and the mark.
5. Add the best straight line (also called linear regression). `confidence_band=True` shades where the true line very likely is (the 95 % confidence interval). `show_equation=True` writes its equation and R^2 (1 means the dots sit right on the line, 0 means no pattern at all).
6. Save it.

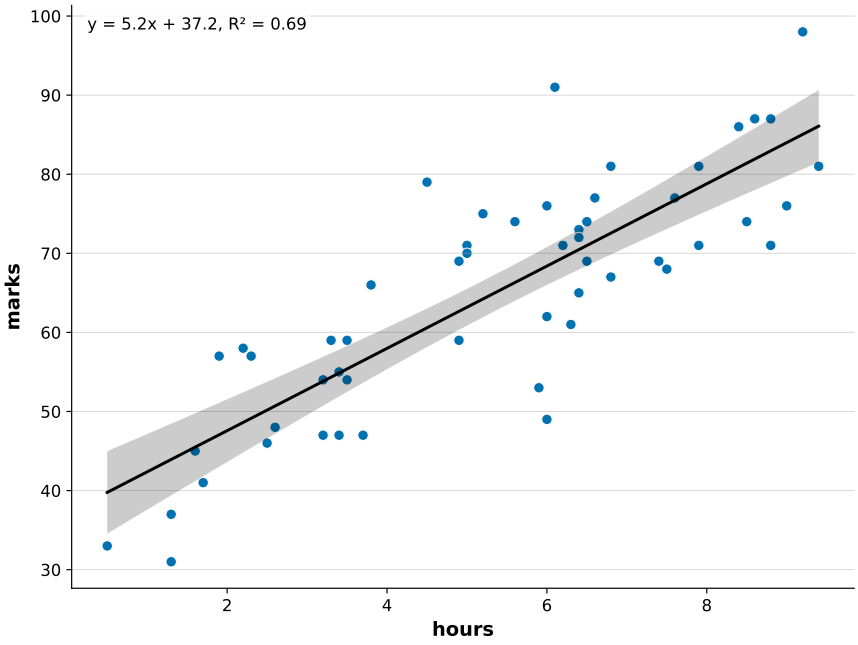


Figure 5.2: Lesson 5, example 2: `lesson5_fit.png`.

Example 3

A density curve is a smooth histogram: high where many values are.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.density(x="marks", colour="class")
chart.save("lesson5_density.png")
```

What each line does

1. Load simplydata.
2. Get the marks table.
3. Start a chart.
4. One smooth curve for each class (a kernel density estimate).
5. Save it.

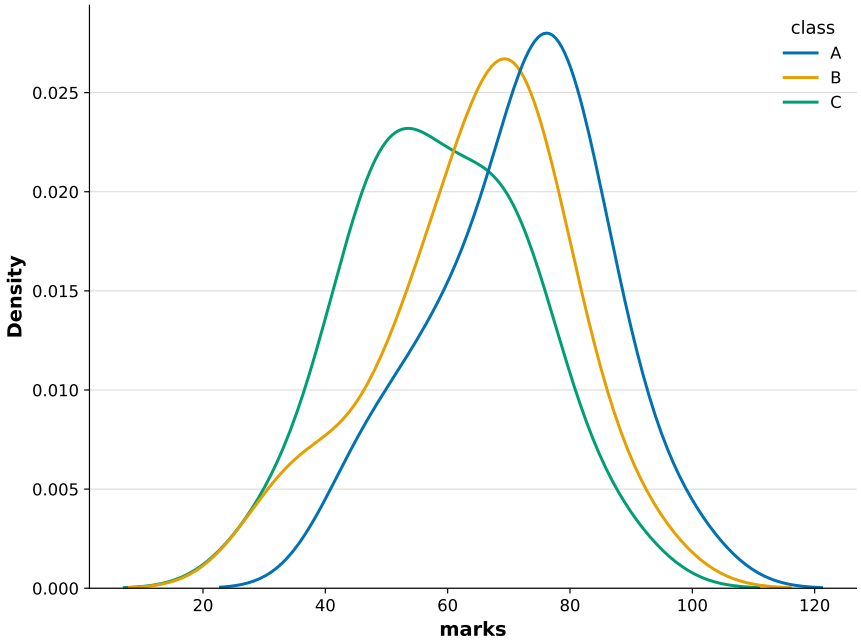


Figure 5.3: Lesson 5, example 3: `lesson5_density.png`.

Good to know

- Average or median? The median is the middle value when the values are put in order. One very big or very small value pulls the average towards it, but hardly moves the median, so the median is safer when a few values are far from the rest (outliers).
- `show=` can be `"average"`, `"median"`, `"total"` (all added up), `"count"` (how many rows), `"smallest"`, `"biggest"` or `"standard deviation"`: the same names as the table words `average_of`, `median_of` and so on.
- Standard deviation or standard error? The standard deviation describes the values: how spread out they are. The standard error describes the average: how sure we are of it. It gets smaller as you collect more values, even when the values spread just as much.
- `error="confidence interval"` draws the 95 % confidence interval: the range the true average is very likely to be in. The short forms `"sd"`, `"se"` and `"ci"` work too. Always say in your caption which error bars you drew.
- `fit_curve(degree=2)` fits a curve with one bend; `smooth_line()` follows the dots freely.

Try it yourself

1. Change `show="average"` to `show="median"`. Do the bars change much?
2. Change `error="standard deviation"` to `error="standard error"`. Are the error bars bigger or smaller? Why?
3. Make a chart of the weather table, draw dots with `x="month_number"`, `y="temperature"` and add `chart.fit_curve(degree=2)`.

Worked solutions: page 166, or type `sdata.lesson(5, solutions=True)`.

Words in this lesson: `show=`, `error=`, `chart.numbers_on_bars()`, `chart.fit_line()`, `chart.density()`, `chart.fit_curve()`, `chart.smooth_line()`.

LESSON 6

Axes and reference lines

In this lesson you will learn

- how to choose where an axis starts and stops
- how to draw a line at a value, like a pass mark, and shade a band
- how to choose the tick marks, and what a log scale is

Example 1

The axes are the two edges with numbers: x along the bottom, y up the side.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="name", y="marks")
chart.range_y(0, 100)
chart.ticks_y(0, 25, 50, 75, 100)
chart.line_at_y(50, label="pass mark")
chart.shade_between_y(40, 60)
chart.rotate_labels_x()
chart.save("lesson6_marks.png")
```

What each line does

1. Load simplydata.
2. Get the marks table.
3. Start a chart.
4. One bar for each child.
5. Make the y-axis go from 0 to 100, the lowest and highest mark you can get.
6. Put the numbers on the y-axis at these places.
7. Draw a dashed line across at 50, with a name.
8. Shade the band between 40 and 60.
9. Turn the names so they fit.
10. Save it.

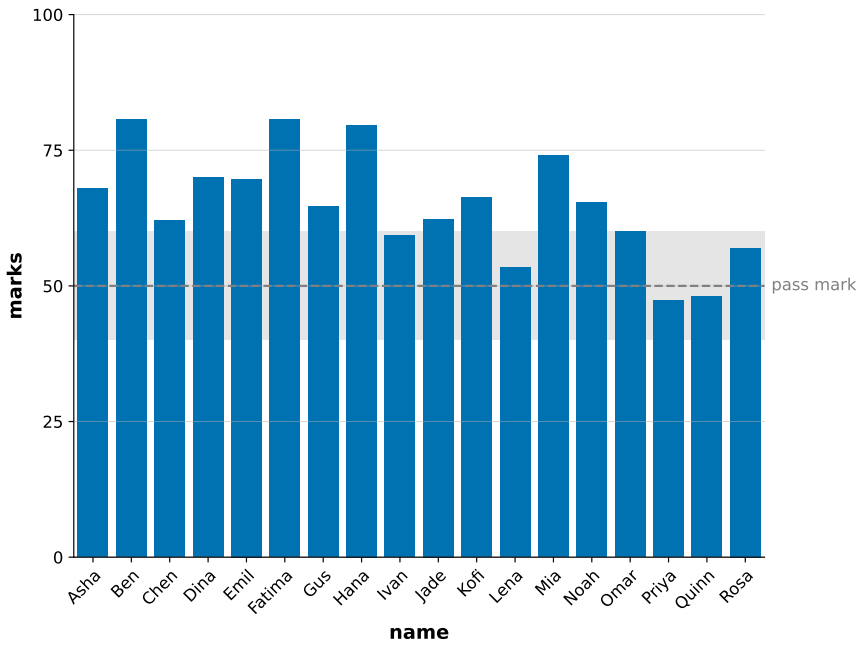


Figure 6.1: Lesson 6, example 1: `lesson6_marks.png`.

Example 2

On a log scale 1, 10, 100 and 1000 are the same distance apart. It helps when some numbers are much bigger than others.

```
import simplydata as sdata
growth = sdata.example("growth")
chart = sdata.chart(growth)
chart.lines(x="year", y="population", colour="country")
chart.log_scale_y()
chart.line_at_x(2010, label="2010")
chart.label_y("People (millions)")
chart.save("lesson6_log.png")
```

What each line does

1. Load simplydata.
2. Get the growth table: people in four made-up countries.
3. Start a chart.
4. One line per country.
5. Use a log scale on the y-axis. A country that grows by the same share each year now gets a straight line.
6. Draw an up-and-down line at the year 2010.
7. Name the y-axis.
8. Save it.

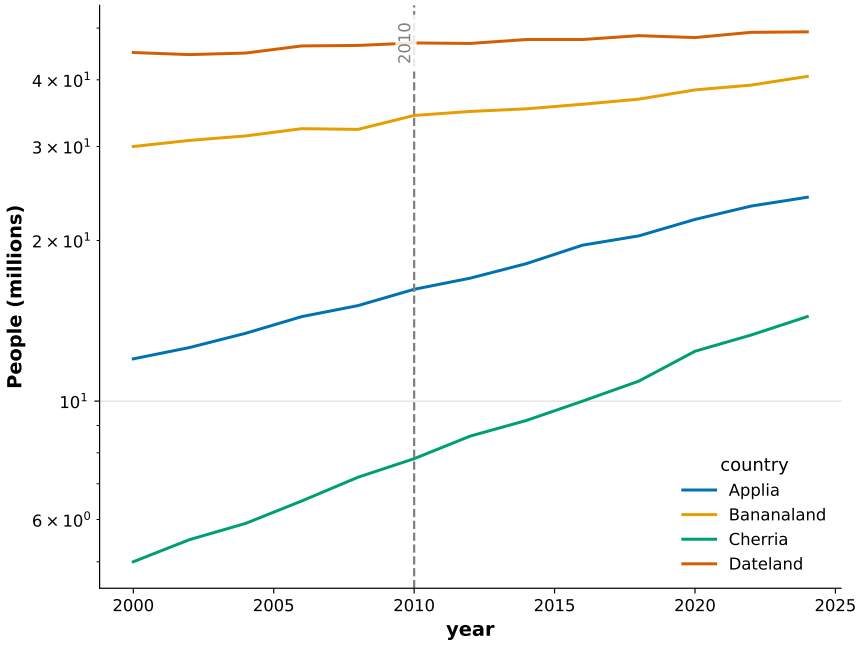


Figure 6.2: Lesson 6, example 2: `lesson6_log.png`.

Good to know

- Every axis word ends in _x or _y: `range_x`, `ticks_x`, `log_scale_x`, `line_at_x`, `shade_between_x`, `flip_x` ...
- `range_y(0, None)` fixes only the bottom and lets the top choose itself.
- Bars should start at 0: a bar is read by its length, so a y-axis that starts at 40 makes small differences look big.

Try it yourself

1. Move the pass mark to 60 and shade between 0 and 40 instead.
2. Take away `chart.log_scale_y()`. Which countries look different?
3. Try `chart.flip_y()` on any chart. What happens?

Worked solutions: page 168, or type `sdata.lesson(6, solutions=True)`.

Words in this lesson: `chart.range_y()`, `chart.ticks_y()`, `chart.line_at_y()`, `chart.shade_between_y()`, `chart.log_scale_y()`, `chart.line_at_x()`, `chart.flip_y()`.

LESSON 7

Several charts together

In this lesson you will learn

- how to make one small chart for each group
- how to put different charts on one page: side by side, one above the other, or in a grid
- how to write A, B, C on the charts, as figures in papers have
- how to compare every number column with every other

Example 1

`one_chart_per` draws the same chart once for each group.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.fit_line()
chart.one_chart_per("class")
chart.save("lesson7_one_chart_per.png")
```

What each line does

1. Load simplydata.
2. Get the marks table.
3. Start a chart.
4. Dots of hours and marks.
5. Add a fit line.
6. Make one small chart for each class, with the same axes so they are easy to compare.
7. Save it.

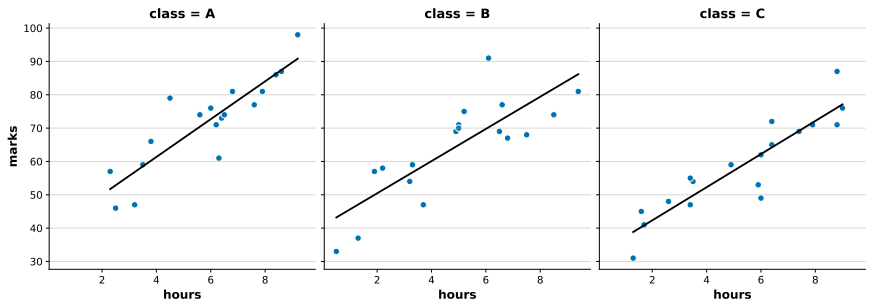


Figure 7.1: Lesson 7, example 1: `lesson7_one_chart_per.png`.

Example 2

Each chart is an object of its own, so different charts can share one page.

```
import simplydata as sdata
weather = sdata.example("weather")
a = sdata.chart(weather).lines(x="month", y="temperature", colour="city")
b = sdata.chart(weather).bars(x="month", y="rain", colour="city")
page = sdata.one_above_other(a, b)
page.title_top("A year of weather")
page.save("lesson7_page.png")
```

What each line does

1 Load simplydata.
2 Get the weather table.
3 Make a first chart, called a. Steps can also be joined with dots on one line.
4 Make a second chart, called b.
5 Put a above b on one page.
6 Give the whole page a title.
7 Save the page.

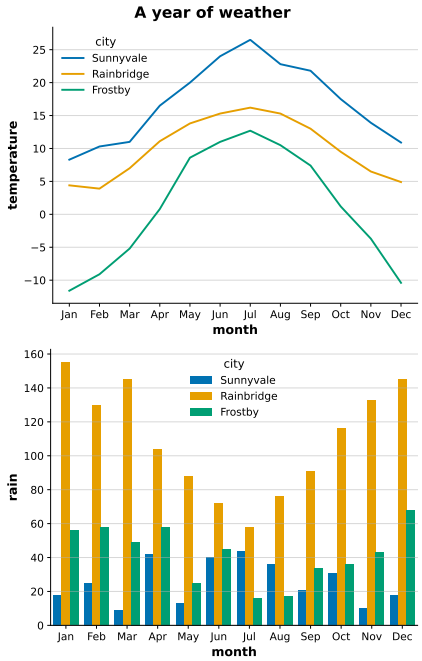


Figure 7.2: Lesson 7, example 2: `lesson7_page.png`.

Example 3

A grid holds many charts in rows and columns. `letters=True` writes A, B, C and D on them, as figures in papers have.

```
import simplydata as sdata
marks = sdata.example("marks")
a = sdata.chart(marks).bars(x="class", y="marks").title_top("Average")
b = sdata.chart(marks).histogram(x="marks").title_top("All marks")
c = sdata.chart(marks).dots(x="height", y="marks").title_top("Height")
d = sdata.chart(marks).boxes(x="subject", y="marks").title_top("Subjects")
page = sdata.grid(a, b, c, d, per_row=2, letters=True)
page.save("lesson7_grid.png")
```

What each line does

1 Load simplydata.
2 Get the marks table.
3 Chart a.
4 Chart b.
5 Chart c.
6 Chart d.
7 All four in a grid, 2 charts per row, with a letter on each.
8 Save it.

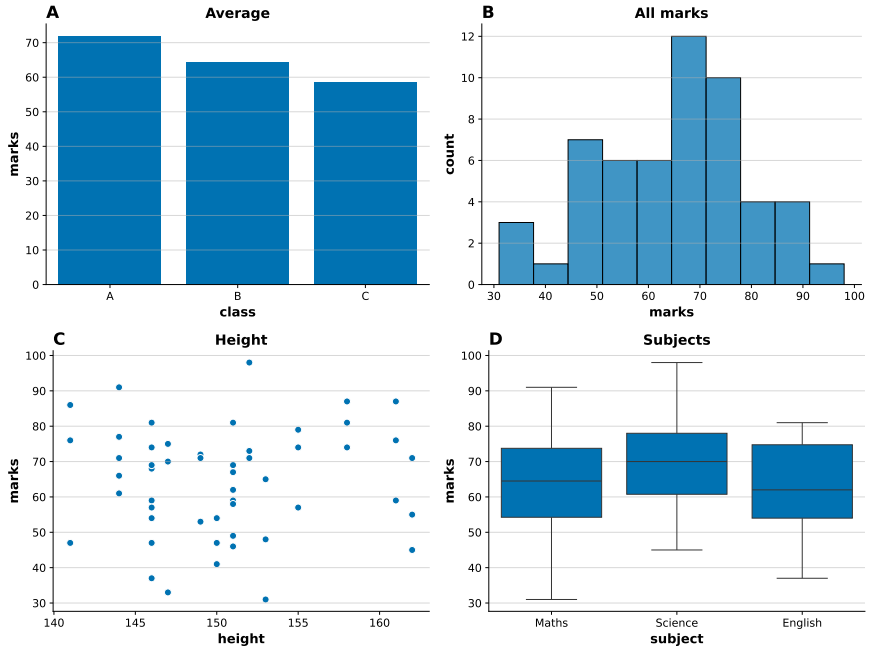


Figure 7.3: Lesson 7, example 3: `lesson7_grid.png`.

Example 4

Charts from different tables can share a page too.

```
import simplydata as sdata
rain = sdata.chart(sdata.example("weather")).lines(x="month", y="rain", colour="city")
forest = sdata.chart(sdata.example("growth")).lines(x="year", y="forest", colour="country")
page = sdata.side_by_side(rain, forest, letters=True)
page.save("lesson7_two_tables.png")
```

What each line does

1 Load simplydata.
2 A chart of the weather table, called rain.
3 A chart of the growth table, called forest.
4 Put the two charts next to each other, with A and B.
5 Save the page.

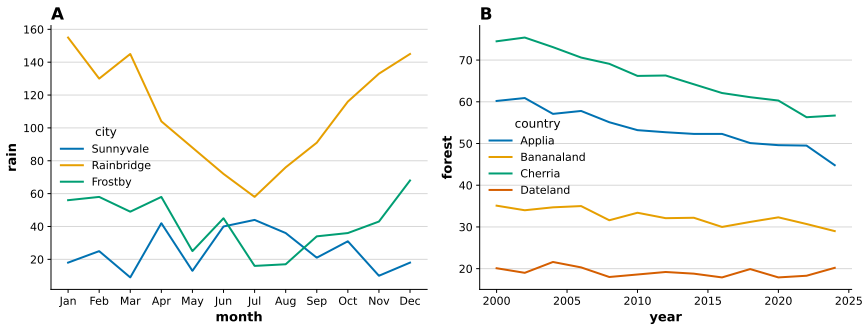


Figure 7.4: Lesson 7, example 4: `lesson7_two_tables.png`.

Example 5

`sdata.pairs` draws every number column against every other, in one go.

```
import simplydata as sdata
pairs = sdata.pairs(sdata.example("marks"), colour="class")
pairs.save("lesson7_pairs.png")
```

What each line does

1. Load simplydata.
2. Hours, marks and height, each against each other. The middle line shows how each column is spread out.
3. Save it. Which two columns go together?

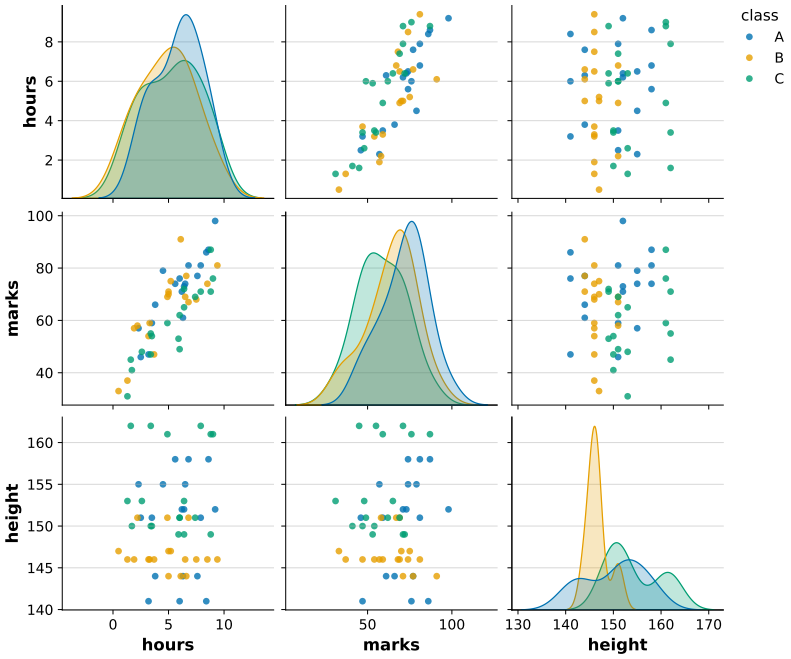


Figure 7.5: Lesson 7, example 5: `lesson7_pairs.png`.

Good to know

- `same_y=True` gives all the charts on a page the same y-axis, so heights can be compared fairly: `sdata.side_by_side(a, b, same_y=True)`.

- `sdata.side_by_side`, `sdata.one_above_other` and `sdata.grid` all take `letters=True`.
- `chart.one_chart_per(down="class", across="subject")` makes a table of small charts: the classes go down, the subjects go across. `per_row=3` puts 3 small charts in a row.

Try it yourself

1. Change `chart.one_chart_per("class")` to `chart.one_chart_per("subject")`.
2. Use `sdata.side_by_side(a, b)` instead of `sdata.one_above_other(a, b)` for the weather charts.
3. Make `sdata.pairs` of `sdata.example("weather")` with `colour="city"`.

Worked solutions: page 170, or type `sdata.lesson(7, solutions=True)`.

Words in this lesson: `chart.one_chart_per()`, `sdata.one_above_other()`, `sdata.grid()`, `sdata.side_by_side()`, `sdata.pairs()`, `chart.fit_line()`, `chart.histogram()`.

LESSON 8

Hard tricks, and your own program

In this lesson you will learn

- a second y-axis, a zoom box and a moving chart
- a heatmap that sorts itself
- how to write your own program from start to end

Example 1

A second y-axis shows a column measured in other units on the same chart.

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month", y="temperature")
chart.second_y("rain", drawn_as="bars", label="Rain (mm)")
chart.label_y("Temperature (°C)")
chart.save("lesson8_second_y.png")
```

What each line does

1. Load simplydata.
2. Get the weather table.
3. Start a chart.
4. A line of temperature (average of the three cities).
5. Add rain, drawn as bars, with its own y-axis on the right.
6. Name the left y-axis.
7. Save it.

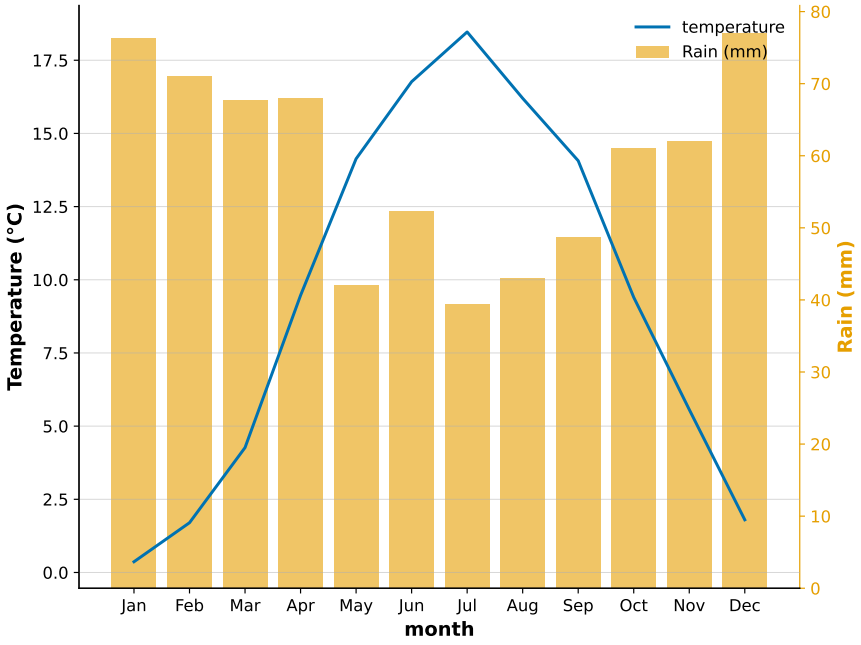


Figure 8.1: Lesson 8, example 1: `lesson8_second_y.png`.

Example 2

A zoom box shows a small part of the chart close up.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.zoom_box(x_between=[2, 4], y_between=[40, 60], where="top left")
chart.save("lesson8_zoom.png")
```

What each line does

1. Load simplydata.
2. Get the marks table.
3. Start a chart.
4. Dots of hours and marks.
5. Show the part between 2 and 4 hours and between 40 and 60 marks, bigger, in a box at the top left.
6. Save it.

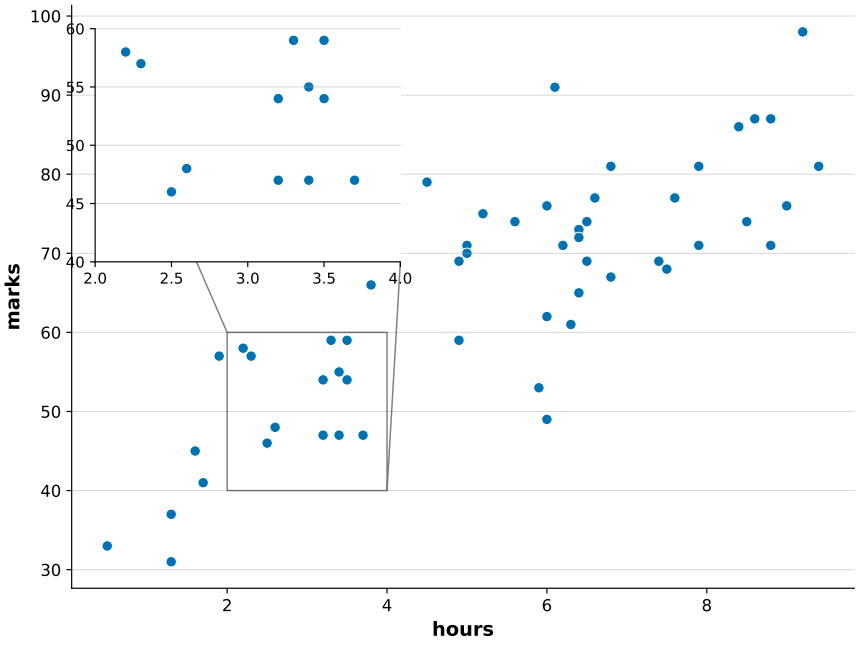


Figure 8.2: Lesson 8, example 2: `lesson8_zoom.png`.

Example 3

A heatmap colours a grid of numbers. `cluster=True` puts rows that are alike next to each other.

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.heatmap(x="month", y="city", value="rain", numbers=True, cluster=True)
chart.label_colour_bar("Rain (mm)")
chart.size(width=12, height=4)
chart.save("lesson8_heatmap.png")
```

What each line does

1. Load simplydata.
2. Get the weather table.
3. Start a chart.
4. One square per city and month, coloured by rain, with the number written in. `cluster=True` sorts the rows and columns and draws a tree of how alike they are (hierarchical clustering).
5. Name the colour bar.
6. Make the chart wide and low.
7. Save it.

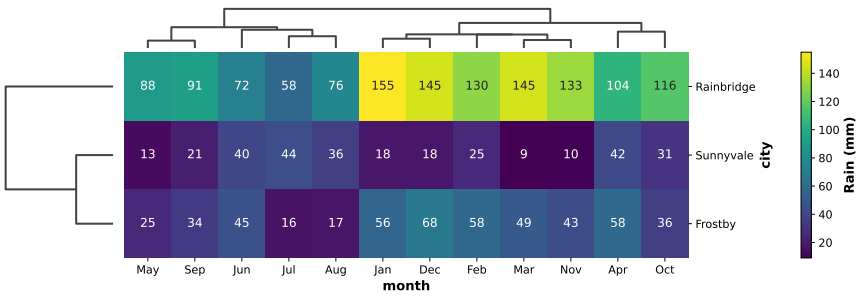


Figure 8.3: Lesson 8, example 3: `lesson8_heatmap.png`.

Example 4

An animation is a chart that moves: one picture for each value of a column.

```
import simplydata as sdata
growth = sdata.example("growth")
chart = sdata.chart(growth)
chart.bars(x="country", y="population")
chart.range_y(0, 50)
chart.animate(over="year")
chart.save("lesson8_growth.gif")
```

What each line does

1. Load simplydata.
2. Get the growth table.
3. Start a chart.
4. One bar per country.
5. Keep the y-axis still, from 0 to 50.
6. Make one picture for each year.
7. Save it as a `.gif`, a moving picture. Open it in a web browser to watch.

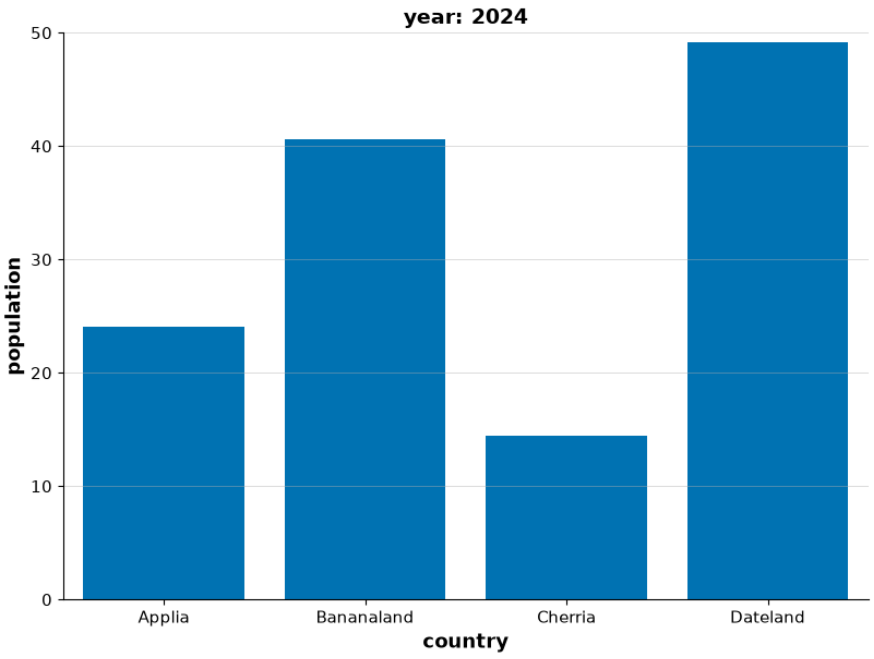


Figure 8.4: Lesson 8, example 4: `lesson8_growth.gif` (the last picture of the animation).

Example 5

Your own program. Every chart follows the same plan: get the data, look, start, draw, words, look, save.

```
import simplydata as sdata
island = sdata.example("island")
print(island)
chart = sdata.chart(island)
chart.contour(x="east", y="north", z="height")
chart.title_top("Treasure Island")
chart.label_x("East (km)")
chart.label_y("North (km)")
chart.label_colour_bar("Height (m)")
chart.note("X marks the spot", x=10, y=4, arrow=True)
chart.style("presentation")
chart.save("lesson8_my_program.png")
```

What each line does

1. Load simplydata.
2. Get the data. For your own file, skip this line and write chart = `sdata.chart("my_file.xlsx")` below.
3. Look at the data before you draw it.
4. Start a chart.
5. Draw: a map where each colour band is one height.
6. Words: a title ...
7. ... and the axis names ...
9. ... and the name of the colour bar.
10. A note with an arrow.
11. 7. Look: big and clear for showing to others.
12. 8. Save.

What it prints

```
     east  north  height
0       0      0      -6
1       0      1      -5
2       0      2      -4
3       0      3      -3
4       0      4      -1
..    ...    ...     ...
220    14     10      -2
221    14     11      -3
222    14     12      -4
223    14     13      -5
224    14     14      -6

[225 rows x 3 columns]
```

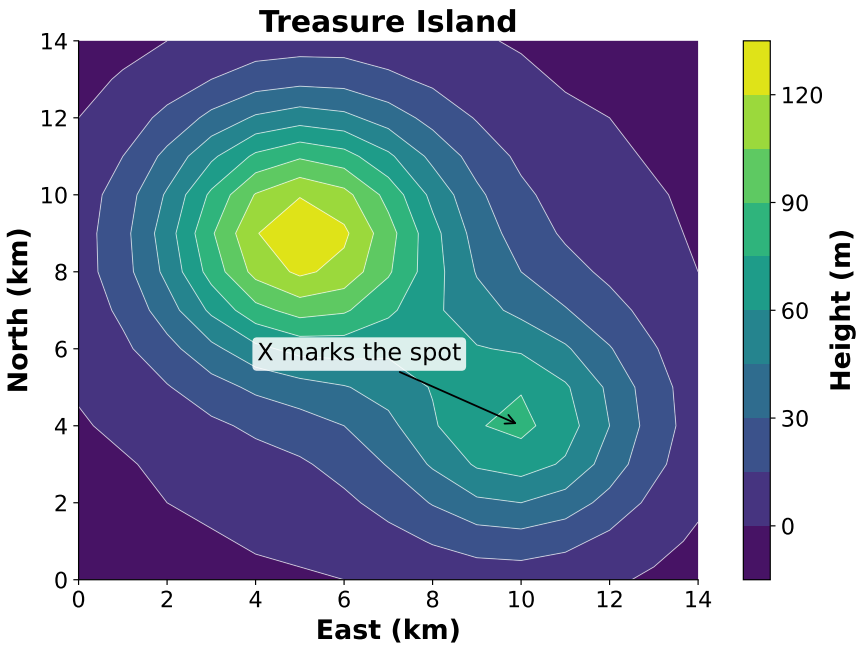


Figure 8.5: Lesson 8, example 5: `lesson8_my_program.png`.

Good to know

- Those are all the charts! `sdata.words()` lists every word and `sdata.help("word")` explains any of them. Lessons 9 to 12 teach tables: how to look at your data, clean it and sum it up before you draw it.
- Your own data: save it from Excel as `.xlsx` or `.csv`, then chart = `sdata.chart("my_file.xlsx")`. If a column name is spelt wrong, simplydata tells you the right one.

Try it yourself

1. Write your own program, from an empty file, for a question you care about.
2. Use any example table, or your own. Give it a title, axis names and a note.
3. Make a second chart of the same table, put the two charts side by side and save the page as a PDF.

Worked solutions: page 172, or type `sdata.lesson(8, solutions=True)`.

Words in this lesson: `chart.second_y()`, `chart.zoom_box()`, `chart.heatmap()`, `chart.label_colour_bar()`, `chart.size()`, `chart.animate()`, `chart.contour()`, `chart.note()`, `chart.style()`, `sdata.words()`, `sdata.help()`.

LESSON 9

Tables: open one and look at it

In this lesson you will learn

- how to turn data into a simplydata table
- how to look at the first and last rows, and the names of the columns
- how to describe every column and count what is in it

Example 1

Before you draw, look at your data. `sdata.table` gives you plain words for that.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
print(table)
table.show_first_rows(5)
table.show_last_rows(3)
print(table.column_names())
table.describe_columns()
table.count_rows()
```

What each line does

1. Load simplydata.
2. Make a table from the practice marks. For your own file, write `sdata.table("my_file.xlsx")`.
3. Print the table. The first line says how many rows and columns it has. Then come the first 10 rows.
4. Show only the first 5 rows.
5. Show the last 3 rows.
6. Print the names of the columns.
7. Describe every column: is it numbers or text, how many cells are empty, and the smallest, biggest and average number.
8. Count the rows, and say the answer in a sentence. One row is one child in one test.

What it prints

```
simplydata table: 54 rows, 6 columns
   name class  subject  hours  marks  height
0  Asha     A    Maths    6.3     61     144
1  Asha     A  Science    3.8     66     144
2  Asha     A  English    7.6     77     144
3   Ben     A    Maths    6.4     73     152
4   Ben     A  Science    9.2     98     152
5   Ben     A  English    6.2     71     152
6  Chen     A    Maths    7.9     81     151
7  Chen     A  Science    3.5     59     151
8  Chen     A  English    2.5     46     151
9  Dina     A    Maths    6.5     74     155
   name class  subject  hours  marks  height
```

```
0  Asha     A    Maths    6.3     61     144
1  Asha     A  Science    3.8     66     144
2  Asha     A  English    7.6     77     144
3   Ben     A    Maths    6.4     73     152
4   Ben     A  Science    9.2     98     152
    name class  subject  hours  marks  height
51  Rosa     C    Maths    3.4     55     162
52  Rosa     C  Science    1.6     45     162
53  Rosa     C  English    7.9     71     162
['name', 'class', 'subject', 'hours', 'marks', 'height']
The table has 54 rows and 6 columns.

 column   kind  empty cells smallest biggest average different values
   name   text            0                                       18
  class   text            0                                        3
subject   text            0                                        3
  hours number            0      0.5     9.4    5.33
  marks number            0       31      98   64.91
 height number            0      141     162  150.39
The table has 54 rows.
```

Example 2

`count_each` counts how often each different value comes up.

```
import simplydata as sdata
table = sdata.table(sdata.example("fruit"))
table.show_first_rows()
table.count_each("fruit")
table.count_each("shop")
```

What each line does

1. Load simplydata.
2. Make a table of fruit sold by two shops.
3. Show the first rows. With nothing in the brackets you get 5.
4. For each fruit: in how many rows it comes up. It says the answer in a sentence, and gives back a small table that also has the percent of all the rows.
5. The same for each shop.

What it prints

```
   day          shop   fruit  count  price
0  Mon   Corner shop   apple     39   0.44
1  Mon  Market stall  banana     22   0.23
2  Mon  Market stall   grape     13   0.05
3  Mon  Market stall    pear      6   0.41
4  Tue   Corner shop   apple     39   0.44
How many rows have each "fruit" (6 different values): apple 9, banana 8, grape 5, cherry 4, orange 4, pear
    ↪ 3.
How many rows have each "shop" (2 different values): Market stall 19, Corner shop 14.
```

Good to know

- Looking never changes the table. Only steps that say what they change (`keep_rows`, `sort_by` ...) do, and lesson 10 shows them.
- In Jupyter, VS Code or Spyder, write table on the last line of a cell to see it as a nice

table.

- `sdata.table` reads the same files as `sdata.chart`: `.xlsx`, `.csv`, `.tsv`, `.parquet` and `.json`. For one sheet of an Excel file: `sdata.table("results.xlsx", sheet="2026")`.
- The row numbers on the left start at 0, as everywhere in Python.

Try it yourself

1. Make a table of `sdata.example("weather")`. How many rows does it have?
2. Describe its columns. Which column has the biggest numbers?
3. Count each `"city"`. Does every city have the same number of rows?

Worked solutions: page 174, or type `sdata.lesson(9, solutions=True)`.

Words in this lesson: `sdata.table()`, `table.show_first_rows()`, `table.show_last_rows()`, `table.column_names()`, `table.describe_columns()`, `table.count_rows()`, `table.count_each()`.

LESSON 10

Tables: choose rows and columns, and undo

In this lesson you will learn

- how to keep only the columns and rows you want
- how to sort the rows
- how to take a step back, and start again
- how to save your table

Example 1

A table is changed in small steps, like a chart. Each step says what it did.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_columns("name", "subject", "marks")
table.keep_rows("subject", equal_to="Maths")
table.keep_rows("marks", above=60)
table.sort_by("marks", biggest_first=True)
table.show_first_rows(10)
table.steps()
table.undo()
table.start_again()
print(table)
```

What each line does

1. Load simplydata.
2. Make a table of the marks.
3. Keep only these three columns, in this order.
4. Keep only the rows where the subject is Maths. simplydata says how many rows it kept.
5. Of those, keep only marks above 60.
6. Put the rows in order, the biggest mark first.
7. Show the first 10 rows: the best Maths marks.
8. List every step you did, in order.
9. Take back the last step (the sorting).
10. Go back to the whole table, as it was at the start.
11. All 54 rows and 6 columns are back.

What it prints

```
Kept 18 of 54 rows.
Kept 11 of 18 rows.
     name subject  marks
0    Hana   Maths     91
1    Chen   Maths     81
2    Jade   Maths     81
3    Dina   Maths     74
```

```
4  Fatima   Maths     74
5     Ben   Maths     73
6    Noah   Maths     72
7    Kofi   Maths     68
8     Gus   Maths     67
9    Omar   Maths     62
1. keep columns name, subject, marks
2. keep rows where subject is "Maths"
3. keep rows where marks is above 60
4. sort by marks (biggest first)
Undone: sort by marks (biggest first)
simplydata table: 54 rows, 6 columns
   name class  subject  hours  marks  height
0  Asha     A     Maths    6.3     61     144
1  Asha     A  Science    3.8     66     144
2  Asha     A  English    7.6     77     144
3   Ben     A     Maths    6.4     73     152
4   Ben     A  Science    9.2     98     152
5   Ben     A  English    6.2     71     152
6  Chen     A     Maths    7.9     81     151
7  Chen     A  Science    3.5     59     151
8  Chen     A  English    2.5     46     151
9  Dina     A     Maths    6.5     74     155
```

Example 2

`keep_rows` and `remove_rows` know many conditions. Then save what you made.

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.keep_rows("month_number", between=[6, 8])
table.remove_rows("city", equal_to="Frostby")
table.remove_columns("wind_direction", "wind_speed")
table.rename_column("temperature", to="temp")
print(table)
table.save("lesson10_summer.xlsx")
```

What each line does

1. Load simplydata.
2. Make a table of the weather.
3. Keep the summer months: 6, 7 and 8. `between=` keeps both ends.
4. Remove the rows of the city Frostby.
5. Remove two columns we do not need.
6. Give the column `"temperature"` the shorter name `"temp"`.
7. Look at what is left: 6 rows and 5 columns.
8. Save the table as an Excel file. A name ending in `.csv` saves a CSV file.

What it prints

```
Kept 9 of 36 rows.
Removed 3 of 9 rows. 6 rows left.
simplydata table: 6 rows, 5 columns
  month  month_number        city  temp  rain
0   Jun             6   Sunnyvale  24.0    40
1   Jul             7   Sunnyvale  26.5    44
2   Aug             8   Sunnyvale  22.8    36
3   Jun             6  Rainbridge  15.3    72
```

```
4   Jul             7  Rainbridge  16.2    58
5   Aug             8  Rainbridge  15.3    76
```

Good to know

- Conditions for `keep_rows` and `remove_rows`: above, below, `at_least`, `at_most`, `equal_to`, `not_equal_to`, between, `one_of`, contains, `starts_with`, `ends_with` and `is_empty`.
- `one_of=` takes a list: `table.keep_rows("subject", one_of=["Maths", "Science"])`.
- `undo()` takes back one step at a time. `start_again()` is a step too, so `undo()` can take it back.
- Your file is never changed: only `table.save(...)` writes a file.

Try it yourself

1. In the marks table, keep only class `"B"` and sort by `"hours"`. Who studied the longest?
2. Undo the sorting, then sort by name instead.
3. Keep 5 random rows with `table.keep_random_rows(5, seed=1)` and save them as `"five.csv"`.

Worked solutions: page 175, or type `sdata.lesson(10, solutions=True)`.

Words in this lesson: `table.keep_columns()`, `table.keep_rows()`, `table.sort_by()`, `table.steps()`, `table.undo()`, `table.start_again()`, `table.remove_rows()`, `table.remove_columns()`, `table.rename_column()`, `table.save()`, `table.keep_random_rows()`.

LESSON 11

Tables: clean the data and add new columns

In this lesson you will learn

- how to find and fill empty cells
- how to fix columns: numbers written as text, extra spaces, wrong values
- how to add new columns, in plain words or like a sum in maths
- what long and wide tables are

Example 1

Real data often has empty cells. The practice tables have none, so we make some first.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.replace_values("marks", [61, 73], with_value=None)
table.count_empty_cells()
table.fill_empty_cells("marks", with_the="average")
table.undo()
table.remove_rows_with_empty_cells("marks")
print(table)
```

What each line does

1. Load simplydata.
2. Make a table of the marks.
3. Make every mark of 61 or 73 empty. `with_value=None` means an empty cell.
4. Show how many empty cells each column has.
5. Fill the empty marks with the average of the other marks.
6. Changed your mind? Take it back: the cells are empty again.
7. This time remove the rows with an empty cell in marks.
8. Some rows fewer than at the start.

What it prints

```
Replaced 2 cells in marks.
Empty cells in each column:
  name     0
  class    0
  subject  0
  hours    0
  marks    2
  height   0
Total: 2 cells empty.
Filled 2 cells in marks.
Undone: fill empty cells in marks with the average
Removed 2 of 54 rows. 52 rows left.
simplydata table: 52 rows, 6 columns
   name class  subject  hours  marks  height
```

```
0  Asha     A  Science    3.8   66.0     144
1  Asha     A  English    7.6   77.0     144
2   Ben     A  Science    9.2   98.0     152
3   Ben     A  English    6.2   71.0     152
4  Chen     A    Maths    7.9   81.0     151
5  Chen     A  Science    3.5   59.0     151
6  Chen     A  English    2.5   46.0     151
7  Dina     A    Maths    6.5   74.0     155
8  Dina     A  Science    4.5   79.0     155
9  Dina     A  English    2.3   57.0     155
```

Example 2

Data typed by people is often untidy. These words fix whole columns at once.

```
import simplydata as sdata
table = sdata.table({"name": ["  Asha", "ben ", "Chen"], "height": ["144", "152", "unknown"
    ↪ ]})
table.remove_extra_spaces("name")
table.replace_values("name", "ben", with_value="Ben")
table.make_numbers("height")
table.describe_columns()
```

What each line does

1. Load simplydata.
2. Make a small table by hand: each column name, then its values in a list.
3. Take away the spaces before and after the names.
4. Write `"ben"` with a capital letter.
5. Turn the heights from text into numbers. `"unknown"` is not a number, so that cell becomes empty, and simplydata tells you.
6. Now height is a column of numbers.

What it prints

```
Replaced 1 cell in name.
1 cell could not become numbers and is now empty, for example: "unknown"
height is now numbers.
The table has 3 rows and 2 columns.

column   kind  empty cells smallest biggest average different values
  name   text            0                                        3
height number            1     144     152     148
```

Example 3

A new column can be worked out from other columns, in plain words.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.add_column("marks_per_hour", by_dividing=["marks", "hours"])
table.add_group_column("grade", from_column="marks", cut_at=[40, 60, 80], names=["D", "C", "B
    ↪ ", "A"])
table.show_first_rows()
table.count_each("grade")
```

What each line does

1 Load simplydata.
2 Make a table of the marks.
3 A new column called `"marks_per_hour"`: the marks divided by the hours.
4 A grade for each mark: below 40 is D, 40 to 59 is C, 60 to 79 is B, 80 and more is A.
5 Look at the two new columns on the right.
6 How many of each grade?

What it prints

```
   name class  subject  hours  marks  height  marks_per_hour grade
0  Asha     A    Maths    6.3     61     144        9.682540     B
1  Asha     A  Science    3.8     66     144       17.368421     B
2  Asha     A  English    7.6     77     144       10.131579     B
3   Ben     A    Maths    6.4     73     152       11.406250     B
4   Ben     A  Science    9.2     98     152       10.652174     A
How many rows have each "grade" (4 different values): B 25, C 18, A 8, D 3.
```

Example 4

Or write the new column like a sum in maths. Here each subject first gets its own column: the table goes from long to wide.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_columns("name", "class", "subject", "marks")
table.turn_rows_into_columns(names_from="subject", values_from="marks")
table.add_column("total", "Maths + Science + English")
table.add_column("average", "total / 3")
table.add_rank_column("place", from_column="total")
table.sort_by("place")
table.show_first_rows()
```

What each line does

1 Load simplydata.
2 Make a table of the marks.
3 Keep the columns we need.
4 Make one column for each subject: Maths, Science and English. Each child now has one row.
5 A new column `"total"`: the three marks added up.
6 The average of the three marks.
7 Each child's place: 1 for the biggest total.
8 Put the rows in order of place.
9 Who came first?

What it prints

```
     name class  Maths  Science  English  total    average  place
0     Ben     A     73       98       71    242  80.666667      1
1  Fatima     A     74       87       81    242  80.666667      1
2    Hana     B     91       71       77    239  79.666667      3
3     Mia     C     59       87       76    222  74.000000      4
4    Dina     A     74       79       57    210  70.000000      5
```

Good to know

- Long or wide? A long table has one row for each measurement: the marks table has a row for each child in each subject. A wide table puts the measurements of one child side by side, one column per subject. Charts and tests like long tables (also called tidy data); sums across columns, like the total above, need wide ones.
- Filling empty cells is called imputation. Filling with the average makes the values look less spread out than they really are, so say in your report how you filled them. `with_the="median"`, `"value above"` or `"value below"` and `with_value=0` work too.
- `remove_rows_with_empty_cells()` with nothing in the brackets removes every row that has an empty cell in any column.
- More fixing words: `make_text`, `make_dates`, `make_lowercase` and `make_uppercase`.
- Plain words for new columns: `by_adding=`, `by_subtracting=`, `by_multiplying=` and `by_dividing=`, or `from_column=` with `plus=`, `minus=`, `times=` or `divide_by=`.
- In a maths-style column, a column name with spaces goes in backticks: `"‘max temp‘ - 2"`.

Try it yourself

1. In the first example, fill the empty marks with the median instead of the average.
2. Add a column `"height_m"` with the height in metres: `from_column="height"`, `divide_by=100`.
3. In the last example, add a column `"best_two"` with the formula `"total - English"`.

Worked solutions: page 176, or type `sdata.lesson(11, solutions=True)`.

Words in this lesson: `table.replace_values()`, `table.count_empty_cells()`, `table.fill_empty_cells()`, `table.remove_rows_with_empty_cells()`, `table.remove_extra_spaces()`, `table.make_numbers()`, `table.add_column()`, `table.add_group_column()`, `table.add_rank_column()`, `table.turn_rows_into_columns()`.

LESSON 12

Tables: summaries, and from a table to a chart

In this lesson you will learn

- how to work out averages, totals and counts, for each group
- how to tell if two columns go together, and why that is not the same as a cause
- how to draw a table, and which rows a chart uses
- how to join two tables and turn columns into rows

Example 1

A summary turns many rows into a few numbers. It never changes your table, and it says its answer in a sentence.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.average_of("marks")
table.average_of("marks", for_each="class")
table.biggest_of("marks", for_each="subject")
table.correlation_between("hours", "marks")
```

What each line does

1. Load simplydata.
2. Make a table of the marks.
3. The average of all the marks: just one number.
4. One average for each class. `for_each=` also gives back a new, small table.
5. The best mark in each subject.
6. Do more hours go with more marks? The answer is a number r from -1 to 1 (Pearson's correlation), explained in words.

What it prints

```
The average of "marks" is 64.91 (54 values).
The average of "marks" for each "class": A 71.83, B 64.28, C 58.61.
The biggest of "marks" for each "subject": English 81, Maths 91, Science 98.
The correlation between "hours" and "marks" is r = 0.83 (Pearson, 54 rows): very strong, going up, and very
    ↪ unlikely to be chance (p < 0.0001).
```

Example 2

A summary for each group is a table too, so you can sort it and draw it.

```
import simplydata as sdata
table = sdata.table(sdata.example("fruit"))
totals = table.total_of("count", for_each="fruit")
totals.sort_by("total of count", biggest_first=True)
```

```
chart = totals.chart()
chart.bars(x="fruit", y="total of count")
chart.numbers_on_bars()
chart.save("lesson12_fruit.png")
```

What each line does

1. Load simplydata.
2. Make a table of fruit sales.
3. A new table called totals: one row per fruit, with a column `"total of count"`. table itself does not change.
4. Best seller first.
5. Start a chart from the summary table.
6. One bar per fruit.
7. Write the total on each bar.
8. Save the chart.

What it prints

```
The total of "count" for each "fruit": apple 337, banana 205, cherry 59, grape 117, orange 73, pear 20.
```

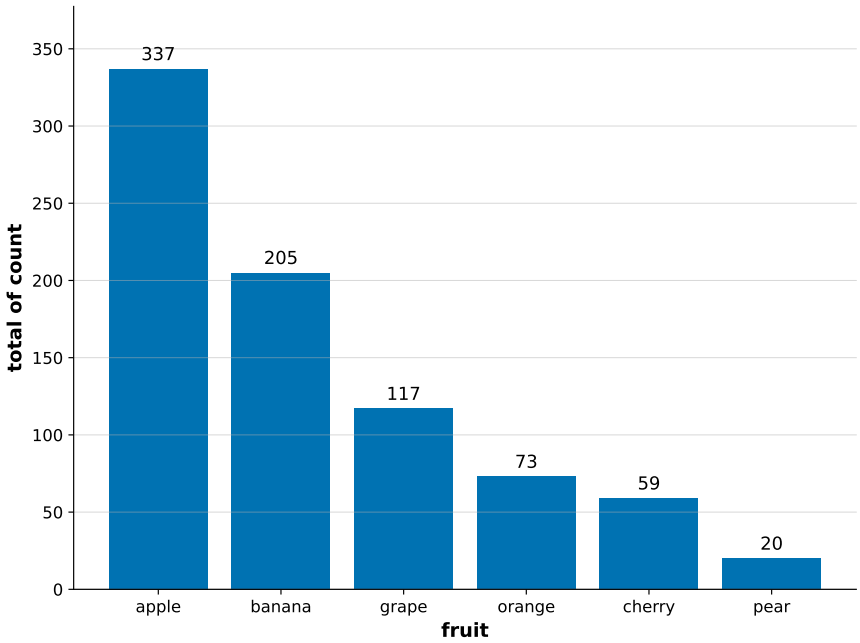


Figure 12.1: Lesson 12, example 2: `lesson12_fruit.png`.

Example 3

A chart uses the table as it was when you made the chart. Table steps after it are for the next chart.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_rows("class", equal_to="A")
all_tests = table.chart()
all_tests.dots(x="hours", y="marks")
table.keep_rows("subject", equal_to="Maths")
maths = table.chart()
```

```
maths.dots(x="hours", y="marks")
page = sdata.side_by_side(all_tests, maths, same_y=True)
page.save("lesson12_maths.png")
```

What each line does

1. Load simplydata.
2. Make a table of the marks.
3. Keep class A.
4. A chart of the table as it is now: every test of class A. `sdata.chart(table)` does the same.
5. Draw hours against marks.
6. A table step after the chart. The chart `all_tests` does not change.
7. A second chart, of the table as it is now: only the Maths tests of class A.
8. Draw the same thing again.
9. Both charts on one page, with the same y-axis, so you can compare them.
10. Save the page.

What it prints

```
Kept 18 of 54 rows.
Kept 6 of 18 rows.
```

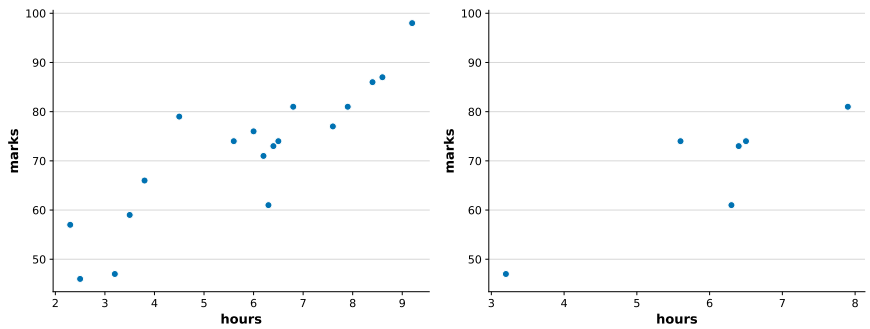


Figure 12.2: Lesson 12, example 3: `lesson12_maths.png`.

Example 4

Two tables can be joined: their rows are matched on a column they share.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
ages = {"name": ["Asha", "Ben", "Chen"], "age": [12, 13, 12]}
table.join_with(ages, using="name")
table.show_first_rows()
table.average_of("marks", for_each="age")
```

What each line does

1. Load simplydata.
2. Make a table of the marks.
3. A small table of ages, typed by hand: each column name, then its values.
4. Bring in the age, matching the rows on the name. Only children found in both tables stay.
5. Look: there is a new column, age.

6 The average mark at each age.

What it prints

```
9 of 54 rows in this table found a match in the other table.
   name class  subject  hours  marks  height  age
0  Asha     A    Maths    6.3     61     144   12
1  Asha     A  Science    3.8     66     144   12
2  Asha     A  English    7.6     77     144   12
3   Ben     A    Maths    6.4     73     152   13
4   Ben     A  Science    9.2     98     152   13
The average of "marks" for each "age": 12 65, 13 80.67.
```

Example 5

Turning columns into rows makes a wide table long: the shape most charts like.

```
import simplydata as sdata
table = sdata.table({"name": ["Asha", "Ben", "Chen"], "test_1": [61, 73, 81], "test_2": [70,
    ↪ 68, 90]})
table.turn_columns_into_rows(["test_1", "test_2"], names_to="test", values_to="mark")
table.show_first_rows(6)
chart = table.chart()
chart.bars(x="name", y="mark", colour="test")
chart.save("lesson12_tests.png")
```

What each line does

1 Load simplydata.
2 A small, wide table: two tests, one column for each.
3 Put the two tests under each other. The new column `"test"` says which test, and `"mark"` holds the mark. Each child now has two rows.
4 Look at the new, long shape.
5 Start a chart from it.
6 Both tests side by side for each child.
7 Save it.

What it prints

```
    name    test  mark
0  Asha  test_1    61
1   Ben  test_1    73
2  Chen  test_1    81
3  Asha  test_2    70
4   Ben  test_2    68
5  Chen  test_2    90
```

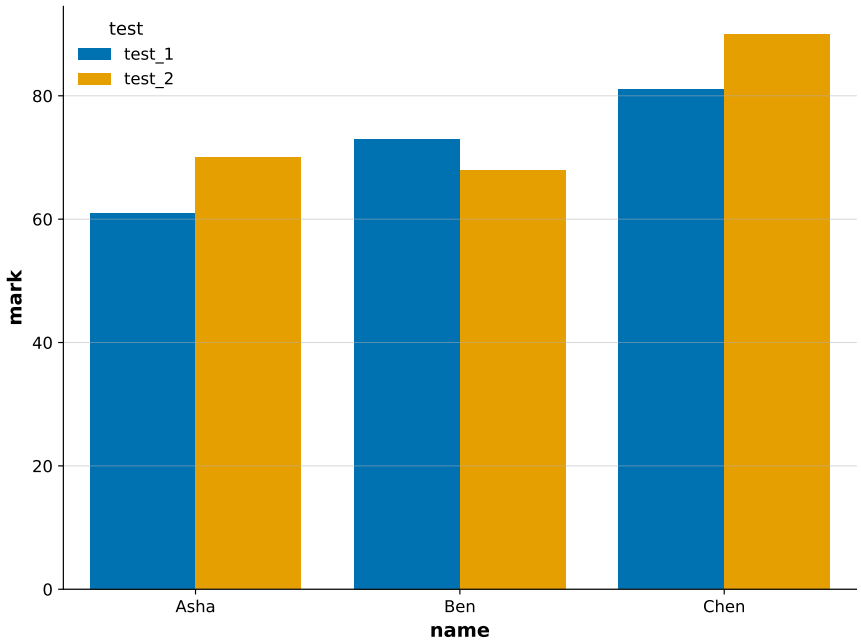


Figure 12.3: Lesson 12, example 5: `lesson12_tests.png`.

Good to know

- Correlation is not cause. Children who study more might also sleep more, or like the subject more; r only says that two columns go up and down together, not which one makes the other change.
- More summaries: `median_of`, `total_of`, `smallest_of`, `biggest_of`, `standard_deviation_of` and `count_rows`. Each one can take `for_each=`.
- A chart never changes its table, and a table step never changes a chart you already made. So one cleaned table can feed many charts.
- `table.add_rows_from(other)` puts the rows of another table under this one. `turn_rows_into_columns` does the opposite of `turn_columns_into_rows`.
- Lesson 13 teaches numbers, and lesson 14 the statistical tests: is a difference real, or could it be chance?

Try it yourself

1. Make the fruit totals `for_each="shop"` instead. Which shop sold more?
2. Work out the average temperature `for_each="city"` in the weather table and draw it as bars.
3. Is there a link between height and marks? Use `correlation_between` to find out.

Worked solutions: page 178, or type `sdata.lesson(12, solutions=True)`.

Words in this lesson: `table.average_of()`, `table.biggest_of()`, `table.correlation_between()`, `table.total_of()`, `table.chart()`, `sdata.side_by_side()`, `table.join_with()`, `table.turn_columns_into_rows()`, `table.add_rows_from()`.

LESSON 13

Numbers: lists and grids of numbers

In this lesson you will learn

- how to make a list of numbers, and a grid of them
- two ways to do maths: symbols make new numbers, words change the numbers you have
- how to keep some numbers, sum them up, and draw them

Example 1

Numbers are for when you have just numbers, not a table with named columns.

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
print(nums)
bigger = nums * 2
print(bigger)
nums.add(1)
nums.undo()
nums.keep_numbers(above=10)
nums.average()
nums.standard_deviation()
nums.steps()
```

What each line does

1. Load simplydata.
2. Make a list of five numbers and call it nums.
3. Look at them. The first line says how many there are.
4. Maths with symbols makes NEW numbers: every number times 2. nums itself does not change.
5. 24, 30, 18, 40 and 28.
6. Maths with words CHANGES the numbers you have: add 1 to each one.
7. Like a table, numbers can take back every word.
8. Keep only the numbers above 10. The same conditions as `keep_rows`: above, below, between ...
9. The average of what is left, said in a sentence.
10. How spread out they are (divide by n - 1, as in pandas and R).
11. List the steps you did.

What it prints

```
simplydata numbers: 5 numbers
[12, 15,  9, 20, 14]
simplydata numbers: 5 numbers
[24, 30, 18, 40, 28]
Undone: add 1
Kept 4 of 5 numbers.
The average of the numbers is 15.25 (4 numbers).
The standard deviation of the numbers is 3.4 (4 numbers).
1. keep numbers above 10
```

Example 2

Counting, random numbers and a chart.

```
import simplydata as sdata
months = sdata.numbers(1, to=12)
temps = sdata.numbers([4, 6, 9, 13, 17, 20, 22, 21, 17, 12, 8, 5])
chart = sdata.chart()
chart.lines(x=months, y=temps, dots=True)
chart.save("lesson13_line.png")
heights = sdata.random_numbers(how_many=200, average=150, standard_deviation=8, seed=1)
heights.median()
heights.percentile(90)
chart = heights.chart()
chart.histogram(x="value", bins=12)
chart.save("lesson13_heights.png")
```

What each line does

1. Load simplydata.
2. Count from 1 to 12. The end is included.
3. A temperature for each month.
4. Start an empty chart: the numbers go straight into the chart type.
5. Draw the temperatures as a line, month by month.
6. Save it.
7. 200 made-up heights that bunch around 150 in a bell shape (a normal distribution). `seed=1` gives the same numbers every time, so others can repeat your work.
8. The middle height.
9. The height that 90 % of them are below.
10. Start a chart from the numbers. Their column is called value.
11. Count how many heights fall in each range.
12. Save it. Can you see the bell shape?

What it prints

```
The median of the numbers is 150.02 (200 numbers).
The 90th percentile of the numbers is 157.2 (200 numbers).
```

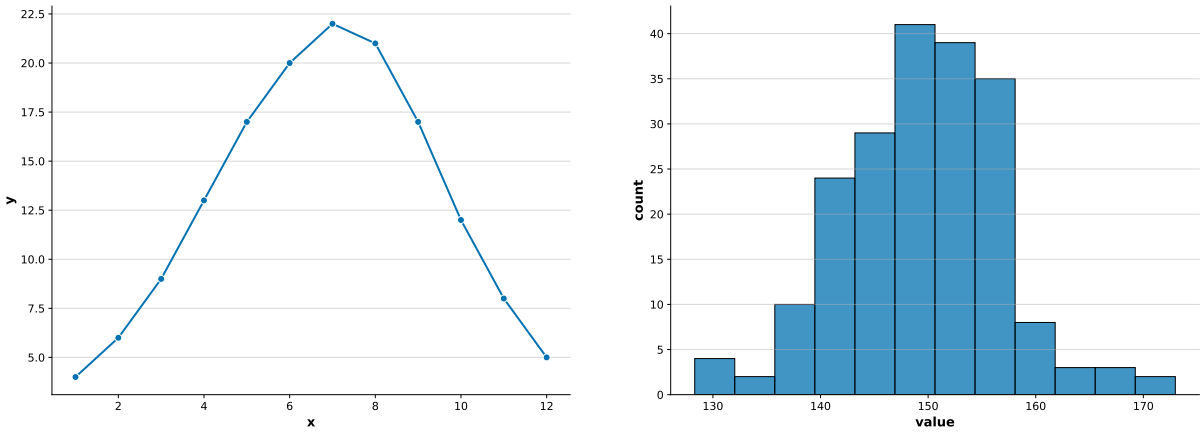


Figure 13.1: Lesson 13, example 2: `lesson13_line.png` and `lesson13_heights.png`.

Example 3

A grid holds numbers in rows and columns. Positions count from 1.

```
import simplydata as sdata
grid = sdata.number_grid([[3, 5, 2], [8, 6, 4]])
print(grid.number_at(row=1, column=2))
grid.total(for_each="row")
grid.average(for_each="column")
grid.swap_rows_and_columns()
print(grid)
chart = grid.chart()
chart.heatmap(numbers=True)
chart.save("lesson13_grid.png")
```

What each line does

1. Load simplydata.
2. A grid with 2 rows and 3 columns: one list for each row.
3. The number in row 1, column 2: 5.
4. The total of each row: 10 and 18.
5. The average of each column.
6. Rows become columns (in maths: the transpose).
7. Now 3 rows and 2 columns.
8. Start a chart from the grid.
9. One coloured square for each number, with the number written in.
10. Save it.

What it prints

```
5
The total of each row: 10, 18.
The average of each column: 5.5, 5.5, 3.
simplydata number grid: 3 rows, 2 columns
[[3, 8],
```

```
[5, 6],
[2, 4]]
```

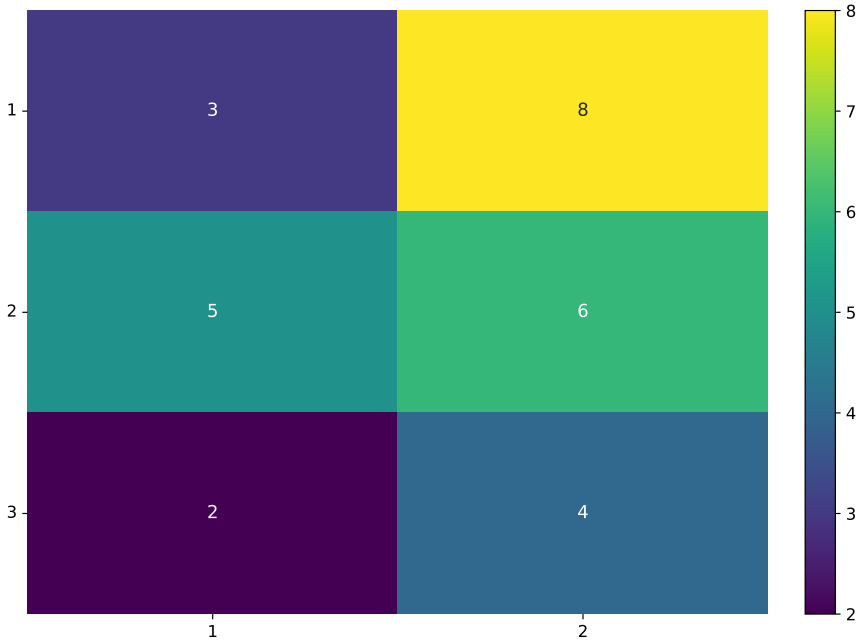


Figure 13.2: Lesson 13, example 3: `lesson13_grid.png`.

Good to know

- The words follow the table words, with numbers in place of rows: `keep_first_numbers`, `remove_empty_numbers`, `fill_empty_numbers`, `show_first_numbers`.
- Summaries have no _of, because there is no column to name: `nums.average()`, not `table.average_of("marks")`.
- A sum with no answer, such as dividing by 0 or the square root of a negative number, gives an empty value and a note, never a crash.
- `nums.table()` turns numbers into a table, and `nums.as_numpy()` gives a plain NumPy array if another Python tool needs one.

Try it yourself

1. Make the numbers 1 to 10, then square them in two ways: with ** 2 and with `power(2)`.
2. Make 6 dice throws with `sdata.random_numbers(how_many=6, between=[1, 6], whole_numbers=True, seed=1)`. What is their total?
3. Make a grid of 3 rows and 4 columns filled with 0: `sdata.number_grid(rows=3, columns=4, filled_with=0)`.

Worked solutions: page 179, or type `sdata.lesson(13, solutions=True)`.

Words in this lesson: `sdata.numbers()`, `sdata.random_numbers()`, `sdata.number_grid()`, `nums.add()`, `nums.undo()`, `nums.keep_numbers()`, `nums.average()`, `nums.standard_deviation()`, `nums.median()`, `nums.percentile()`, `nums.chart()`, `grid.number_at()`, `grid.swap_rows_and_columns()`, `nums.steps()`.

LESSON 14

Statistical tests: real difference, or chance?

In this lesson you will learn

- what a statistical test asks, and what a p-value means
- how to compare the averages of groups, and which groups differ
- how to compare counts, and how to fit a line or a formula

Example 1

Groups almost never have exactly the same average. A test asks: is the difference bigger than chance alone would often give?

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.compare_averages("marks", between="class")
table.compare_each_pair("marks", between="class")
table.compare_averages("marks", between="subject", based_on="ranks")
```

What each line does

1. Load simplydata.
2. Make a table of the marks.
3. Do the three classes really differ in their average mark? The answer comes in words. The p-value is the chance of a difference this big if the classes were really the same: below 0.05 we call it unlikely to be chance. With three or more groups this test is also called Welch's ANOVA; with two groups, a t-test.
4. Which classes differ? Each pair gets its own t-test (Welch's). Testing many pairs makes a false alarm more likely, so the p-values are adjusted (Holm's method).
5. `based_on="ranks"` puts the marks in order and compares the places instead (a Kruskal-Wallis test, or Mann-Whitney for two groups). It is safer when groups are small and the values are not bell-shaped.

What it prints

```
The average "marks" is 71.8 in A, 64.3 in B and 58.6 in C. Differences this big between the groups are
    ↪ unlikely to be chance (p = 0.028). Omega-squared = 0.0967: the groups explain about 10 % of the
    ↪ differences in "marks".
Test: Welch's one-way ANOVA (F = 3.99, df = 2 and 34.0), the usual test for three or more averages; it does
    ↪ not assume the groups have the same spread. To see which groups differ: table.compare_each_pair("
    ↪ marks", between="class").
Each pair of "class" groups compared on "marks" with Welch's t-test; the p-values are adjusted for 3 pairs
    ↪ with Holm's method, so the chance of any false alarm stays at 5 %.
  A vs B: the average of A is 7.56 higher; could easily be chance (Holm-adjusted p = 0.245).
  A vs C: the average of A is 13.2 higher; unlikely to be chance (Holm-adjusted p = 0.023).
  B vs C: the average of B is 5.67 higher; could easily be chance (Holm-adjusted p = 0.250).
The median "marks" is 62.0 in English, 64.5 in Maths and 70.0 in Science. Differences this big between the
    ↪ groups could easily be chance (p = 0.398); this does not show that there is no difference. Epsilon-
    ↪ squared = 0.0348: the groups explain about 3 % of the differences in the ranks.
```

```
Test: Kruskal-Wallis test (H = 1.84, df = 2), which compares ranks rather than averages, so it does not need
    ↪  bell-shaped values. To see which groups differ: table.compare_each_pair("marks", between="subject
    ↪ ", based_on="ranks").
```

Example 2

Counts can be compared too: does how often something happens differ between groups?

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.add_group_column("result", from_column="marks", cut_at=[60], names=["below 60", "60 or
    ↪ more"])
table.compare_counts("result", between="class")
```

What each line does

1. Load simplydata.
2. Make a table of the marks.
3. A new column that says if each mark is below 60 or not.
4. Do the classes differ in how many marks are 60 or more? This counts the rows for each class and result and uses the chi-square test. Cramer's V says how strong the link is, from 0 (none) to 1.

What it prints

```
Comparing how often each "result" turns up in each "class" (2 x 3 table, 54 rows): the differences could
    ↪ easily be chance (p = 0.122); this does not show that there is no difference. Cramer's V = 0.279 (0
    ↪  means no link, 1 a complete link).
Test: chi-square test of independence (chi-square = 4.21, df = 2), the usual test for counts.
```

Example 3

A fit finds the line or formula that follows the data best, and says how sure it is.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.fit_straight_line(x="hours", y="marks")
growth = sdata.table(sdata.example("growth"))
growth.keep_rows("country", equal_to="Applia")
growth.fit_formula("population = a * exp(b * (year - 2000))")
```

What each line does

1. Load simplydata.
2. Make a table of the marks.
3. How much do the marks go up for each extra hour of study? This is linear regression by least squares. R squared says how much of the differences in marks the line explains.
4. A second table, with the growth data.
5. Keep one country.
6. Fit a formula you write yourself. Names that are columns (population, year) are data; the others (a and b) are found by simplydata. Here b is the growth rate each year. This is nonlinear least squares.

What it prints

```
"marks" goes up by 5.20 for each 1 more "hours" (95 % CI 4.24 to 6.17); this slope is very unlikely to be
    ↪ chance (p < 0.0001). R squared = 0.69: "hours" explains about 69 % of the differences in "marks".
Line: marks = 5.20 * hours + 37.2, by least squares over 54 rows. A link does not show that one thing causes
    ↪  the other.
Kept 13 of 52 rows.
Fitted "population = a * exp(b * (year - 2000))" to 13 rows: a = 12.0 +/- 0.0787, b = 0.0297 +/- 0.000392 (
    ↪ value +/- standard error). R squared = 1.00 (only a rough guide for a curve).
Values found: a, b. Data: year. Method: nonlinear least squares, as R nls().
```

Good to know

- The idea behind every test: imagine there is no real difference (this is called the null hypothesis). The p-value says how surprising your data would be then. A small p-value means: hard to explain by chance alone.
- A p-value does not say how big or important the difference is: the effect size does that (omega-squared, Cohen's d, Cramer's V ...). And a large p-value does not show that there is no difference. It only means these data cannot tell.
- A 95 % confidence interval is the range the true value is very likely in. A narrow one means the data pin the value down well; a wide one means you need more data.
- Each test gives back a one-row table with all its numbers, so you can save it: result = `table.compare_averages(...)`, then `result.save("test.csv")`.
- Also: `compare_before_after` for two measurements of the same thing (a paired t-test), and `compare_average_to` for one average against a number you choose.

Try it yourself

1. Compare the average marks `between="subject"`. Could the differences be chance?
2. Is the average temperature different between the cities of `sdata.example("weather")`?
3. Fit a straight line of `"forest"` against `"year"` in `sdata.example("growth")` for one country.

Worked solutions: page 180, or type `sdata.lesson(14, solutions=True)`.

Words in this lesson: `table.compare_averages()`, `table.compare_each_pair()`, `table.compare_counts()`, `table.fit_straight_line()`, `table.fit_formula()`, `table.add_group_column()`.

LESSON 15

A whole analysis: from a question to a figure

In this lesson you will learn

- how to put all the lessons together: open, look, clean, sum up, test, draw and save
- how to make a figure with panels A and B, ready for a report or a paper
- how `steps()` writes down what you did, in plain words
- how to hand your table to pandas or Polars, and back

Example 1

A whole analysis with one question: do children who study longer get better marks, and do the classes differ? One table feeds two charts, and the two charts become one figure.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.describe_columns()
table.remove_rows_with_empty_cells()
table.average_of("marks", for_each="class")
table.compare_averages("marks", between="class")
a = table.chart()
a.boxes(x="class", y="marks")
a.all_dots(x="class", y="marks")
a.label_y("Mark (out of 100)")
b = table.chart()
b.dots(x="hours", y="marks", colour="class")
b.fit_line(confidence_band=True)
b.label_x("Hours of study")
figure = sdata.grid(a, b, per_row=2, letters=True)
figure.title_top("Marks, classes and hours of study")
figure.style("paper")
figure.save("lesson15_figure.pdf")
table.save("lesson15_clean.xlsx")
table.steps()
```

What each line does

1. Load simplydata.
2. Open the data. For your own file: `sdata.table("results.xlsx")`.
3. Look: what is in each column, and are any cells empty?
4. Clean: remove the rows with an empty cell. The practice table has none, so nothing is removed.
5. Sum up: the average mark of each class.
6. Test: could the differences between the classes be chance?
7. Chart A, from the cleaned table.
8. The marks of each class as boxes.

```
Every mark as a dot, on top of the boxes.
Name its y-axis.
Chart B, from the same table.
Hours of study against marks, one colour per class.
A straight line for each class, with its 95 % confidence band.
Name its x-axis.
Both charts on one page, with A and B at the top left.
A title for the whole page.
A quiet look that suits a paper.
Save the figure as a PDF: sharp at any size.
Save the cleaned table that the figure was drawn from.
Every table step in plain words, in order: a good start for the methods part of your report.
```

What it prints

```
The table has 54 rows and 6 columns.

 column   kind  empty cells smallest biggest average different values
   name   text            0                                       18
  class   text            0                                        3
subject   text            0                                        3
  hours number            0      0.5     9.4    5.33
  marks number            0       31      98   64.91
 height number            0      141     162  150.39
Removed 0 of 54 rows. 54 rows left.
The average of "marks" for each "class": A 71.83, B 64.28, C 58.61.
The average "marks" is 71.8 in A, 64.3 in B and 58.6 in C. Differences this big between the groups are
     ↪ unlikely to be chance (p = 0.028). Omega-squared = 0.0967: the groups explain about 10 % of the
     ↪ differences in "marks".
Test: Welch's one-way ANOVA (F = 3.99, df = 2 and 34.0), the usual test for three or more averages; it does
     ↪ not assume the groups have the same spread. To see which groups differ: table.compare_each_pair("
     ↪ marks", between="class").
1. remove rows with an empty cell
```

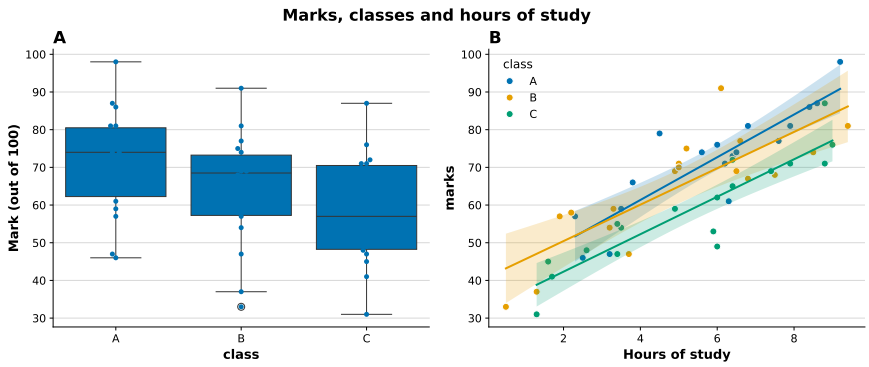


Figure 15.1: Lesson 15, example 1: `lesson15_figure.pdf`.

Example 2

Two questions about two tables give two charts. `sdata.grid` puts them on one page.

```
import simplydata as sdata
weather = sdata.table(sdata.example("weather"))
weather.keep_rows("city", equal_to="Rainbridge")
rain = weather.chart()
```

```
rain.bars(x="month", y="rain")
rain.line_at_y(100, label="100 mm")
growth = sdata.table(sdata.example("growth"))
forest = growth.chart()
forest.lines(x="year", y="forest", colour="country")
forest.label_y("Forest (% of the land)")
page = sdata.grid(rain, forest, per_row=2, letters=True)
page.save("lesson15_two_tables.png")
```

What each line does

1. Load simplydata.
2. Table 1: the weather.
3. Keep one city.
4. A chart of it, called rain.
5. Its rain through the year.
6. A line at 100 mm.
7. Table 2: four countries.
8. A chart of it, called forest.
9. Their forest over the years.
10. Name the y-axis.
11. Both charts on one page, with letters.
12. Save the page.

What it prints

```
Kept 12 of 36 rows.
```

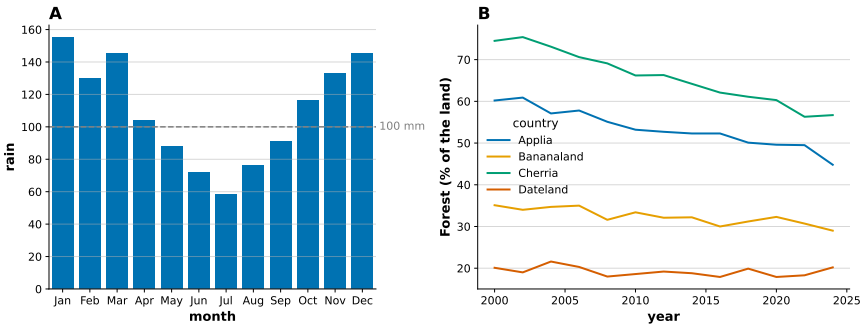


Figure 15.2: Lesson 15, example 2: `lesson15_two_tables.png`.

Example 3

A table goes to pandas or Polars and back whenever another Python tool needs it.

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.keep_rows("city", equal_to="Rainbridge")
df = table.as_pandas()
print(df["rain"].max())
again = sdata.table(df)
chart = sdata.chart(df)
chart.bars(x="month", y="rain")
chart.save("lesson15_rain.png")
```

What each line does

1 Load simplydata.
2 A table of the weather.
3 Keep one city.
4 A plain pandas DataFrame (`table.as_polars()` gives a Polars one).
5 Any pandas code works on it.
6 And back: `sdata.table` takes a pandas or Polars table you already have.
7 `sdata.chart` takes one too.
8 Draw it.
9 Save the chart.

What it prints

```
Kept 12 of 36 rows.
155
```

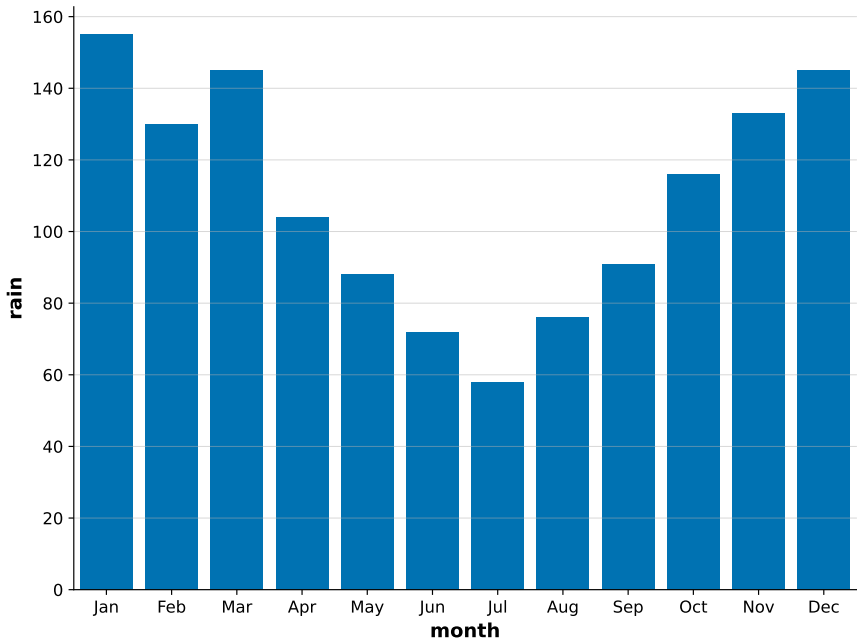


Figure 15.3: Lesson 15, example 3: `lesson15_rain.png`.

Good to know

- Every analysis follows the same plan: open, look, clean, sum up, test, draw, words, save. Lessons 1 to 14 taught each part; this lesson puts them together.
- The table and the charts are separate objects, as in pandas and matplotlib: one cleaned table can feed many charts, and each chart keeps the rows the table had when the chart was made.
- `table.save` saves the table, and `chart.save` or `page.save` the picture. Save both, so anyone can check the figure against its data.
- `table.steps()` lists the table steps. Together with the chart lines of your program, it says everything you did.
- simplydata words never need `as_pandas()`; it is only for other tools.

Try it yourself

1. Do a whole analysis of `sdata.example("weather")`: look at it, compare the average rain `between="city"`, draw the rain of each city as boxes, and save the chart and the table.
2. Make a second chart of the temperature, then put the two charts together with `sdata.side_by_side(..., letters=True)`.
3. Read the `steps()` of your table out loud. Would someone else understand what you did?

Worked solutions: page 181, or type `sdata.lesson(15, solutions=True)`.

Words in this lesson: `sdata.table()`, `table.describe_columns()`, `table.remove_rows_with_empty_cells()`, `table.compare_averages()`, `table.chart()`, `chart.boxes()`, `chart.all_dots()`, `chart.dots()`, `chart.fit_line()`, `sdata.grid()`, `chart.style()`, `table.save()`, `table.steps()`, `table.as_pandas()`, `sdata.chart()`.

PART III

Dictionary of words

How to read the dictionary

Each entry gives the word as you type it, what it does (in the words of `sdata.help("word")`), the options you can put in its brackets and what they normally are, the library that does the work underneath, and an example you can run as it is. The same entries are in Python: `sdata.help("bars")`. The words come in the same groups as `sdata.words()`; the index of words at the end of the book lists them A to Z.

Chart words

Steps on a chart: `chart = sdata.chart(table)`, then `chart.bars(...)`. Words that start with `sdata.` are called on simplydata itself, and words that end in `=` are options inside a chart step.

Start, show and save

`sdata.chart()`

Start a chart. `sdata.chart(sdata.example("marks"))`

`sdata.chart(table)` a pandas or Polars table, like `sdata.example("marks")`
`sdata.chart("my_file.xlsx")` your own file (Excel, CSV, TSV, Parquet, JSON)
`sdata.chart("my_file.xlsx", sheet="2026")` one sheet of an Excel file
`sdata.chart()` no table: give lists to each step instead
`fast=True` always use the fast engine (Polars) to read the file

Options

`data=`
`sheet=`
`fast=` normally `False`

Underneath pandas reads the file (Polars for big files); the chart is drawn with matplotlib and seaborn

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.save("my_chart.png")
```

`sdata.example()`

A small made-up table to practise with. `sdata.example("marks")`

`sdata.example()` with no name lists every table and its columns.
The tables are: marks, weather, fruit, growth, island.

Options

`name=`

Underneath pandas (`read_csv` of a small file inside simplydata)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
sdata.example()
```

```
marks = sdata.example("marks")
print(marks)
```

chart.show()

Show the chart in a window. `chart.show()`

In Jupyter, VS Code and Spyder charts appear by themselves.

Options none: just write the empty brackets.

Good to know Not the same as the `show=` option inside bars and lines: for that, type `sdata.help("show=")`.

Underneath matplotlib (pyplot.show)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="name", y="marks")
chart.show()
chart.save("marks.png")
```

chart.save()

Save the chart. `chart.save("marks.png")`

The end of the file name decides the format: `"marks.png"`, `"marks.pdf"`.

Options

`file_name`	needed
`quality=`	normally `"normal"`

Good to know File types: `.png` (a picture), `.pdf` and `.svg` (sharp at any size, good for printing), `.jpg`, `.tif`, and `.gif` for animations. `quality="high"` saves a bigger, sharper picture.

Underneath matplotlib (Figure.savefig); Pillow writes `.gif` animations

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="name", y="marks")
chart.save("marks.png")
chart.save("marks.pdf")
chart.save("marks_sharp.png", quality="high")
```

Chart types

chart.bars()

Also written `chart.bar()`

Bars. `chart.bars(x="name", y="marks")` (with `sdata.example("marks")`)

`colour="class"` gives each group its own colour, or `colour="blue"` one colour. `show=` says what each bar stands for when a group has many rows: `"average"` (default), `"median"`, `"total"`, `"count"`, `"smallest"`, `"biggest"` or `"standard deviation"`. `error="standard deviation"`, `"standard error"` or `"confidence interval"` adds error bars (with `show="average"`); a line says which one was drawn. `stacked=True` puts groups on top of each other; `sideways=True` lays bars down.

Options

`x=`	
`y=`	
`colour=`	
`show=`	normally `"average"`
`error=`	
`stacked=`	normally `False`
`sideways=`	normally `False`

Underneath seaborn (barplot; countplot for `show="count"`)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="name", y="marks", colour="class")
chart.rotate_labels_x()
chart.save("bars.png")
```

chart.lines()

Also written `chart.line()`

Lines. `chart.lines(x="month", y="temperature", colour="city")` (with `sdata.example("weather")`)

`colour="city"` draws one line per group. `dots=True` marks each point. `dashed=True` draws a dashed line. `show=` works as in bars (not `"count"`). `error="standard deviation"`, `"standard error"` or `"confidence interval"` adds a shaded band; a line says which one was drawn.

Options

`x=`	
`y=`	
`colour=`	
`show=`	normally `"average"`
`error=`	
`dots=`	normally `False`
`dashed=`	normally `False`

Underneath seaborn (lineplot)

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month", y="temperature", colour="city", dots=True)
chart.save("lines.png")
```

chart.dots()

Also written `chart.scatter()`

Dots (a scatter chart). `chart.dots(x="hours", y="marks")` (with `sdata.example("marks")`)

`colour="class"` colours dots by group. `size="height"` makes dots bigger for bigger values, or `size=10` makes all dots that big. `shape="subject"` gives each group a shape.

Options

`x=`
`y=`
`colour=`
`size=`
`shape=`

Underneath seaborn (scatterplot)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks", colour="class", size="height")
chart.save("dots.png")
```

chart.area()

A line with the space under it coloured in.
`chart.area(x="year", y="population", colour="country")` (with `sdata.example("growth")`)

`colour="country"` gives each group its own area; `stacked=True` piles the groups on top of each other, so the top edge shows the total.

Options

`x=`
`y=`
`colour=`
`stacked=` normally `False`

Underneath matplotlib (`fill_between`, or stackplot with `stacked=True`)

Example you can run (it uses the practice table `growth`)

```
import simplydata as sdata
growth = sdata.example("growth")
chart = sdata.chart(growth)
chart.area(x="year", y="population", colour="country", stacked=True)
chart.save("area.png")
```

chart.histogram()

Bars that count how many values fall in each range. `chart.histogram(x="marks", bins=10)` (with `sdata.example("marks")`)

`bins=` is how many bars (or a list of where each bar starts and ends). `colour="class"` draws one see-through histogram per group.

Options

`x=`	
`bins=`	normally `10`
`colour=`	

Underneath seaborn (histplot)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.histogram(x="marks", bins=10)
chart.save("histogram.png")
```

chart.boxes()

Also written `chart.box()`

Boxes that show the middle half of the values and the median.
`chart.boxes(x="class", y="marks")` (with `sdata.example("marks")`)

The line in the box is the median, the box holds the middle half of the values, the whiskers reach the rest, and lone dots are values far away from the others.

Options

`x=`	
`y=`	
`colour=`	

Underneath seaborn (boxplot)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.boxes(x="class", y="marks")
chart.save("boxes.png")
```

chart.violins()

Also written `chart.violin()`

Shapes that are wide where many values are. `chart.violins(x="class", y="marks")` (with `sdata.example("marks")`)

Each violin stops at the smallest and the largest value; the small box inside shows the middle half of the values.

Options

`x=`
`y=`
`colour=`

Underneath seaborn (violinplot)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.violins(x="subject", y="marks")
chart.save("violins.png")
```

chart.heatmap()

A grid of coloured squares, one colour for each number.
`chart.heatmap(x="month", y="city", value="rain", numbers=True)` (with `sdata.example("weather")`)

With no columns named, every number column of the table is used (rows are the table's rows): `chart.heatmap(numbers=True)`. `numbers=True` writes the number in each square. `cluster=True` puts similar rows and columns next to each other and draws a tree of how alike they are.

Options

`x=`	
`y=`	
`value=`	
`numbers=`	normally `False`
`cluster=`	normally `False`

Underneath seaborn (heatmap); SciPy (scipy.cluster.hierarchy) for `cluster=True`

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.heatmap(x="month", y="city", value="rain", numbers=True)
chart.save("heatmap.png")
```

chart.pie()

A circle cut into slices, one slice per group. `chart.pie(labels="fruit", values="count")` (with `sdata.example("fruit")`)

Without `values=` each slice shows how many rows have that label.

Options

`labels=`	
`values=`	

Underneath matplotlib (Axes.pie)

Example you can run (it uses the practice table `fruit`)

```
import simplydata as sdata
fruit = sdata.example("fruit")
chart = sdata.chart(fruit)
chart.pie(labels="fruit", values="count")
chart.save("pie.png")
```

chart.polar()

Dots on a round chart: how far round, and how far out. `chart.polar(angle="wind_direction", distance="wind_speed", colour="city")` (with `sdata.example("weather")`)

Angles are in degrees (a full turn is 360), going round from the right, the way maths measures angles. Use `degrees=False` if your angles are in radians.

Options

`angle=`	
`distance=`	
`colour=`	
`degrees=`	normally `True`

Underneath matplotlib (polar axes)

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.polar(angle="wind_direction", distance="wind_speed", colour="city")
chart.save("polar.png")
```

chart.contour()

A map where each colour band joins places with the same height.
`chart.contour(x="east", y="north", z="height")` (with `sdata.example("island")`)

`z=` is the height (or depth, or temperature). The points can be a full grid or scattered; scattered points are joined up with straight lines in between. `filled=False` draws only the lines, with their numbers written on them.

Options

`x=`
`y=`
`z=`
`filled=` normally `True`

Underneath matplotlib (contourf); SciPy (interpolate.griddata) for scattered points

Example you can run (it uses the practice table `island`)

```
import simplydata as sdata
island = sdata.example("island")
chart = sdata.chart(island)
chart.contour(x="east", y="north", z="height")
chart.save("contour.png")
```

chart.surface()

A 3D landscape: `z=` is how high the ground is.
`chart.surface(x="east", y="north", z="height")` (with `sdata.example("island")`)

The points can be a full grid or scattered; scattered points are joined up.

Options

`x=`
`y=`
`z=`

Underneath matplotlib (mplot3d `plot_surface`); SciPy (interpolate.griddata) for scattered points

Example you can run (it uses the practice table `island`)

```
import simplydata as sdata
island = sdata.example("island")
chart = sdata.chart(island)
chart.surface(x="east", y="north", z="height")
chart.save("surface.png")
```

chart.dots_3d()

Dots floating in a 3D box. `chart.dots_3d(x="hours", y="height", z="marks", colour="class")` (with `sdata.example("marks")`)

`colour="class"` gives each group its own colour.

Options

```
x=
y=
z=
colour=
```

Underneath matplotlib (mplot3d scatter)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots_3d(x="hours", y="height", z="marks", colour="class")
chart.save("dots_3d.png")
```

Statistics

show=

What each bar or point on a line stands for, when a group has many rows.

`show="average"` (the normal choice) adds up the values and divides by how many there are. `show="median"` is the middle value when they are put in order. `show="total"` adds them all up. `show="count"` counts the rows (bars only). `show="smallest"`, `"biggest"` and `"standard deviation"` work too. The names are the same as the table words: `average_of`, `median_of`, `total_of`, `smallest_of` ...

Works in `chart.bars()`, `chart.lines()`

Underneath seaborn (the estimator of barplot and lineplot)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="class", y="marks", show="median")
chart.save("median.png")
```

error=

Error bars on top of averages: how spread out, or how sure, the values are.

`error="standard deviation"` shows how far the values usually are from the average: how spread out they are. `error="standard error"` shows how sure we are about the average itself; it gets smaller as you collect more values. `error="confidence interval"` is the 95 % confidence interval: the range the true average is very likely to be in. Lines get a shaded band instead of bars. A line is printed saying which one was drawn, so your figure caption can say it too. The short forms `"sd"`, `"se"` and `"ci"` also work.

Works in `chart.bars()`, `chart.lines()`

Underneath seaborn (`errorbar="sd"` or `"se"`); the confidence interval from the t distribution (SciPy)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="class", y="marks", error="standard deviation")
chart.save("error.png")
```

chart.error_bars()

Error bars on dots or lines. `chart.error_bars("standard error")`

For every x value the bar shows the average of y and how sure or spread it is: `"standard deviation"` (n - 1): how spread out the values are around the average; `"standard error"`: how sure we are about the average itself; `"confidence interval"`: the 95 % range the true average is very likely in (t distribution). A line says which one was drawn. Empty cells are skipped. `column="error"` uses error sizes from your own column instead. With colour groups, bars at the same x move a little apart (with their lines and dots) so they do not hide each other; `dodge=False` keeps them on the true x.

Options

`kind=`	normally `"standard deviation"`
`column=`	
`dodge=`	normally `True`

Good to know Standard deviation or standard error? The standard deviation describes the values (how spread out they are). The standard error describes the average (how sure we are of it). Say in your caption which one you drew.

Underneath pandas (group averages) and matplotlib (errorbar); SciPy for the t-based confidence interval

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.lines(x="subject", y="marks", colour="class")
chart.error_bars("standard error")
chart.save("error_bars.png")
```

chart.fit_line()

A straight line that fits the dots best (linear regression). `chart.fit_line()`

The line is found by least squares: it makes the squared distances from the dots to the line as small as possible. `confidence_band=True` shades the 95 % confidence band: where the true line very likely is. `show_equation=True` writes the equation and R^2 on the chart. Each colour group gets its own line. `colour="red"` makes every line red.

Options

`confidence_band=`	normally `False`
`show_equation=`	normally `False`
`colour=`	

Underneath SciPy (stats.linregress), drawn with matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.fit_line(confidence_band=True, show_equation=True)
chart.save("fit_line.png")
```

chart.fit_curve()

A curve that fits the dots best. `chart.fit_curve(degree=2)`

`degree=2` is a curve with one bend (x^2), `degree=3` can bend twice (x^3), and so on. `show_equation=True` writes the equation and R^2 on the chart.

Options

`degree=`	normally `2`
`show_equation=`	normally `False`
`colour=`	

Underneath NumPy (polyfit), drawn with matplotlib

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.dots(x="month_number", y="temperature", colour="city")
chart.fit_curve(degree=2)
chart.save("fit_curve.png")
```

chart.smooth_line()

A smooth line that follows the dots (LOWESS). `chart.smooth_line()`

Good for seeing a trend that is not straight. Each colour group gets its own line.

Options

`colour=`

Underneath NumPy (LOWESS, written out in simplydata as in R's `lowess()`), drawn with matplotlib

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.dots(x="month_number", y="rain", colour="city")
chart.smooth_line()
chart.save("smooth_line.png")
```

chart.density()

A smooth curve that is high where many values are. `chart.density(x="marks", colour="class")` (with `sdata.example("marks")`)

`colour="class"` draws one curve per group; each curve has an area of 1.

Options

`x=`
`colour=`

Underneath seaborn (kdeplot)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.density(x="marks", colour="class")
chart.save("density.png")
```

chart.count_bars()

Bars that count how many rows each group has. `chart.count_bars(x="fruit", colour="shop")` (with `sdata.example("fruit")`)

`colour="shop"` splits each bar by group; `sideways=True` lays the bars down.

Options

`x=`
`colour=`
`sideways=` normally `False`

Underneath seaborn (countplot)

Example you can run (it uses the practice table `fruit`)

```
import simplydata as sdata
fruit = sdata.example("fruit")
chart = sdata.chart(fruit)
chart.count_bars(x="fruit", colour="shop")
chart.save("count_bars.png")
```

chart.all_dots()

Every value as a dot, spread out so no dot hides another.
`chart.all_dots(x="class", y="marks")` (with `sdata.example("marks")`)

Good on top of .boxes() or .violins(), to show each value as well.

Options

`x=`
`y=`
`colour=`

Underneath seaborn (swarmplot; stripplot for groups of more than 100 dots)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.boxes(x="class", y="marks")
chart.all_dots(x="class", y="marks")
chart.save("all_dots.png")
```

chart.side_histograms()

Small histograms along the top and right edges of a dots chart. `chart.side_histograms()`

Options none: just write the empty brackets.

Underneath matplotlib (hist), with NumPy choosing the bins

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.side_histograms()
chart.save("side_histograms.png")
```

`sdata.pairs()`

Every number column against every other. `sdata.pairs(sdata.example("marks"), colour="class")`

Histograms on the diagonal (density curves when `colour=` makes groups), dots everywhere else. `colour="class"` colours each group. `columns=["hours", "marks"]` chooses the columns. Empty cells are skipped. Then use it like any chart: .title_top(`"..."`), .save(`"pairs.png"`).

Options

`data`	needed
`colour=`	
`columns=`	

Underneath seaborn (scatterplot, histplot and kdeplot)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.pairs(marks, colour="class")
chart.title_top("Every number column against every other")
chart.save("pairs.png")
```

Lines and shading

`chart.line_at_x()`

A straight up-and-down line at one x value. `chart.line_at_x(2010, label="2010")`

Options

`value`	needed
`dashed=`	normally `True`
`colour=`	normally `"grey"`
`label=`	

Underneath matplotlib

Example you can run (it uses the practice table `growth`)

```
import simplydata as sdata
growth = sdata.example("growth")
chart = sdata.chart(growth)
chart.lines(x="year", y="population", colour="country")
chart.line_at_x(2010, label="2010")
chart.save("line_at_x.png")
```

`chart.line_at_y()`

A straight side-to-side line at one y value. `chart.line_at_y(50, label="pass mark")`

Options

`value`	needed
`dashed=`	normally `True`
`colour=`	normally `"grey"`
`label=`	

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="name", y="marks")
chart.line_at_y(50, label="pass mark")
chart.rotate_labels_x()
chart.save("line_at_y.png")
```

`chart.shade_between_x()`

Shade the band between two x values, behind the data. `chart.shade_between_x(6, 8)`

Options

`low`	needed
`high`	needed
`colour=`	normally `"grey"`

Underneath matplotlib

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month_number", y="temperature", colour="city")
chart.shade_between_x(6, 8)
chart.save("shade_between_x.png")
```

`chart.shade_between_y()`

Shade the band between two y values, behind the data. `chart.shade_between_y(40, 60)`

Options

`low`	needed
`high`	needed
`colour=`	normally `"grey"`

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.shade_between_y(40, 60)
chart.save("shade_between_y.png")
```

Words on the chart

`chart.title_top()`

A title above the chart. `chart.title_top("My class marks")`

Options

`text`	needed

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="class", y="marks")
chart.title_top("Marks in each class")
chart.save("title_top.png")
```

`chart.caption_bottom()`

A small line of text under the whole chart. `chart.caption_bottom("Data from 2026")`

Options

`text`	needed

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="class", y="marks")
chart.caption_bottom("Made-up data from sdata.example")
chart.save("caption_bottom.png")
```

chart.label_x()

The name under the x-axis. `chart.label_x("Student")`

Options

`text`	needed

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.label_x("Hours of study")
chart.save("label_x.png")
```

chart.label_y()

The name beside the y-axis. `chart.label_y("Marks")`

Options

`text`	needed

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.label_y("Marks (out of 100)")
chart.save("label_y.png")
```

chart.note()

A note on the chart at a place x, y. `chart.note("Biggest", x=2024, y=49)`

`arrow=True` puts the note a little away, with an arrow pointing at x, y. On bar charts x can be a name: `chart.note("Best", x="Ben", y=90)` on bars of names.

Options

`text`	needed
`x=`	
`y=`	
`arrow=`	normally `False`

Underneath matplotlib

Example you can run (it uses the practice table `growth`)

```
import simplydata as sdata
growth = sdata.example("growth")
```

```
chart = sdata.chart(growth)
chart.lines(x="year", y="population", colour="country")
chart.note("Cherria grows fastest", x=2020, y=12, arrow=True)
chart.save("note.png")
```

chart.numbers_on_bars()

Write the value of each bar on it. `chart.numbers_on_bars()` · `decimals=1` shows 12.5

Options

`decimals=`	normally `0`

Underneath matplotlib

Example you can run (it uses the practice table `fruit`)

```
import simplydata as sdata
fruit = sdata.example("fruit")
chart = sdata.chart(fruit)
chart.bars(x="fruit", y="count", show="total")
chart.numbers_on_bars()
chart.save("numbers_on_bars.png")
```

chart.font_size()

Make all the text bigger or smaller. `chart.font_size(14)` (normal is 12)

Options

`size`	needed

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="class", y="marks")
chart.title_top("Big text")
chart.font_size(18)
chart.save("font_size.png")
```

Axes

chart.range_x()

From where to where the x-axis goes. `chart.range_x(0, 10)`

Use None to fix only one end: `chart.range_x(0, None)` starts at 0 and lets the top choose itself.

Options

```
low=
high=
```

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.range_x(0, 10)
chart.save("range_x.png")
```

chart.range_y()

From where to where the y-axis goes. `chart.range_y(0, 100)`

Use None to fix only one end: `chart.range_y(None, 100)` stops at 100.

Options

```
low=
high=
```

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="class", y="marks")
chart.range_y(0, 100)
chart.save("range_y.png")
```

chart.log_scale_x()

A log scale on the x-axis: 1, 10, 100, 1000 are evenly spaced. `chart.log_scale_x()`

Options none: just write the empty brackets.

Underneath matplotlib

Example you can run (it uses the practice table `growth`)

```
import simplydata as sdata
growth = sdata.example("growth")
chart = sdata.chart(growth)
chart.dots(x="population", y="forest", colour="country")
chart.log_scale_x()
```

```
chart.save("log_scale_x.png")
```

chart.log_scale_y()

A log scale on the y-axis: 1, 10, 100, 1000 are evenly spaced. `chart.log_scale_y()`

Options none: just write the empty brackets.

Underneath matplotlib

Example you can run (it uses the practice table `growth`)

```
import simplydata as sdata
growth = sdata.example("growth")
chart = sdata.chart(growth)
chart.lines(x="year", y="population", colour="country")
chart.log_scale_y()
chart.save("log_scale_y.png")
```

chart.ticks_x()

Where the marks on the x-axis go. `chart.ticks_x(1, 4, 7, 10)`

`labels=["Jan", "Apr", "Jul", "Oct"]` writes your own words at those places.

Options

`values...`	as many as you like, separated by commas
`labels=`	

Underneath matplotlib

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month_number", y="rain", colour="city")
chart.ticks_x(1, 4, 7, 10, labels=["Jan", "Apr", "Jul", "Oct"])
chart.save("ticks_x.png")
```

chart.ticks_y()

Where the marks on the y-axis go. `chart.ticks_y(0, 50, 100)`

`labels=["low", "mid", "high"]` writes your own words at those places.

Options

`values...`	as many as you like, separated by commas
`labels=`	

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="class", y="marks")
chart.ticks_y(0, 25, 50, 75)
chart.save("ticks_y.png")
```

chart.rotate_labels_x()

Turn the names under the x-axis so long ones do not bump into each other.

`chart.rotate_labels_x()` turns them 45 degrees; `chart.rotate_labels_x(90)` stands them up.

Options

`angle=`	normally `45`

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="name", y="marks")
chart.rotate_labels_x(45)
chart.save("rotate_labels_x.png")
```

chart.rotate_labels_y()

Turn the names beside the y-axis. `chart.rotate_labels_y(30)`

Options

`angle=`	normally `45`

Underneath matplotlib

Example you can run (it uses the practice table `fruit`)

```
import simplydata as sdata
fruit = sdata.example("fruit")
chart = sdata.chart(fruit)
chart.bars(x="fruit", y="count", sideways=True)
chart.rotate_labels_y(30)
chart.save("rotate_labels_y.png")
```

chart.second_y()

A second y-axis on the right, for a column measured in other units.

```
chart.lines(x="month", y="temperature")
chart.second_y("rain", drawn_as="bars", label="Rain (mm)")
```

(with `sdata.example("weather")`)

`drawn_as=` can be `"lines"`, `"bars"` or `"dots"`. It shares the x-axis with the chart before it.

Options

`y`	needed
`drawn_as=`	normally `"lines"`
`label=`	
`colour=`	

Underneath matplotlib

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month", y="temperature")
chart.second_y("rain", drawn_as="bars", label="Rain (mm)")
chart.save("second_y.png")
```

chart.flip_x()

Turn the x-axis around, so numbers grow from right to left. `chart.flip_x()`

Options none: just write the empty brackets.

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.flip_x()
chart.save("flip_x.png")
```

chart.flip_y()

Turn the y-axis upside down, for example for depth going down. `chart.flip_y()`

Options none: just write the empty brackets.

Underneath matplotlib

Example you can run (it uses the practice table `island`)

```
import simplydata as sdata
island = sdata.example("island")
chart = sdata.chart(island)
```

```
chart.contour(x="east", y="north", z="height")
chart.flip_y()
chart.save("flip_y.png")
```

Colours and look

chart.colour()

Also written `chart.color()`

One colour for the whole chart. `chart.colour("blue")` or `chart.colour("#0072B2")`

Options

`name`	needed

Good to know Inside a chart step, `colour=` does more: `colour="class"` gives each class its own colour (and a legend), and `colour="red"` makes that one step red.

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="class", y="marks")
chart.colour("purple")
chart.save("colour.png")
```

chart.colours()

Also written `chart.colors()`

Your own colours for the groups, in order. `chart.colours("blue", "orange", "green")`

Options

`names...`	as many as you like, separated by commas

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="subject", y="marks", colour="class")
chart.colours("orange", "teal", "purple")
chart.save("colours.png")
```

chart.style()

The overall look. `chart.style("school")`

`"paper"` (default), `"school"`, `"presentation"` or `"dark"`.

Options

`name`	needed

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="subject", y="marks", colour="class")
chart.style("school")
chart.save("style.png")
```

chart.size()

The size in inches. `chart.size(width=8, height=6)`

Options

`width=`	normally `8`
`height=`	normally `6`

Underneath matplotlib

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month", y="rain", colour="city")
chart.size(width=10, height=4)
chart.save("size.png")
```

chart.show_grid()

Grid lines across the chart in both directions. `chart.show_grid()`

Options none: just write the empty brackets.

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.show_grid()
chart.save("show_grid.png")
```

`chart.hide_grid()`

No grid lines. `chart.hide_grid()`

Options none: just write the empty brackets.

Underneath matplotlib

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.hide_grid()
chart.save("hide_grid.png")
```

`chart.legend_top()`

The legend above the chart, in one row. `chart.legend_top()`

Options none: just write the empty brackets.

Underneath matplotlib

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month", y="temperature", colour="city")
chart.legend_top()
chart.save("legend_top.png")
```

`chart.legend_bottom()`

The legend below the chart, in one row under the x-axis name. `chart.legend_bottom()`

Options none: just write the empty brackets.

Underneath matplotlib

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month", y="temperature", colour="city")
chart.legend_bottom()
chart.save("legend_bottom.png")
```

chart.legend_left()

The legend to the left of the chart. `chart.legend_left()`

Options none: just write the empty brackets.

Underneath matplotlib

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month", y="temperature", colour="city")
chart.legend_left()
chart.save("legend_left.png")
```

chart.legend_right()

The legend to the right of the chart, outside the plot. `chart.legend_right()`

Options none: just write the empty brackets.

Underneath matplotlib

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month", y="temperature", colour="city")
chart.legend_right()
chart.save("legend_right.png")
```

chart.hide_legend()

No legend. `chart.hide_legend()`

Options none: just write the empty brackets.

Underneath matplotlib

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month", y="temperature", colour="city")
chart.hide_legend()
chart.save("hide_legend.png")
```

chart.show_legend()

The legend in its usual place, for example after `hide_legend()`. `chart.show_legend()`

A legend only appears when `colour=` (or `shape=`) makes groups.

Options none: just write the empty brackets.

Underneath matplotlib

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month", y="temperature", colour="city")
chart.hide_legend()
chart.show_legend()
chart.save("show_legend.png")
```

chart.label_colour_bar()

Also written `chart.label_color_bar()`

The name beside the colour bar of a heatmap or contour. `chart.label_colour_bar("Height (m)")`

Options

`text`	needed

Underneath matplotlib

Example you can run (it uses the practice table `island`)

```
import simplydata as sdata
island = sdata.example("island")
chart = sdata.chart(island)
chart.contour(x="east", y="north", z="height")
chart.label_colour_bar("Height (m)")
chart.save("label_colour_bar.png")
```

Several charts

chart.one_chart_per()

One small chart for each group. `chart.one_chart_per("class", per_row=3)` (with `sdata.example("marks")`)

`per_row=3` puts 3 small charts in each row. `down="class"`, `across="subject"` makes a table of charts: the classes go down (one row of charts each) and the subjects go across (one column of charts each). `same_x=False` / `same_y=False` lets each small chart choose its own axis range.

Options

`by=`	
`down=`	
`across=`	
`per_row=`	
`same_x=`	normally `True`
`same_y=`	normally `True`

Underneath matplotlib (a grid of subplots)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.one_chart_per("class", per_row=3)
chart.save("one_chart_per.png")
```

sdata.side_by_side()

Charts next to each other. `sdata.side_by_side(a, b).save("both.png")`

`same_y=True` gives them all the same y range, so heights can be compared. `letters=True` writes A, B, C ... at the top left of each chart, as in papers.

Options

`charts...`	as many as you like, separated by commas
`same_x=`	normally `False`
`same_y=`	normally `False`
`letters=`	normally `False`

Underneath matplotlib (subplots)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
a = sdata.chart(marks).bars(x="class", y="marks").title_top("Average")
b = sdata.chart(marks).boxes(x="class", y="marks").title_top("Spread")
page = sdata.side_by_side(a, b, same_y=True)
page.save("side_by_side.png")
```

sdata.one_above_other()

Charts on top of each other. `sdata.one_above_other(a, b).save("both.png")`

`same_x=True` gives them all the same x range, which suits time series. `letters=True` writes A, B, C ... at the top left of each chart, as in papers.

Options

`charts...`	as many as you like, separated by commas
`same_x=`	normally `False`
`same_y=`	normally `False`
`letters=`	normally `False`

Underneath matplotlib (subplots)

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
weather = sdata.example("weather")
a = sdata.chart(weather).lines(x="month", y="temperature", colour="city")
b = sdata.chart(weather).bars(x="month", y="rain", colour="city")
page = sdata.one_above_other(a, b)
page.save("one_above_other.png")
```

sdata.grid()

Charts in rows and columns. `sdata.grid(a, b, c, d, per_row=2)`

`per_row=2` puts 2 charts in each row. `letters=True` writes A, B, C ... at the top left of each chart, as figures in papers do.

Options

`charts...`	as many as you like, separated by commas
`per_row=`	normally `2`
`same_x=`	normally `False`
`same_y=`	normally `False`
`letters=`	normally `False`

Underneath matplotlib (subplots)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
a = sdata.chart(marks).bars(x="class", y="marks")
b = sdata.chart(marks).histogram(x="marks")
c = sdata.chart(marks).dots(x="hours", y="marks")
d = sdata.chart(marks).boxes(x="subject", y="marks")
page = sdata.grid(a, b, c, d, per_row=2)
page.title_top("Four ways to look at the marks")
page.save("grid.png")
```

chart.zoom_box()

A small box showing part of the chart close up.

`chart.zoom_box(x_between=[2, 4], y_between=[40, 60])` (with `sdata.example("marks")`)
`x_between=` and `y_between=` are the part to show: [low, high] on each axis. `where="top right"` (default), `"top left"`, `"bottom right"` or `"bottom left"`.

Options

`x_between=`	
`y_between=`	
`where=`	normally `"top right"`

Underneath matplotlib (`inset_axes` and `indicate_inset_zoom`)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.zoom_box(x_between=[2, 4], y_between=[40, 60], where="top left")
chart.save("zoom_box.png")
```

chart.animate()

A moving chart with one picture for each value of a column.

`chart.animate(over="year")` (with `sdata.example("growth")`), then `chart.save("growth.gif")`
`keep=True` builds the chart up: each picture also keeps the values before it.

Options

`over`	needed
`seconds_per_frame=`	normally `0.8`
`keep=`	normally `False`

Underneath matplotlib (animation) and Pillow (the `.gif` file)

Example you can run (it uses the practice table `growth`)

```
import simplydata as sdata
growth = sdata.example("growth")
chart = sdata.chart(growth)
chart.bars(x="country", y="population")
chart.range_y(0, 50)
chart.animate(over="year")
chart.save("growth.gif")
```

Table words

Steps on a table: `table = sdata.table(...)`, then `table.keep_rows(...)`. Table words work the same on pandas and Polars tables.

Tables: start and look

`sdata.table()`

Start a table.

`sdata.table(sdata.example("marks"))` a practice table
`sdata.table("my_file.xlsx")` a file (Excel, CSV, TSV, Parquet, JSON)
`sdata.table("my_file.xlsx", sheet="2026")` one sheet of an Excel file
`fast=True` use the fast engine (Polars), for big data

Options

`data=`
`sheet=`
`fast=` normally `False`

Underneath pandas or Polars read the file; narwhals lets every table word run on both

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.show_first_rows()
```

`table.show_first_rows()`

Show the first rows. `table.show_first_rows(5)`

Options

`rows=` normally `5`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.show_first_rows(10)
```

`table.show_last_rows()`

Show the last rows. `table.show_last_rows(5)`

Options

`rows=`	normally `5`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.show_last_rows(3)
```

`table.column_names()`

The names of all columns. `table.column_names()`

Options none: just write the empty brackets.

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
print(table.column_names())
```

`table.describe_columns()`

Describe every column: its kind, empty cells, smallest, biggest and average.

`table.describe_columns()` (with `sdata.example("marks")`)

Options none: just write the empty brackets.

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.describe_columns()
```

Choose columns

`table.keep_columns()`

Keep only these columns, in the order you list them.

`table.keep_columns("name", "marks")` (with `sdata.example("marks")`)

Options

`names...`	as many as you like, separated by commas

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_columns("name", "marks")
table.show_first_rows()
```

`table.remove_columns()`

Remove these columns.

`table.remove_columns("height")` (with `sdata.example("marks")`)

Options

`names...`	as many as you like, separated by commas

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.remove_columns("height")
print(table.column_names())
```

`table.rename_column()`

Give a column a new name.

`table.rename_column("marks", to="score")` (with `sdata.example("marks")`)

Options

`old=`	
`to=`	

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.rename_column("marks", to="score")
table.show_first_rows()
```

Choose rows

`table.keep_rows()`

Keep only the rows that match one condition.

`table.keep_rows("marks", above=50)` (with `sdata.example("marks")`)
Conditions: above, below, `at_least`, `at_most`, `equal_to`, `not_equal_to`, `between=[low, high]` (both ends included), `one_of=[...]`, contains, `starts_with`, `ends_with` (capital letters matter), `is_empty=True`. Empty cells only match `is_empty`.

Options

```
column=
above=
below=
at_least=
at_most=
equal_to=
not_equal_to=
between=
one_of=
contains=
starts_with=
ends_with=
is_empty=
```

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_rows("marks", above=50)
table.keep_rows("subject", one_of=["Maths", "Science"])
table.show_first_rows()
```

table.remove_rows()

Remove the rows that match one condition (the opposite of `keep_rows`).

`table.remove_rows("subject", equal_to="English")` (with `sdata.example("marks")`)
Same conditions as `keep_rows`; `between=[low, high]` includes both ends.

Options

```
column=
above=
below=
at_least=
at_most=
equal_to=
not_equal_to=
between=
one_of=
contains=
starts_with=
ends_with=
is_empty=
```

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```python
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.remove_rows("subject", equal_to="English")
table.show_first_rows()
```

table.keep_first_rows()

Keep only the first rows.

`table.keep_first_rows(10)` (with `sdata.example("marks")`)

Options

`n=`	normally `5`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```python
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_first_rows(10)
print(table)
```

table.keep_last_rows()

Keep only the last rows.

`table.keep_last_rows(10)` (with `sdata.example("marks")`)

Options

`n=`	normally `5`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_last_rows(10)
print(table)
```

table.keep_random_rows()

Keep some rows picked at random; give a seed to get the same rows every time.

`table.keep_random_rows(10, seed=1)` (with `sdata.example("marks")`)

Options

`n=`	normally `5`
`seed=`	

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_random_rows(10, seed=1)
print(table)
```

table.remove_duplicate_rows()

Remove rows that repeat an earlier row, or that repeat it in the columns you name.

`table.remove_duplicate_rows(columns=["name"])` (with `sdata.example("marks")`)

Options

`columns=`	

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.remove_duplicate_rows(columns=["name"])
print(table)
```

Empty cells

`table.count_empty_cells()`

Show how many empty cells each column has.

`table.count_empty_cells()` (with `sdata.example("marks")`)

Options none: just write the empty brackets.

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.replace_values("marks", [61, 73], with_value=None)
table.count_empty_cells()
```

`table.remove_rows_with_empty_cells()`

Remove rows with an empty cell (in any column, or only in the column you name).

`table.remove_rows_with_empty_cells("marks")` (with `sdata.example("marks")`)

Options

`column=`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.replace_values("marks", [61, 73], with_value=None)
table.remove_rows_with_empty_cells("marks")
print(table)
```

`table.fill_empty_cells()`

Fill the empty cells of a column with a value, or with the average, median, value above or value below.

`table.fill_empty_cells("marks", with_the="average")` (with `sdata.example("marks")`)

Options

`column=`
`with_value=`
`with_the=`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.replace_values("marks", [61, 73], with_value=None)
table.fill_empty_cells("marks", with_the="average")
table.show_first_rows()
```

Fix columns

`table.make_numbers()`

Turn a column into numbers; cells that are not numbers become empty.

`table.make_numbers("marks")` (with `sdata.example("marks")`)
Understands 12.5, -3, 1e3, 1 200 and 1,200 (thousands) and 12,5 (decimal comma).

Options

`column=`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.make_numbers("marks")
table.describe_columns()
```

`table.make_text()`

Turn a column into text.

`table.make_text("month_number")` (with `sdata.example("weather")`)

Options

`column=`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.make_text("month_number")
table.describe_columns()
```

table.make_dates()

Turn a column into dates; cells that are not dates become empty.

`table.make_dates("year")` (with `sdata.example("growth")`)
Understands 2026-03-15, 15/03/2026, 15.03.2026, 15 March 2026, March 15, 2026, 2026-03 and 2026. 03/04/2026 is read day first, as 3 April.

Options

`column=`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `growth`)

```
import simplydata as sdata
table = sdata.table(sdata.example("growth"))
table.make_dates("year")
table.show_first_rows()
```

table.make_lowercase()

Write all the text in a column in small letters.

`table.make_lowercase("name")` (with `sdata.example("marks")`)

Options

`column=`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.make_lowercase("name")
table.show_first_rows()
```

table.make_uppercase()

Write all the text in a column in capital letters.

`table.make_uppercase("name")` (with `sdata.example("marks")`)

Options

`column=`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.make_uppercase("name")
table.show_first_rows()
```

`table.remove_extra_spaces()`

Remove spaces at the start and end of the text, and double spaces inside it.

`table.remove_extra_spaces("name")` (with `sdata.example("marks")`)

Options

`column=`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.remove_extra_spaces("name")
table.show_first_rows()
```

`table.replace_values()`

Replace a value (or a list of values) in a column; `with_value=None` makes those cells empty.

`table.replace_values("subject", "Maths", with_value="Math")` (with `sdata.example("marks")`)

Options

`column=`
`old=`
`with_value=`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.replace_values("subject", "Maths", with_value="Math")
table.count_each("subject")
```

Sort and new columns

`table.sort_by()`

Put the rows in order of one or more columns; empty cells go last.

Numbers go smallest first and text goes A to Z. `biggest_first=True` turns this round: biggest number first, and text Z to A. Rows that tie keep the order they had.

`table.sort_by("marks", biggest_first=True)` (with `sdata.example("marks")`)

Options

`columns...`	as many as you like, separated by commas
`biggest_first=`	normally `False`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.sort_by("marks", biggest_first=True)
table.show_first_rows()
```

table.add_column()

Add a new column, worked out from other columns in plain words or with a formula.

Choose one way:

`table.add_column("study_and_marks", by_adding=["hours", "marks"])`
`table.add_column("half", from_column="marks", divide_by=2)`
`table.add_column("school", value="Hill School")` same value in every row
`table.add_column("marks_per_hour", "marks / hours")` calculator style

(all with `sdata.example("marks")`). `by_subtracting=["a", "b"]` means a - b and `by_dividing=["a", "b"]` means a / b, in the order written. Dividing by zero gives an empty cell. In a formula, a column name with spaces goes in backticks: `"`max temp` - 2"`.

Options

`name=`
`formula=`
`by_adding=`
`by_subtracting=`
`by_multiplying=`
`by_dividing=`
`from_column=`
`plus=`
`minus=`
`times=`
`divide_by=`
`value=`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.add_column("study_and_marks", by_adding=["hours", "marks"])
table.add_column("marks_per_hour", "marks / hours")
table.show_first_rows()
```

table.add_group_column()

Add a column that puts each number into a group, like turning marks into grades.

`cut_at=` says where one group ends and the next begins; `names=` has one more name than `cut_at=` has numbers. A value exactly on a cut goes into the higher group: with `cut_at=[40, 60, 80]`, a mark of 60 is a `"B"`. Empty cells stay empty.

`table.add_group_column("grade", from_column="marks", cut_at=[40, 60, 80], names=["D", "C", "B", "A"])`

(with `sdata.example("marks")`)

Options

`name=`
`from_column=`
`cut_at=`
`names=`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.add_group_column("grade", from_column="marks", cut_at=[40, 60, 80], names=["D", "C", "B
    ↪ ", "A"])
table.count_each("grade")
```

`table.add_rank_column()`

Add a column with each row's place, where 1 is the best.

The biggest value gets place 1 (`biggest_first=False` gives 1 to the smallest). Rows that tie share the best place and the next place is skipped, as in sports tables: 1, 2, 2, 4. Empty cells get no place.

`table.add_rank_column("place", from_column="marks")` (with `sdata.example("marks")`)

Options

`name=`
`from_column=`
`biggest_first=` normally `True`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.add_rank_column("place", from_column="marks")
table.sort_by("place")
table.show_first_rows()
```

Summaries

`table.count_rows()`

How many rows the table has, or how many in each group.

`table.count_rows(for_each="class")` (with `sdata.example("marks")`)

Options

`for_each=`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.count_rows()
table.count_rows(for_each="class")
```

table.count_each()

Each different value in a column, how many times it appears, and its percent.

Biggest count first. Empty cells count as a value of their own, so the percents are out of all rows. Percents are rounded to 1 decimal place.

`table.count_each("fruit")` (with `sdata.example("fruit")`)

Options

`column`	needed

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `fruit`)

```
import simplydata as sdata
table = sdata.table(sdata.example("fruit"))
table.count_each("fruit")
```

table.average_of()

The average (mean) of a column; empty cells are skipped.

`table.average_of("marks", for_each="class")` (with `sdata.example("marks")`)

Options

`column`	needed
`for_each=`	

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.average_of("marks")
by_class = table.average_of("marks", for_each="class")
print(by_class)
```

table.median_of()

The middle number of a column once it is sorted; empty cells are skipped.

`table.median_of("marks", for_each="class")` (with `sdata.example("marks")`)

Options

`column`	needed
`for_each=`	

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.median_of("marks", for_each="class")
```

table.total_of()

All the numbers in a column added up; empty cells are skipped.

`table.total_of("count", for_each="shop")` (with `sdata.example("fruit")`)

Options

`column`	needed
`for_each=`	

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `fruit`)

```
import simplydata as sdata
table = sdata.table(sdata.example("fruit"))
table.total_of("count", for_each="shop")
```

table.smallest_of()

The smallest number in a column; empty cells are skipped.

`table.smallest_of("temperature", for_each="city")` (with `sdata.example("weather")`)

Options

`column`	needed
`for_each=`	

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.smallest_of("temperature", for_each="city")
```

table.biggest_of()

The biggest number in a column; empty cells are skipped.

`table.biggest_of("temperature", for_each="city")` (with `sdata.example("weather")`)

Options

`column`	needed
`for_each=`	

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.biggest_of("temperature", for_each="city")
```

table.standard_deviation_of()

How spread out the numbers are: the sample standard deviation (divide by n - 1).

It is the same as pandas .std() and R `sd()`. Empty cells are skipped.

`table.standard_deviation_of("marks", for_each="class")` (with `sdata.example("marks")`)

Options

`column`	needed
`for_each=`	

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.standard_deviation_of("marks", for_each="class")
```

table.correlation_between()

How strongly two columns of numbers go up or down together, from -1 to 1.

`method="pearson"` (a straight-line link, the usual one) or `"spearman"` (uses ranks, for any link that keeps going one way). Prints the strength in words and the p-value from SciPy, and gives back the number r. Rows with an empty cell in either column are skipped. Strength words for the size of r: below 0.2 very weak, from 0.2 weak, from 0.4 medium, from 0.6 strong, from 0.8 very strong (after Evans 1996). p below 0.001: `"very unlikely to be chance"`; below 0.05: `"unlikely to be chance"`; otherwise `"could easily be chance"`. A correlation does not show that one thing causes the other.

`table.correlation_between("hours", "marks")` (with `sdata.example("marks")`)

Options

`column_a`	needed
`column_b`	needed
`method=`	normally `"pearson"`

Underneath SciPy (stats.pearsonr or stats.spearmanr)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.correlation_between("hours", "marks")
```

Statistical tests

`table.compare_averages()`

Are the averages of the groups really different, or could it be chance?

`table.compare_averages("marks", between="class")` (with `sdata.example("marks")`)
Two groups: Welch's t-test (the R `t.test()` default). Three or more: Welch's one-way ANOVA (the R `oneway.test()` default). Neither needs the groups to have the same spread. `based_on="ranks"` uses ranks instead (Mann-Whitney U or Kruskal-Wallis): safer when groups are small and their values are not bell-shaped. `test=` chooses another test: `"welch"`, `"student"`, `"welch anova"`, `"anova"`, `"mann-whitney"` or `"kruskal-wallis"`. Prints the answer in words (with the test used and why) and gives back a one-row table of results. The effect size says how big the difference is: Hedges' g (the difference in standard deviations) for 2 groups, omega-squared (the share of the differences explained by the groups) for 3 or more. Each group is checked for a bell shape (Shapiro-Wilk) and a note says if it does not look like one; with 30 or more values that matters little (the central limit theorem). To see which groups differ, use `compare_each_pair()`.

Options

`column=`	
`between=`	
`based_on=`	normally `"averages"`
`test=`	

Underneath SciPy (stats.ttest_ind with `equal_var=False`; Welch's ANOVA written out with the F distribution; mannwhitneyu, kruskal)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
result = table.compare_averages("marks", between="class")
table.compare_averages("marks", between="class", based_on="ranks")
result.save("marks_test.csv")
```

`table.compare_each_pair()`

Which pairs of groups differ? Every pair is compared, with Holm's correction.

`table.compare_each_pair("marks", between="class")` (with `sdata.example("marks")`)
Each pair gets Welch's t-test (or Mann-Whitney U with `based_on="ranks"`). Testing many pairs makes a false alarm more likely, so the p-values are adjusted with Holm's method: the chance of at least one false alarm stays at 5 %. This is the same as R `pairwise.t.test(pool.sd = FALSE, p.adjust.method = "holm")`. Gives back a table with one row per pair.

Options

`column=`
`between=`
`based_on=` normally `"averages"`

Underneath SciPy (stats.ttest_ind or mannwhitneyu), with Holm's correction

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.compare_each_pair("marks", between="class")
```

`table.compare_before_after()`

Did the values change from one column to the other, row by row, more than chance?

`table.compare_before_after("Maths", "Science")` (with `sdata.example("marks")`)
For two measurements of the same thing, such as each child's mark before and after a course. Uses the change in each row (after minus before): a paired t-test, or the Wilcoxon signed-rank test with `based_on="ranks"`. Rows with an empty cell in either column are skipped. Effect size: Cohen's `d_z`, the average change in standard deviations of the changes. Gives back a one-row table of results.

Options

`before=`
`after=`
`based_on=` normally `"averages"`

Underneath SciPy (stats.ttest_rel or stats.wilcoxon)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_columns("name", "subject", "marks")
table.turn_rows_into_columns(names_from="subject", values_from="marks")
table.compare_before_after("Maths", "Science")
```

`table.compare_average_to()`

Is the average of a column different from a number you choose, or could it be chance?

`table.compare_average_to("marks", 60)` (with `sdata.example("marks")`)
A one-sample t-test, or the Wilcoxon signed-rank test (on the median) with `based_on="ranks"`. Empty cells are skipped. Effect size: Cohen's d, the difference in standard deviations. Gives back a one-row table of results.

Options

`column=`
`value=`
`based_on=` normally `"averages"`

Underneath SciPy (stats.ttest_1samp or stats.wilcoxon)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.compare_average_to("marks", 60)
```

table.compare_counts()

Does how often each value turns up differ between groups, more than chance?

`table.add_group_column("result", from_column="marks", cut_at=[60], names=["below 60", "60 or more"])`
`table.compare_counts("result", between="class")` (with `sdata.example("marks")`)
(Do the classes differ in how many children score 60 or more?) Counts the rows for each pair of values and uses the chi-square test of independence (as R `chisq.test()`: Yates' continuity correction for 2 x 2 tables only). If an expected count is below 5 a note says so; for a 2 x 2 table `test="fisher"` then gives Fisher's exact test. Empty cells are skipped. Effect size: Cramer's V, the strength of the link from 0 (none) to 1 (complete). Gives back a one-row table.

Options

`column=`
`between=`
`test=`

Underneath SciPy (stats.chi2_contingency, or `fisher_exact` with `test="fisher"`)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.add_group_column("result", from_column="marks", cut_at=[60], names=["below 60", "60 or
    ↪ more"])
table.compare_counts("result", between="class")
```

table.fit_straight_line()

Fit the straight line y = slope * x + intercept that best follows the points.

`table.fit_straight_line(x="hours", y="marks")` (with `sdata.example("marks")`)
Ordinary least squares (the same as R `lm(y ~ x)`). Prints how much y goes up for each 1 more of x, with its 95 % confidence interval and p-value, and R squared: the share of the differences in y that the line explains. Rows with an empty cell in x or y are skipped. A link does not show that x causes y. Gives back a one-row table.

Options

`x=`
`y=`

Underneath SciPy (stats.linregress)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.fit_straight_line(x="hours", y="marks")
```

`table.fit_formula()`

Find the values in a formula that make it follow the data best.

`table.fit_formula("population = a * exp(b * (year - 2000))")` (with `sdata.example("growth")`)
Write the column to fit on the left of =. On the right, names that are columns are data; any other name (such as a, b or rate) is a value to find. Uses nonlinear least squares (SciPy `curve_fit`; standard errors as in R `nls()`). Every value to find starts at 1; give better start values with `start={"a": 10, "b": 0.03}` if the fit fails. Prints each value with its standard error, and R squared (only a rough guide for a curve). Rows with an empty cell are skipped. Gives back a one-row table.

Options

```
formula=
start=
```

Underneath SciPy (optimize.curve_fit)

Example you can run (it uses the practice table `growth`)

```
import simplydata as sdata
table = sdata.table(sdata.example("growth"))
table.keep_rows("country", equal_to="Applia")
table.fit_formula("population = a * exp(b * (year - 2000))")
```

Combine and reshape

`table.join_with()`

Bring in the columns of another table, matching rows on a column such as a name.

`keep=` says which rows stay:
`"matching rows"` only rows found in both tables (the normal choice)
`"all rows from this table"` every row of this table; no match gives empty cells
`"all rows from both"` every row of both tables
other can be a simplydata table, a pandas or Polars table, or a file name. A column found in both tables (other than `using=`) gets `"_other"` added to the other table's copy.

`table.join_with({"name": ["Asha", "Ben"], "age": [12, 13]}, using="name")` (with `sdata.example("marks")`)

Options

```
other=
using=
keep=                                   normally "matching rows"
```

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
ages = {"name": ["Asha", "Ben"], "age": [12, 13]}
table.join_with(ages, using="name", keep="all rows from this table")
table.show_first_rows()
```

`table.add_rows_from()`

Put the rows of another table under this one; columns are matched by their names.

`table.add_rows_from(sdata.example("marks"))` (with `sdata.example("marks")`: the class twice)

Options

`other=`

Underneath pandas (concat) or Polars (concat), whichever holds the table

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.add_rows_from(sdata.example("marks"))
print(table)
```

`table.turn_columns_into_rows()`

Turn some columns into rows: each value gets its own row, with its old column name beside it.

This makes a `"long"` table, which most charts like. The new column `names_to=` holds the old column names and `values_to=` holds the values.

`table.turn_columns_into_rows(["temperature", "rain"])` (with `sdata.example("weather")`)

Options

`columns=`	
`names_to=`	normally `"column"`
`values_to=`	normally `"value"`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.keep_columns("month", "city", "temperature", "rain")
table.turn_columns_into_rows(["temperature", "rain"])
table.show_first_rows()
```

`table.turn_rows_into_columns()`

Turn rows into columns: each different value in `names_from=` becomes a new column, filled from `values_from=`.

This makes a `"wide"` table, the opposite of `turn_columns_into_rows()`. All the other columns say which row is which, so rows only merge when those all agree.

`table.turn_rows_into_columns(names_from="city", values_from="temperature")` (with `sdata.example("weather")`)

Options

`names_from=`
`values_from=`

Underneath narwhals, which runs the step in pandas or Polars (whichever holds the table)

Example you can run (it uses the practice table `weather`)

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.turn_rows_into_columns(names_from="city", values_from="temperature")
table.show_first_rows()
```

Undo and finish

`table.undo()`

Take the last step back. `table.undo()`

Options none: just write the empty brackets.

Underneath simplydata itself: it keeps every version of the table

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_rows("marks", above=50)
table.undo()
print(table)
```

`table.steps()`

List every step you did to this table, in order. `table.steps()`

Options none: just write the empty brackets.

Underneath simplydata itself: it keeps every version of the table

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_rows("marks", above=50)
table.sort_by("marks")
table.steps()
```

`table.start_again()`

Go back to the table as it was at the start. `table.start_again()`

Options none: just write the empty brackets.

Underneath simplydata itself: it keeps every version of the table

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_rows("marks", above=50)
table.remove_columns("height")
table.start_again()
print(table)
```

table.save()

Save the table; the end of the file name decides the type. `table.save("clean.xlsx")`

File types: `.xlsx` (Excel), `.csv`, `.tsv`, `.parquet` and `.json`.

Options

`file_name`	needed

Underneath pandas (`to_excel`, `to_csv` ...) or Polars, whichever holds the table

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_rows("marks", above=50)
table.save("good_marks.xlsx")
table.save("good_marks.csv")
```

table.chart()

Start a chart from this table. chart = `table.chart()`

Options none: just write the empty brackets.

Underneath matplotlib and seaborn (the chart is drawn from a pandas copy of the table)

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_rows("subject", equal_to="Maths")
chart = table.chart()
chart.bars(x="name", y="marks")
chart.rotate_labels_x()
chart.save("maths_marks.png")
```

table.as_pandas()

Give back the table as a plain pandas table, for other Python tools. df = `table.as_pandas()`

Only needed when another library wants pandas: simplydata words never need it.

Options none: just write the empty brackets.

Underneath pandas

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_rows("marks", above=50)
df = table.as_pandas()
print(type(df))
```

table.as_polars()

Give back the table as a plain Polars table, for other Python tools. df = `table.as_polars()`

Needs Polars (pip install simplydata[fast]). simplydata words never need it.

Options none: just write the empty brackets.

Underneath Polars

Example you can run (it uses the practice table `marks`)

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_rows("marks", above=50)
df = table.as_polars()
print(type(df))
```

Number words

Steps on numbers, `nums = sdata.numbers([12, 15, 9, 20, 14])`, and on grids of numbers, `grid = sdata.number_grid([[3, 5, 2], [8, 6, 4]])`.

Numbers: make and look

sdata.numbers()

Make a list of numbers.

`sdata.numbers([3, 5, 2, 8])` these numbers (or `sdata.numbers(3, 5, 2, 8)`)
`sdata.numbers(1, to=10)` 1, 2, 3 ... 10 (the end is included)
`sdata.numbers(0, to=100, step=25)` 0, 25, 50, 75, 100
`sdata.numbers(0, to=1, how_many=5)` 5 numbers evenly spaced: 0, 0.25, 0.5, 0.75, 1
`sdata.numbers(10, to=1)` counting down: 10, 9 ... 1
None or NaN in a list is an empty (missing) value.

Options

`values...` as many as you like, separated by commas
`to=`
`step=`
`how_many=`

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([3, 5, 2, 8])
print(nums)
print(sdata.numbers(1, to=10))
print(sdata.numbers(0, to=1, how_many=5))
```

sdata.random_numbers()

Make random numbers; give `seed=` to get the same numbers every time.

`sdata.random_numbers(how_many=10, between=[0, 1], seed=1)`
any number from 0 up to 1, each equally likely (a uniform distribution)
`sdata.random_numbers(how_many=10, between=[1, 6], whole_numbers=True, seed=1)`
whole numbers from 1 to 6, both ends included, like rolling a dice
`sdata.random_numbers(how_many=100, average=50, standard_deviation=10, seed=1)`
numbers that bunch around the average in a bell shape (a normal distribution); about 2 in 3 land within one standard deviation of the average `rows=` and `columns=` in place of `how_many=` give a grid.

Options

`how_many=`	
`between=`	
`average=`	
`standard_deviation=`	
`whole_numbers=`	normally `False`
`seed=`	
`rows=`	
`columns=`	

Underneath NumPy (random.default_rng)

Example you can run

```
import simplydata as sdata
dice = sdata.random_numbers(how_many=10, between=[1, 6], whole_numbers=True, seed=1)
dice.show()
heights = sdata.random_numbers(how_many=100, average=150, standard_deviation=8, seed=1)
heights.average()
```

`nums.show()`

Show every number. `nums.show()`

Options none: just write the empty brackets.

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.show()
```

`nums.show_first_numbers()`

Show the first numbers. `nums.show_first_numbers(5)`

Options

`how_many=`	normally `5`

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.show_first_numbers(3)
```

nums.show_last_numbers()

Show the last numbers. `nums.show_last_numbers(5)`

Options

`how_many=` normally `5`

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.show_last_numbers(2)
```

nums.number_at()

The number at a position, counting from 1: `number_at(1)` is the first number.

```
nums.number_at(1)
```

Positions count from 1, the way people count and the way charts number the x axis.

Options

`position=`

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
print(nums.number_at(1))
print(nums.number_at(nums.count()))
```

nums.count()

How many numbers there are (empty ones included). `nums.count()`

Options none: just write the empty brackets.

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
print(nums.count())
```

nums.count_empty_numbers()

How many values are empty (missing). `nums.count_empty_numbers()`

Options none: just write the empty brackets.

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, None, 9, None, 14])
print(nums.count_empty_numbers())
```

Numbers: maths in words

nums.add()

Add a number to every number (or add numbers one by one). `nums.add(5)`

For new numbers, write bigger = nums + 5 instead.

Options

`number=`

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.add(5)
nums.show()
```

nums.subtract()

Take a number away from every number. `nums.subtract(2)`

For new numbers, write smaller = nums - 2 instead.

Options

`number=`

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.subtract(2)
nums.show()
```

`nums.times()`

Multiply every number by a number. `nums.times(2)`

For new numbers, write double = nums * 2 instead.

Options

`number=`

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.times(2)
nums.show()
```

`nums.divide_by()`

Divide every number by a number. `nums.divide_by(4)`

Dividing by 0 has no answer, so those values become empty and a note says how many. For new numbers, write quarter = nums / 4 instead.

Options

`number=`

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.divide_by(4)
nums.show()
```

`nums.power()`

Raise every number to a power: `power(2)` squares, `power(3)` cubes. `nums.power(2)`

`power(0.5)` is the same as the square root. For new numbers, write squares = nums ** 2.

Options

`number=`

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.power(2)
nums.show()
```

`nums.square_root()`

The square root of every number: 9 becomes 3. `nums.square_root()`

Negative numbers have no square root among ordinary numbers, so they become empty.

Options none: just write the empty brackets.

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([9, 16, 25])
nums.square_root()
nums.show()
```

`nums.absolute_value()`

Every number without its minus sign: -3 becomes 3. `nums.absolute_value()`

Options none: just write the empty brackets.

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([-3, 5, -2])
nums.absolute_value()
nums.show()
```

`nums.round_to()`

Round every number to this many decimal places. `nums.round_to(2)`

A number exactly halfway goes to the even neighbour (2.5 becomes 2, 3.5 becomes 4), like R and NumPy; this stops a run of halves from always pushing totals up. `round_to(0)` gives whole numbers; `round_to(-1)` rounds to tens.

Options

`decimals=`	normally `0`

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([2.345, 3.14159, 7.07])
nums.round_to(1)
nums.show()
```

nums.log()

The logarithm of every number: natural (base e) unless you give `base=10` or `base=2`.

```
nums.log() nums.log(base=10) nums.log(base=2)
```

`log(base=10)` of 1000 is 3, because 10 x 10 x 10 = 1000. The log only exists for numbers above 0, so 0 and negative numbers become empty.

Options

`base=`	normally `"e"`

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([1, 10, 100, 1000])
nums.log(base=10)
nums.show()
```

nums.exp()

e (about 2.718) to the power of every number; it undoes `log()`. `nums.exp()`

Options none: just write the empty brackets.

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([0, 1, 2])
nums.exp()
nums.show()
```

nums.running_total()

Each number becomes the total so far (the cumulative sum): 1, 2, 3 becomes 1, 3, 6.

```
nums.running_total()
```

Empty values stay empty and are skipped in the total, like pandas `cumsum()`.

Options none: just write the empty brackets.

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([1, 2, 3, 4])
nums.running_total()
nums.show()
```

Numbers: choose, sort and fill

`nums.keep_numbers()`

Keep only the numbers that match one condition. `nums.keep_numbers(above=50)`

Conditions (the same as `table.keep_rows`): above, below, `at_least`, `at_most`, `equal_to`, `not_equal_to`, `between=[low, high]` (both ends included), `one_of=[...]`, `is_empty=True`. Empty values only match `is_empty`.

Options

```
above=
below=
at_least=
at_most=
equal_to=
not_equal_to=
between=
one_of=
is_empty=
```

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.keep_numbers(above=10)
nums.show()
```

`nums.remove_numbers()`

Remove the numbers that match one condition (the opposite of `keep_numbers`).

```
nums.remove_numbers(below=0)
```

Same conditions as `keep_numbers`; `between=[low, high]` includes both ends.

Options

```
above=
below=
at_least=
at_most=
equal_to=
not_equal_to=
between=
one_of=
is_empty=
```

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.remove_numbers(below=10)
nums.show()
```

nums.keep_first_numbers()

Keep only the first numbers. `nums.keep_first_numbers(10)`

Options

how_many=	normally 5

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.keep_first_numbers(3)
nums.show()
```

nums.keep_last_numbers()

Keep only the last numbers. `nums.keep_last_numbers(10)`

Options

how_many=	normally 5

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.keep_last_numbers(3)
nums.show()
```

nums.keep_random_numbers()

Keep some numbers picked at random (they keep their order); a seed gives the same pick every time.

```
nums.keep_random_numbers(10, seed=1)
```

Options

how_many=	normally 5
seed=	

Underneath NumPy (random.default_rng)

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.keep_random_numbers(3, seed=1)
nums.show()
```

`nums.remove_duplicate_numbers()`

Remove numbers that repeat an earlier number, so each number is left once, in its first place.

```
nums.remove_duplicate_numbers()
```

Options none: just write the empty brackets.

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([3, 5, 3, 8, 5])
nums.remove_duplicate_numbers()
nums.show()
```

`nums.remove_empty_numbers()`

Remove the empty (missing) values. `nums.remove_empty_numbers()`

Options none: just write the empty brackets.

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, None, 9, 14])
nums.remove_empty_numbers()
nums.show()
```

`nums.fill_empty_numbers()`

Fill the empty values with a number, or with the average, median, value above or value below.

`nums.fill_empty_numbers(with_value=0)` or `nums.fill_empty_numbers(with_the="average")`

Options

```
with_value=
with_the=
```

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, None, 9, 14])
nums.fill_empty_numbers(with_the="average")
nums.show()
```

`nums.sort()`

Put the numbers in order, smallest first; empty values go last. `nums.sort(biggest_first=True)`

Options

`biggest_first=`	normally `False`

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.sort(biggest_first=True)
nums.show()
```

`nums.reverse()`

Turn the order round: the last number comes first. `nums.reverse()`

Options none: just write the empty brackets.

Underneath NumPy

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.reverse()
nums.show()
```

Numbers: summaries

`nums.average()`

The average (mean) of the numbers; empty values are skipped.

`nums.average()`

Options none: just write the empty brackets.

Underneath NumPy (functions that skip empty values, such as nanmean)

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.average()
```

nums.median()

The middle number once the numbers are sorted; empty values are skipped.

```
nums.median()
```

Options none: just write the empty brackets.

Underneath NumPy (functions that skip empty values, such as nanmean)

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.median()
```

nums.total()

All the numbers added up; empty values are skipped.

```
nums.total()
```

Options none: just write the empty brackets.

Underneath NumPy (functions that skip empty values, such as nanmean)

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.total()
```

nums.smallest()

The smallest number; empty values are skipped.

```
nums.smallest()
```

Options none: just write the empty brackets.

Underneath NumPy (functions that skip empty values, such as nanmean)

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.smallest()
```

nums.biggest()

The biggest number; empty values are skipped.

```
nums.biggest()
```

Options none: just write the empty brackets.

Underneath NumPy (functions that skip empty values, such as nanmean)

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.biggest()
```

nums.standard_deviation()

How spread out the numbers are: the sample standard deviation (divide by n - 1).

It is the same as pandas .std() and R `sd()`. Empty values are skipped.

```
nums.standard_deviation()
```

Options none: just write the empty brackets.

Underneath NumPy (functions that skip empty values, such as nanmean)

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.standard_deviation()
```

nums.percentile()

The number that this percent of the numbers are below: `percentile(50)` is the median.

```
nums.percentile(90)
```

Uses NumPy's usual `"linear"` method, the same as pandas and R. Empty values are skipped.

Options

```
percent=
```

Underneath NumPy (nanpercentile, the usual `"linear"` method, as pandas and R)

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.percentile(90)
```

nums.count_each()

Each different number, how many times it appears, and its percent, as a table.

Biggest count first. Empty values count as a value of their own, so the percents are out of all numbers. Percents are rounded to 1 decimal place.

```
nums.count_each()
```

Options none: just write the empty brackets.

Underneath NumPy (functions that skip empty values, such as nanmean)

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([3, 5, 3, 8, 5, 3])
nums.count_each()
```

Grids of numbers

sdata.number_grid()

Make a grid of numbers in rows and columns.

`sdata.number_grid([[1, 2, 3], [4, 5, 6]])` from rows of numbers (2 rows, 3 columns)
`sdata.number_grid(rows=3, columns=4, filled_with=0)` a grid of 0s (filled_with=1 for 1s, or any number)
`sdata.number_grid([1, 2, 3, 4, 5, 6], columns=3)` a list cut into rows of 3

Options

```
values=
rows=
columns=
filled_with=                    normally 0
```

Underneath NumPy (a two-dimensional array)

Example you can run

```
import simplydata as sdata
grid = sdata.number_grid([[1, 2, 3], [4, 5, 6]])
print(grid)
print(sdata.number_grid(rows=2, columns=3, filled_with=0))
```

nums.number_grid()

Make a NEW grid from these numbers, filled row by row. grid = `nums.number_grid(columns=3)`

Give `rows=` or `columns=` (or both); the numbers must fill the grid exactly.

Options

```
rows=
columns=
```

Underneath NumPy (a two-dimensional array)

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([1, 2, 3, 4, 5, 6])
grid = nums.number_grid(columns=3)
print(grid)
```

grid.count_rows()

How many rows the grid has. `grid.count_rows()`

Options none: just write the empty brackets.

Underneath NumPy (a two-dimensional array)

Example you can run

```
import simplydata as sdata
grid = sdata.number_grid([[3, 5, 2], [8, 6, 4]])
print(grid.count_rows())
```

grid.count_columns()

How many columns the grid has. `grid.count_columns()`

Options none: just write the empty brackets.

Underneath NumPy (a two-dimensional array)

Example you can run

```
import simplydata as sdata
grid = sdata.number_grid([[3, 5, 2], [8, 6, 4]])
print(grid.count_columns())
```

grid.shape()

How many rows and columns, as (rows, columns). `grid.shape()`

Options none: just write the empty brackets.

Underneath NumPy (a two-dimensional array)

Example you can run

```
import simplydata as sdata
grid = sdata.number_grid([[3, 5, 2], [8, 6, 4]])
print(grid.shape())
```

`grid.show_first_rows()`

Show the first rows. `grid.show_first_rows(5)`

Options

`rows=`	normally `5`

Underneath NumPy (a two-dimensional array)

Example you can run

```
import simplydata as sdata
grid = sdata.number_grid([[3, 5, 2], [8, 6, 4]])
grid.show_first_rows(1)
```

`grid.show_last_rows()`

Show the last rows. `grid.show_last_rows(5)`

Options

`rows=`	normally `5`

Underneath NumPy (a two-dimensional array)

Example you can run

```
import simplydata as sdata
grid = sdata.number_grid([[3, 5, 2], [8, 6, 4]])
grid.show_last_rows(1)
```

`grid.number_at()`

The number at a row and column, counting from 1. `grid.number_at(row=1, column=2)`

Options

`row=`	
`column=`	

Underneath NumPy (a two-dimensional array)

Example you can run

```
import simplydata as sdata
grid = sdata.number_grid([[3, 5, 2], [8, 6, 4]])
print(grid.number_at(row=1, column=2))
```

grid.row()

One row of the grid as new numbers, counting from 1: `row(1)` is the top row. `grid.row(1)`

Options

`position=`

Underneath NumPy (a two-dimensional array)

Example you can run

```
import simplydata as sdata
grid = sdata.number_grid([[3, 5, 2], [8, 6, 4]])
top = grid.row(1)
top.show()
```

grid.column()

One column of the grid as new numbers, counting from 1: `column(1)` is the left one. `grid.column(1)`

Options

`position=`

Underneath NumPy (a two-dimensional array)

Example you can run

```
import simplydata as sdata
grid = sdata.number_grid([[3, 5, 2], [8, 6, 4]])
left = grid.column(1)
left.show()
```

grid.numbers()

All the numbers of the grid as a list, row by row (flatten). nums = `grid.numbers()`

Options none: just write the empty brackets.

Underneath NumPy (a two-dimensional array)

Example you can run

```
import simplydata as sdata
grid = sdata.number_grid([[3, 5, 2], [8, 6, 4]])
nums = grid.numbers()
nums.show()
```

`grid.swap_rows_and_columns()`

Turn rows into columns and columns into rows (the transpose in maths).

```
grid.swap_rows_and_columns()
```

A grid of 2 rows and 3 columns becomes 3 rows and 2 columns. Matrix multiplication and other linear algebra are planned for simplydata v0.7.

Options none: just write the empty brackets.

Underneath NumPy (a two-dimensional array)

Example you can run

```
import simplydata as sdata
grid = sdata.number_grid([[3, 5, 2], [8, 6, 4]])
grid.swap_rows_and_columns()
print(grid)
```

Numbers: undo and finish

`nums.undo()`

Take the last step back. `nums.undo()`

Options none: just write the empty brackets.

Underneath simplydata itself: it keeps every version of the numbers

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.add(5)
nums.undo()
nums.show()
```

`nums.steps()`

List every step you did, in order. `nums.steps()`

Options none: just write the empty brackets.

Underneath simplydata itself: it keeps every version of the numbers

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.add(5)
nums.keep_numbers(above=15)
nums.steps()
```

nums.start_again()

Go back to the numbers as they were at the start. `nums.start_again()`

Options none: just write the empty brackets.

Underneath simplydata itself: it keeps every version of the numbers

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.add(5)
nums.times(2)
nums.start_again()
nums.show()
```

nums.table()

Turn the numbers into a simplydata table. `nums.table(name="marks")`

Options

`name=`	normally `"value"`

Underneath pandas

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
table = nums.table(name="marks")
print(table)
```

nums.chart()

Start a chart from these numbers. chart = `nums.chart()`

Options none: just write the empty brackets.

Underneath matplotlib and seaborn, from a pandas table with one column called value

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
chart = nums.chart()
chart.histogram(x="value", bins=4)
chart.save("nums.png")
```

`nums.save()`

Save the numbers; the end of the file name decides the type. `nums.save("nums.csv")`

File types: `.csv`, `.tsv`, `.xlsx` (Excel), `.parquet` and `.json`, the same as tables.

Options

`file_name`	needed

Underneath pandas (`to_csv`, `to_excel` ...)

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
nums.save("nums.csv")
```

`nums.as_numpy()`

Give back the numbers as a plain NumPy array, for other Python tools. array = `nums.as_numpy()`

Options none: just write the empty brackets.

Underneath simplydata itself: it keeps every version of the numbers

Example you can run

```
import simplydata as sdata
nums = sdata.numbers([12, 15, 9, 20, 14])
array = nums.as_numpy()
print(type(array))
```

Help words

Words that teach simplydata.

Help

`sdata.words()`

Print every simplydata word with a short example. `sdata.words()`

For more about one word, type `sdata.help("bars")`.

Options none: just write the empty brackets.

Underneath plain Python: simplydata reads its own word list and docstrings

Example you can run

```
import simplydata as sdata
sdata.words()
```

`sdata.help()`

Explain one word: what it does, its options and an example you can run. `sdata.help("bars")`

```
sdata.help() on its own explains how to get help. Table words work too: sdata.help("keep_rows").
```

Options

```
word=
```

Underneath plain Python: simplydata reads its own word list and docstrings

Example you can run

```
import simplydata as sdata
sdata.help("bars")
sdata.help("show=")
sdata.help("keep_rows")
sdata.help("average")
```

`sdata.lesson()`

A lesson: an example to run, what each line does, and a task. `sdata.lesson(1)`

`sdata.lesson()` on its own lists all the lessons. `solutions=True` also shows a worked solution for each task: `sdata.lesson(3, solutions=True)`.

Options

`number=`
`solutions=` normally `False`

Underneath plain Python: simplydata reads its own word list and docstrings

Example you can run

```
import simplydata as sdata
sdata.lesson()
sdata.lesson(1)
```

PART IV

Data science ideas

Describing numbers

Before any test or model, a few numbers describe a column well: where its middle is, and how spread out it is. simplydata's summary words say each one in a sentence.

Average or median?

The **average** (formally the **arithmetic mean**, written $\bar{x}$) adds all the values and divides by how many there are:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

The **median** is the middle value when the values are put in order (with an even number of values, the average of the two in the middle). Half the values are below it and half above.

Both describe the "middle", but they react differently to values far from the rest (**outliers**). Five children earn pocket money of 2, 3, 3, 4 and 50 a week: the average is 12.4, though four of the five get 4 or less; the median is 3. One extreme value pulls the average towards it but hardly moves the median. So:

- Use the **average** when the values are roughly symmetric (a bell shape), and when you want to add things up (total = average × count).
- Use the **median** when the values are lopsided (**skewed**), such as incomes, house prices, waiting times or bacterial counts, or when there are outliers you cannot remove.
- When the two are far apart, say so: it tells the reader the values are skewed.

In simplydata: `table.average_of("marks")`, `table.median_of("marks")`, `nums.average()`, `nums.median()`, and on charts `show="average"` or `show="median"`. Lesson 5 draws both.

Spread: range, percentiles and standard deviation

Two classes can have the same average mark, one with every child near 65 and the other with marks from 20 to 100. The **spread** tells them apart.

- The **range** is the biggest value minus the smallest. It is easy to understand but depends on just two values, often the strangest ones.
- A **percentile** is the value that a given share of the values is below: `nums.percentile(90)` is the value that 90 % are below. The median is the 50th percentile. The middle half of the values lies between the 25th and 75th percentiles; that distance is the **interquartile range (IQR)**, and it is the box in a box plot (`chart.boxes`).
- The **standard deviation** (SD, written s) is about how far a typical value is from the average:

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}}.$$

  Its square, s^2, is the **variance**. Dividing by $n - 1$ instead of n (Bessel's correction) makes the sample variance an unbiased estimate of the variance of everything the sample came from;

simplydata always divides by $n - 1$, as pandas and R do (NumPy divides by n unless told otherwise).

The SD goes with the average and the IQR with the median. For all the marks of the practice table: “average 64.9 (SD 15.0)” or “median 68.5 (IQR 54.3 to 74.8)”.

The shape of the values: distributions

The way values are spread out is their **distribution**; a histogram (`chart.histogram`) or a density curve (`chart.density`) shows it. Many measurements, such as heights or measuring errors, follow a bell shape called the **normal** (or Gaussian) **distribution**. For a normal distribution about 68 % of the values lie within one SD of the average and about 95 % within two SD.

Not everything is bell-shaped. Counts and concentrations are often skewed to the right (many small values, a few large ones); a log scale (`chart.log_scale_y()`) often makes them look more symmetric. Look at the distribution before choosing what to report and which test to use.

How sure are we?

We almost never measure everything. We measure a **sample** (18 children, 40 water samples) and want to learn about the **population** it came from (all children of that age, the whole fjord). A number worked out from a sample, such as its average, is an **estimate**, and another sample would give a slightly different one. These ideas say how much.

Standard deviation or standard error?

They sound alike and are often mixed up, but they answer different questions.

	Standard deviation (SD)	**Standard error of the mean (SE, SEM)**
Describes	the *values*: how spread out they are	the *average*: how precisely the sample pins it down
Formula	s (above)	SE = $s/\sqrt{n}$
More data	stays about the same	gets smaller (4 times the data halves it)
Use it to	show the spread of individuals	compare averages, build confidence intervals
simplydata	`error="standard deviation"`	`error="standard error"`

Error bars of the SE are always shorter than those of the SD, and can look reassuringly small. That is why a figure caption must say which one was drawn; simplydata prints a line saying it, ready for your caption.

Confidence intervals

A **95 % confidence interval** (CI) is a range of plausible values for the true number in the population, worked out from the sample. For an average it is

$$\bar{x} \pm t_{0.975,\,n-1} \times \mathrm{SE},$$

where $t_{0.975,\,n-1}$ is a number from the t distribution: about 2 for most samples, a bit more for small ones (2.26 for $n = 10$, 2.09 for $n = 20$).

What "95 %" means, exactly: if you repeated the whole study many times and built an interval each time, about 95 out of 100 of those intervals would contain the true value. It does *not* mean that the true value moves around, nor that 95 % of the individual values lie inside (that would be about $\bar{x} \pm 2$ SD for a bell shape).

- A **narrow** interval means the data pin the value down well; a **wide** one means you need more data.
- An interval says more than a p-value: it gives the size of the effect *and* how sure you are of it. `table.fit_straight_line` prints the slope with its 95 % CI.
- On charts: `error="confidence interval"` on bars and lines, and `fit_line(confidence_band=True)` for a fitted line.

Why averages behave well: the central limit theorem

Even when the values themselves are skewed, the *average* of many of them is close to bell-shaped; this is the **central limit theorem**. It is why tests on averages (t-tests, ANOVA) usually work well with 30 or more values per group, even for values that are not bell-shaped. With small groups of skewed values, rank-based tests are safer (`based_on="ranks"`).

Testing: real difference, or chance?

The null hypothesis and the p-value

Two groups almost never have exactly the same average, even when nothing real separates them: samples differ by chance. A statistical test asks whether a difference is bigger than chance alone would often produce. It works like a court case, where the accused is innocent until proven guilty:

1. Assume there is **no real difference**. This assumption is the **null hypothesis** (H_0). The **alternative hypothesis** (H_1) is that there is one.
2. Work out how surprising your data would be if H_0 were true. The **p-value** is the probability of a difference at least as big as the one you saw, if there were really no difference.
3. A small p-value means the data are hard to explain by chance alone. By custom, below 0.05 is called **statistically significant**; that threshold is the **significance level** α.

simplydata says the p-value in words: below 0.001 "very unlikely to be chance", below 0.05 "unlikely to be chance", otherwise "could easily be chance".

Good to know

What a p-value is **not**:

- It is **not** the probability that H_0 is true, nor the probability that the result is a fluke.
- A small p-value does **not** say the difference is big or important. With thousands of values, tiny, meaningless differences become "significant". The *effect size* says how big it is.
- A large p-value does **not** show that there is no difference. It only says these data cannot tell. "Absence of evidence is not evidence of absence."
- 0.049 and 0.051 are practically the same evidence. Report the exact p-value, not just "$p < 0.05$".

Two kinds of mistake, and power

- A **false alarm** (**type I error**, a false positive): the test says there is a difference when there is none. With $\alpha = 0.05$ this happens in about 1 test in 20 when H_0 is true.
- A **missed difference** (**type II error**, a false negative): there is a real difference, but the test does not find it. Small samples miss differences often.

The **power** of a study is the chance of finding a difference of a given size if it is really there. More data, less noisy measurements and bigger true differences all raise power. A study with low power can only find large effects; plan the sample size before collecting data.

Many tests at once

Every test at $\alpha = 0.05$ carries a 5 % risk of a false alarm, and the risks add up: with 20 tests on pure noise, you expect about one "significant" result. Correcting for **multiple comparisons** keeps the chance of *any* false alarm at 5 %. `table.compare_each_pair` uses **Holm's method**, which is never less powerful than the simpler Bonferroni correction (multiply each p-value by the number of tests). Decide which comparisons you will make before you look at the data.

Effect sizes: how big is the difference?

The p-value answers "could it be chance?". The **effect size** answers "how big is it?", which is usually what matters. Report both, ideally with a confidence interval for the effect.

Effect size	What it says	Given by
Difference in averages	in the units of the data ("7.6 marks higher"): often the clearest	`compare_averages`, `compare_each_pair`
Cohen's d, Hedges' g	the difference in standard deviations; g corrects d for small samples. Rough guide: 0.2 small, 0.5 medium, 0.8 large	`compare_averages` (2 groups), `compare_average_to`, `compare_before_after` (d_z)
Omega-squared ω^2	the share of all the differences that the groups explain (0 to 1)	`compare_averages` (3+ groups)
Cramér's V	how strongly two categories are linked, 0 (none) to 1	`compare_counts`
Correlation r	how closely two columns go up and down together, −1 to 1	`correlation_between`
R^2	the share of the differences in y that a line or curve explains	`fit_straight_line`, `fit_formula`

The “small, medium, large” labels (from Jacob Cohen) are only a last resort: a “small” difference in survival can be important, and a “large” one in a lab assay may not be.

Links between columns

Correlation

A **correlation** measures how closely two columns of numbers go up and down together, from −1 (one always goes down when the other goes up) through 0 (no straight-line link) to 1.

- **Pearson’s** r (the usual one) measures a *straight-line* link. It is sensitive to outliers, and can be near 0 for a strong curved link (a U shape).
- **Spearman’s** ρ (rho) uses ranks: it measures any link that keeps going one way (always up, or always down), and is robust to outliers. `table.correlation_between("hours", "marks", method="spearman")`.
- r^2 is the share of the differences in one column that a straight line through the other explains. $r = 0.5$ explains only a quarter.

Always draw the dots (`chart.dots`) before trusting a correlation: very different patterns can have the same r (Anscombe’s quartet is the famous example).

Correlation is not cause

Ice cream sales and drownings go up together, but ice cream does not cause drowning: hot weather drives both. When two things go together there are several possible reasons:

1. A causes B.
2. B causes A (**reverse causation**): do people exercise because they are healthy, or are they healthy because they exercise?
3. A third thing causes both (a **confounder**): hot weather, age, wealth, or the size of a hospital.
4. Chance: test enough pairs of columns and some will line up by luck.

Only an **experiment** where you change A yourself, ideally by random assignment (a randomised controlled trial), shows cause directly. With observational data, say “is associated with”, not “causes”. simplydata’s link words remind you: “A link does not show that one thing causes the other.”

Fitting a line or a formula

Linear regression finds the straight line $y = a + b\,x$ that goes as close as possible to the dots, by making the sum of the squared vertical distances from the dots to the line as small as possible (**least squares**). The **slope** b says how much y changes for each 1 more of x; its p-value tests whether the slope could be 0, and its confidence interval gives a range of plausible slopes. R^2 says how much of the differences in y the line explains. `table.fit_straight_line(x="hours", y="marks")` prints all of these; `chart.fit_line()` draws the line.

When the relation is curved, such as growth that multiplies by the same share each year, write the formula yourself: `table.fit_formula("population = a * exp(b * (year - 2000))")` finds a and b by **nonlinear least squares**. A fitted line or curve describes the data you have; predicting far outside their range (**extrapolation**) is risky.

The shape of a table

Tidy data: long and wide tables

The same marks can be stored in two shapes:

Long (tidy)			**Wide**			
name	subject	marks	name	Maths	Science	English
Asha	Maths	61	Asha	61	66	77
Asha	Science	66	Ben	73	98	71
Asha	English	77				
Ben	Maths	73				
…						

A table is **tidy** (Hadley Wickham's term) when each **variable** is a column, each **observation** is a row, and each value has its own cell. The long table is tidy: it has one column for what was measured (`marks`) and columns that say which child and which subject.

- **Long tables** suit charts, groups and tests: `chart.bars(x="subject", y="marks")` and `compare_averages("marks", between="subject")` need the subject as a column.
- **Wide tables** suit sums across columns (a total per child) and paired tests (`compare_before_after("Maths", "Science")`), and are how people often type data into a spreadsheet.
- `table.turn_columns_into_rows` makes a wide table long (in pandas: `melt`; in R: `pivot_longer`); `table.turn_rows_into_columns` makes a long table wide (`pivot`).

Missing values and imputation

Real data has empty cells: a sample was lost, a question skipped. What to do depends on *why* they are empty:

- **Missing completely at random**: the reason has nothing to do with the data (a tube dropped). Leaving those rows out loses some power but does not bias the result.
- **Missing at random**: the reason depends on other columns you have (older people skip a question more often). Leaving rows out can bias the result; taking that other column into account helps.
- **Missing not at random**: the reason depends on the missing value itself (very ill patients drop out; concentrations below the detection limit). No simple fix removes the bias; say so in your report.

Removing rows with empty cells (`table.remove_rows_with_empty_cells()`) is simple and honest when few rows are affected. **Filling** them in is called **imputation**: `table.fill_empty_cells("marks", with_the="average")` (also `"median"`, `"value above"`, `"value below"`, or `with_value=0`). Filling with the average keeps the average the same, but makes the values look *less spread out* than they really are, so standard deviations, standard errors and p-values become too small. Filling with the value above or below suits a series in time order. Whatever you choose, count the empty cells first (`table.count_empty_cells()`) and say in your methods how you handled them.

Which test when?

First draw the data: a chart shows outliers, skew and odd values that a single p-value hides. Then answer four questions:

1. **What is the question?** A difference between groups, a change within the same rows, a link between two columns, or the shape of a relation?
2. **What kind of data?** Numbers (marks, heights, concentrations) or categories (pass or fail, species, site)?
3. **How many groups, and are they the same rows?** Two groups or more; independent groups (different children) or paired measurements (the same children twice)?
4. **Are the groups small and the values far from bell-shaped?** Then use the rank-based version: `based_on="ranks"`. simplydata checks each group for a bell shape (Shapiro–Wilk test) and adds a note when it matters.

Question	simplydata word	Test (the usual one)	With `based_on="ranks"`
Is one average different from a number you choose?	`compare_average_to("marks", 60)`	one-sample t-test	Wilcoxon signed-rank test
Do **two** groups differ in their average?	`compare_averages("marks", between="group")`	Welch's t-test	Mann–Whitney U test
Do **three or more** groups differ?	`compare_averages("marks", between="class")`	Welch's one-way ANOVA	Kruskal–Wallis test
Which groups differ?	`compare_each_pair("marks", between="class")`	pairwise Welch t-tests, Holm's correction	pairwise Mann–Whitney tests, Holm's correction
Did the values change, in the **same** rows measured twice?	`compare_before_after("before", "after")`	paired t-test	Wilcoxon signed-rank test
Do **counts** of categories differ between groups?	`compare_counts("result", between="class")`	chi-square test of independence	— (for a 2×2 table with small counts: `test="fisher"`, Fisher's exact test)
Do two columns of numbers go up and down together?	`correlation_between("hours", "marks")`	Pearson's r	`method="spearman"`: Spearman's ρ
How much does y change with x?	`fit_straight_line(x="hours", y="marks")`	linear regression (least squares)	—
Does the data follow a formula (growth, decay, saturation)?	`fit_formula("y = a * exp(b * x)")`	nonlinear least squares	—

Good to know

- **Welch, not Student.** For averages simplydata uses Welch's versions, which do not assume that the groups have the same spread. They are R's defaults and lose almost nothing when the spreads are equal. An expert can still choose: `test="student"` or `test="anova"`.
- **Paired or not?** If each row of one measurement belongs to a row of the other (the same child, the same site), the test must be paired: `compare_before_after`. Treating paired data as independent groups throws away information.
- **Counts, not percentages.** `compare_counts` needs the rows themselves (one row per observation), not percentages worked out by hand.
- **One test per question.** Decide the question and the test before looking at the results; trying several tests and keeping the best p-value makes false alarms much more likely.
- Each test gives back a one-row table with every number in it: `result = table.compare_averages(...)`, then `result.save("test.csv")`.

Writing it up

A good results sentence lets a reader check your work. Give:

1. the summary for each group (average and SD, or median and IQR) and the number of values n;
2. the effect size, ideally with its 95 % confidence interval;
3. the name of the test and the exact p-value;
4. in figure captions, what the bars or bands show (SD, SE or 95 % CI) and n.

For example: "Class A scored on average 13.2 marks more than class C (Welch's t-test with Holm's correction, $p = 0.023$; $n = 18$ per class)." simplydata's printed answers name each test, and `table.steps()` lists every change you made to the table, in order, for your methods.

PART V

Solutions

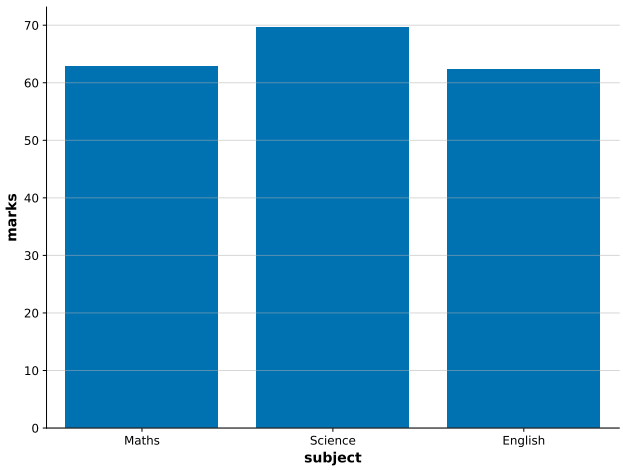
Solutions to the tasks

One worked solution for each “Try it yourself” task. There is often more than one good answer: if yours runs and answers the question, it is right. Each solution runs on its own and was run to make the chart and output shown. In Python, `sdata.lesson(3, solutions=True)` prints the solutions of lesson 3.

Lesson 1: Your first chart

Task 1. Change `x="name"` to `x="subject"`. What does each bar show now?

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="subject", y="marks")
chart.save("lesson1_subjects.png")
```

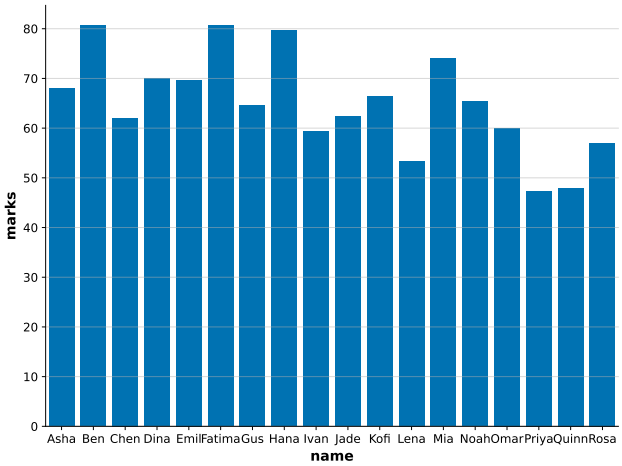


Lesson 1, task 1: `lesson1_subjects.png`.

Answer Each bar is now one subject, as tall as the average mark of all 18 children in that subject.

Task 2. Save the chart as `"my_first_chart.png"` and also as `"my_first_chart.pdf"`.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="name", y="marks")
chart.save("my_first_chart.png")
chart.save("my_first_chart.pdf")
```

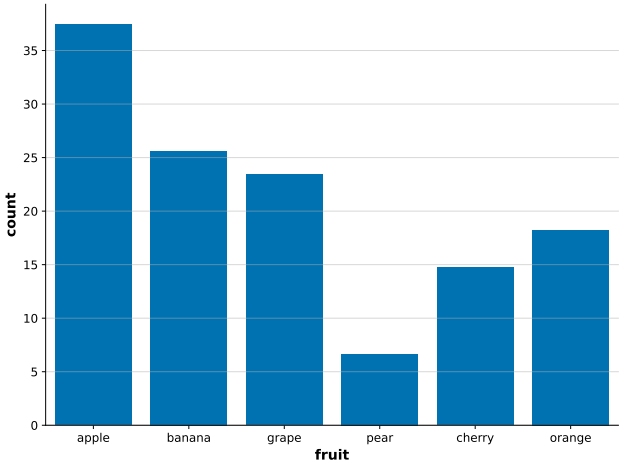


Lesson 1, task 2: `my_first_chart.png`.

Answer One chart can be saved many times; the end of the file name (`.png` or `.pdf`) decides the type.

Task 3. Make a chart of the `"fruit"` table: bars with `x="fruit"` and `y="count"`.

```
import simplydata as sdata
fruit = sdata.example("fruit")
chart = sdata.chart(fruit)
chart.bars(x="fruit", y="count")
chart.save("fruit.png")
```



Lesson 1, task 3: `fruit.png`.

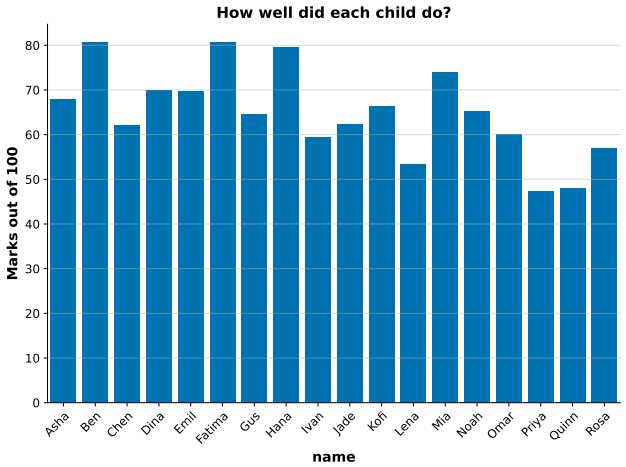
Answer One bar per fruit. Each fruit was sold several times, so a bar is the average count of one sale; `show="total"` would show how many were sold in the whole week.

Back to lesson 1

Lesson 2: Titles and labels

Task 1. Give the chart your own title, and label the y-axis `"Marks out of 100"`.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="name", y="marks")
chart.title_top("How well did each child do?")
chart.label_y("Marks out of 100")
chart.rotate_labels_x(45)
chart.save("lesson2_my_title.png")
```

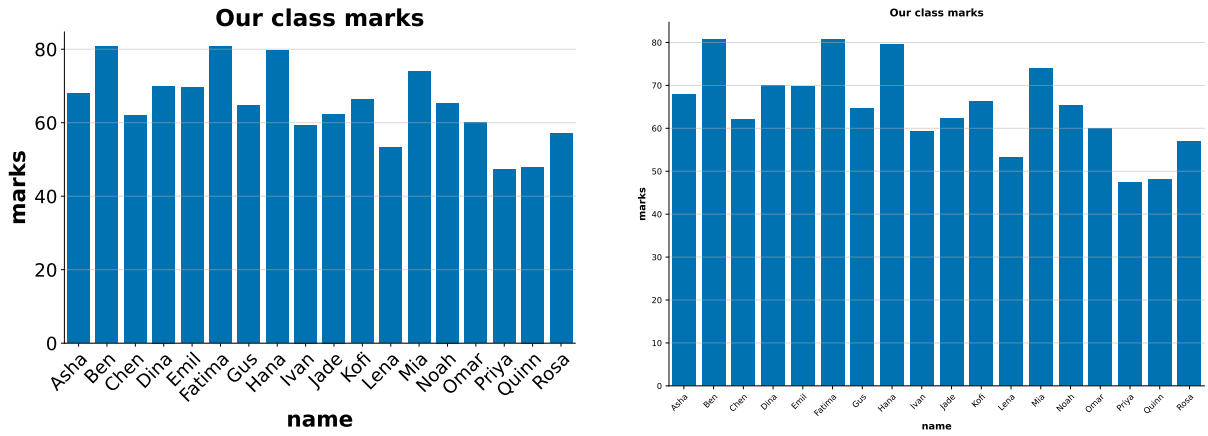


Lesson 2, task 1: `lesson2_my_title.png`.

Answer `title_top` writes any title you like, and `label_y` names the y-axis.

Task 2. Try `chart.font_size(20)`. Then try 9. Which is easier to read?

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="name", y="marks")
chart.title_top("Our class marks")
chart.rotate_labels_x(45)
chart.font_size(20)
chart.save("lesson2_big_text.png")
chart.font_size(9)
chart.save("lesson2_small_text.png")
```

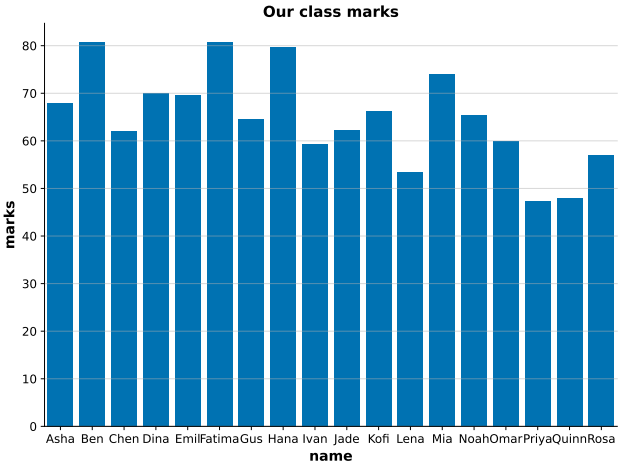


Lesson 2, task 2: `lesson2_big_text.png` and `lesson2_small_text.png`.

Answer The last `font_size` wins, so one chart gives both pictures. Size 20 is easy to read on a slide but crowds the names; 9 fits everything but is small. 12 to 14 suits most charts.

Task 3. Take away the line `chart.rotate_labels_x(45)`. What happens to the names?

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="name", y="marks")
chart.title_top("Our class marks")
chart.save("lesson2_no_rotate.png")
```



Lesson 2, task 3: `lesson2_no_rotate.png`.

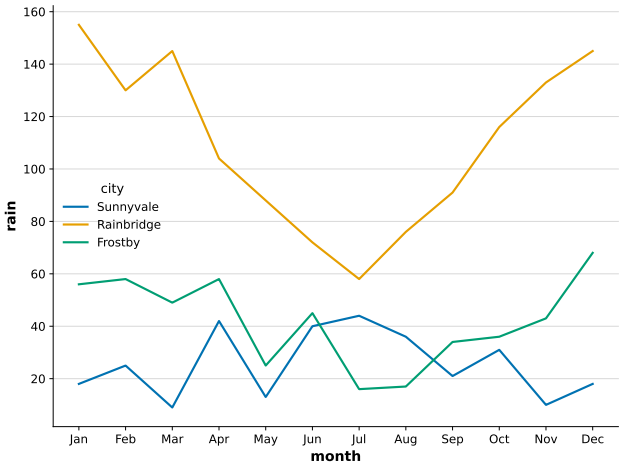
Answer Without `rotate_labels_x` the 18 names lie flat under the bars and run into each other.

Back to lesson 2

Lesson 3: Colours, groups and legends

Task 1. In the weather chart, add `colour="city"` to `chart.lines(...)`. How many lines do you get?

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.lines(x="month", y="rain", colour="city")
chart.save("lesson3_cities.png")
```

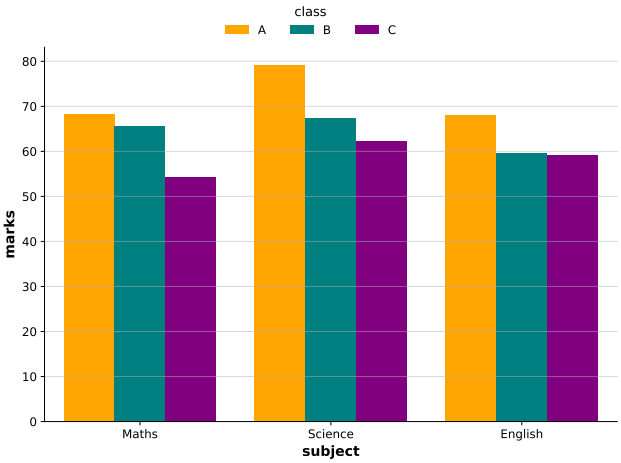


Lesson 3, task 1: `lesson3_cities.png`.

Answer Three lines, one for each city, and a legend that says which colour is which city.

Task 2. Move the legend to the top with `chart.legend_top()`.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="subject", y="marks", colour="class")
chart.colours("orange", "teal", "purple")
chart.legend_top()
chart.save("lesson3_legend_top.png")
```

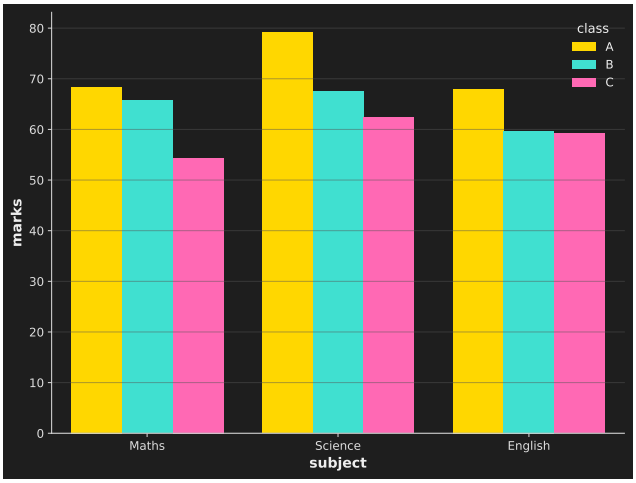


Lesson 3, task 2: `lesson3_legend_top.png`.

Answer `legend_top` puts the legend above the chart, in place of `legend_bottom`.

Task 3. Try `chart.style("dark")` and choose three colours that look good on it.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="subject", y="marks", colour="class")
chart.style("dark")
chart.colours("gold", "turquoise", "hotpink")
chart.save("lesson3_dark.png")
```



Lesson 3, task 3: `lesson3_dark.png`.

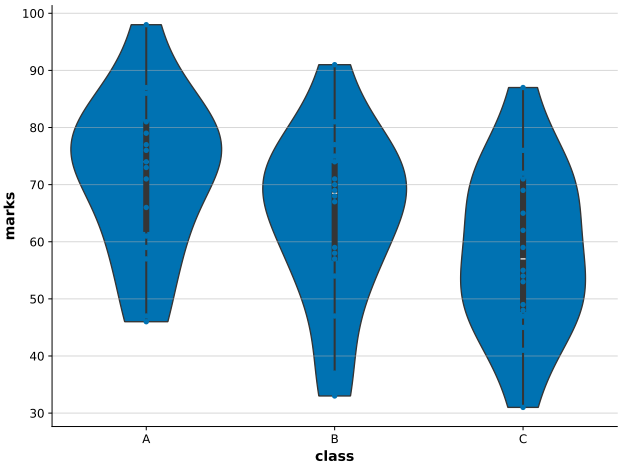
Answer Light, bright colours stand out on a dark background; dark colours such as navy would almost disappear.

Back to lesson 3

Lesson 4: Different chart types

Task 1. Draw `chart.violins(x="class", y="marks")` instead of boxes. What is different?

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.violins(x="class", y="marks")
chart.all_dots(x="class", y="marks")
chart.save("lesson4_violins.png")
```

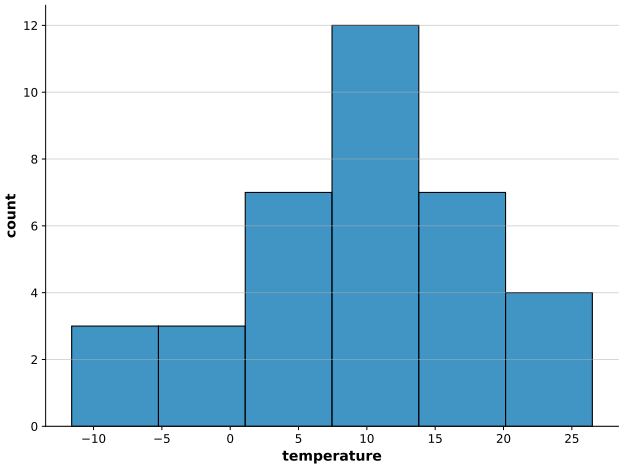


Lesson 4, task 1: `lesson4_violins.png`.

Answer A violin shows the whole shape of the marks: it is wide where many marks are and thin where few are. A box shows only the middle half and the median.

Task 2. Draw a histogram of the weather `"temperature"` column with `bins=6`.

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.histogram(x="temperature", bins=6)
chart.save("lesson4_temperature.png")
```

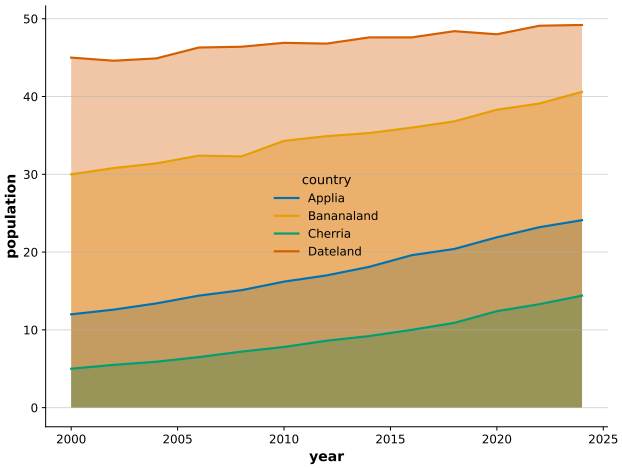


Lesson 4, task 2: `lesson4_temperature.png`.

Answer Six ranges of temperature, each bar as high as the number of months (of all three cities) that fall in it.

Task 3. Make a chart of `sdata.example("growth")` and draw `chart.area(x="year", y="population", colour="country")`.

```
import simplydata as sdata
growth = sdata.example("growth")
chart = sdata.chart(growth)
chart.area(x="year", y="population", colour="country")
chart.save("lesson4_area.png")
```



Lesson 4, task 3: `lesson4_area.png`.

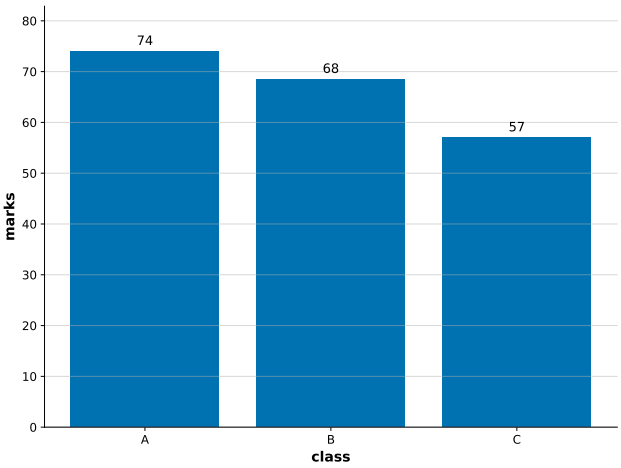
Answer An area chart is a line chart with the space under each line filled in.

Back to lesson 4

Lesson 5: Statistics: averages, error bars and fit lines

Task 1. Change `show="average"` to `show="median"`. Do the bars change much?

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="class", y="marks", show="median")
chart.numbers_on_bars()
chart.save("lesson5_median.png")
```



Lesson 5, task 1: `lesson5_median.png`.

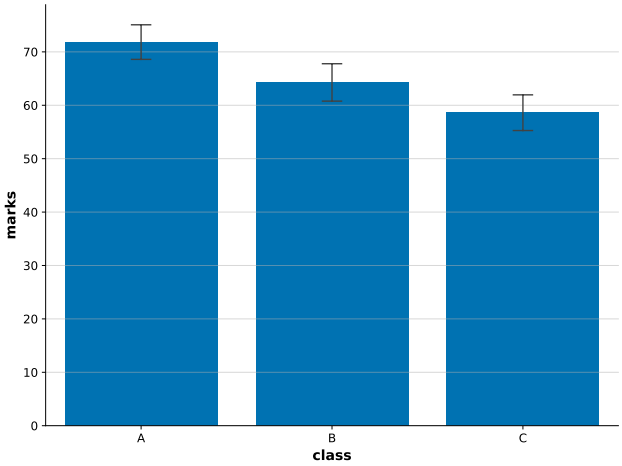
Answer Error bars go with averages, so `error=` comes out. The medians (A 74, B 68.5, C 57) are close to the averages (A 71.8, B 64.3, C 58.6): no single mark pulls them far.

Task 2. Change `error="standard deviation"` to `error="standard error"`. Are the error bars bigger or smaller? Why?

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="class", y="marks", show="average", error="standard error")
chart.save("lesson5_standard_error.png")
```

What it prints

```
Error bars: the standard error of the average (how sure we are about the average itself).
```

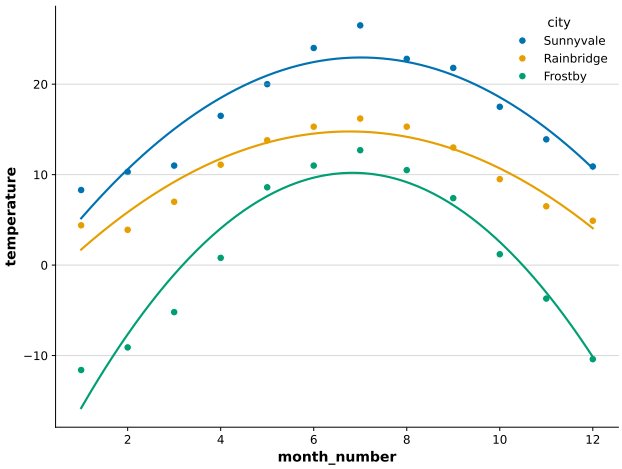


Lesson 5, task 2: `lesson5_standard_error.png`.

Answer Smaller: the standard error is the standard deviation divided by the square root of the number of values (here 18), so it is about a quarter of the standard deviation.

Task 3. Make a chart of the weather table, draw dots with `x="month_number"`, `y="temperature"` and add `chart.fit_curve(degree=2)`.

```
import simplydata as sdata
weather = sdata.example("weather")
chart = sdata.chart(weather)
chart.dots(x="month_number", y="temperature", colour="city")
chart.fit_curve(degree=2)
chart.save("lesson5_fit_curve.png")
```



Lesson 5, task 3: `lesson5_fit_curve.png`.

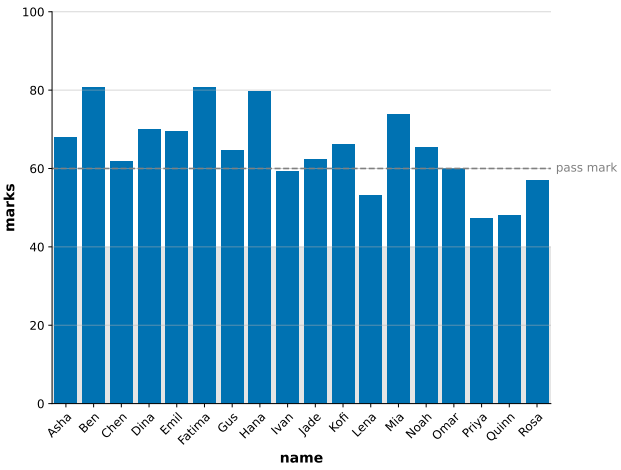
Answer Each city gets a curve with one bend: warm in the middle of the year, cold at both ends.

Back to lesson 5

Lesson 6: Axes and reference lines

Task 1. Move the pass mark to 60 and shade between 0 and 40 instead.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="name", y="marks")
chart.range_y(0, 100)
chart.line_at_y(60, label="pass mark")
chart.shade_between_y(0, 40)
chart.rotate_labels_x()
chart.save("lesson6_pass_60.png")
```

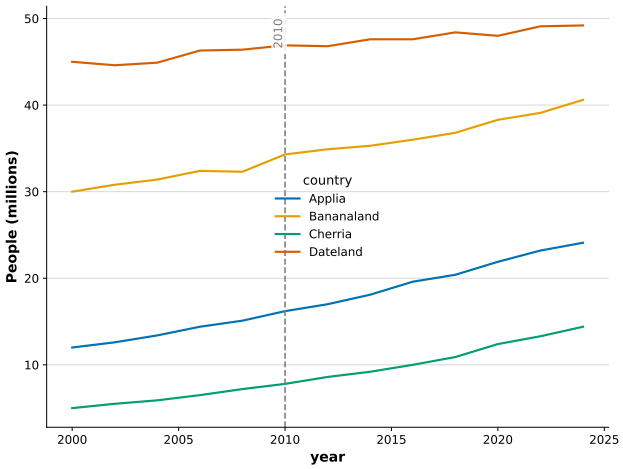


Lesson 6, task 1: `lesson6_pass_60.png`.

Answer `line_at_y(60)` moves the line; `shade_between_y(0, 40)` shades the band from 0 to 40.

Task 2. Take away `chart.log_scale_y()`. Which countries look different?

```
import simplydata as sdata
growth = sdata.example("growth")
chart = sdata.chart(growth)
chart.lines(x="year", y="population", colour="country")
chart.line_at_x(2010, label="2010")
chart.label_y("People (millions)")
chart.save("lesson6_no_log.png")
```

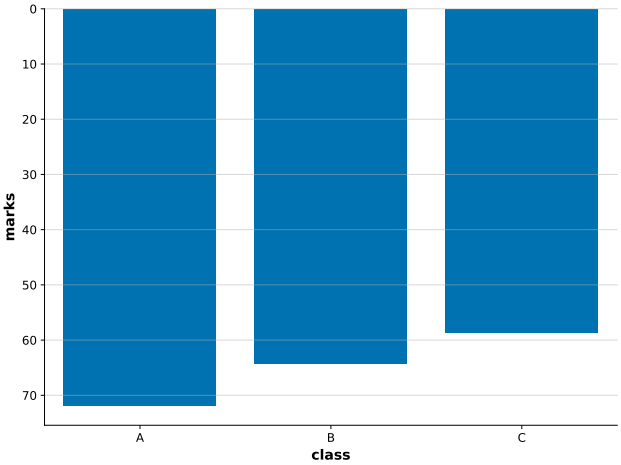


Lesson 6, task 2: `lesson6_no_log.png`.

Answer The lines now bend upwards instead of running straight, and the small countries (Cherria and Applia) sit low and look flat, although Cherria nearly triples.

Task 3. Try `chart.flip_y()` on any chart. What happens?

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.bars(x="class", y="marks")
chart.flip_y()
chart.save("lesson6_flip.png")
```



Lesson 6, task 3: `lesson6_flip.png`.

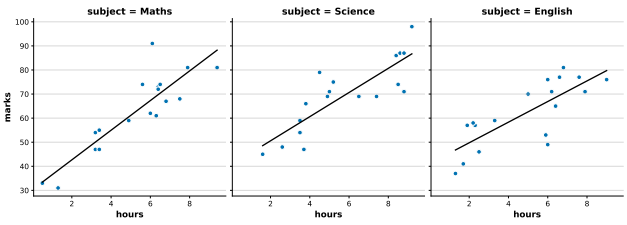
Answer The y-axis turns upside down: 0 is at the top and the bars hang down. Useful for depths, where bigger numbers are further down.

Back to lesson 6

Lesson 7: Several charts together

Task 1. Change `chart.one_chart_per("class")` to `chart.one_chart_per("subject")`.

```
import simplydata as sdata
marks = sdata.example("marks")
chart = sdata.chart(marks)
chart.dots(x="hours", y="marks")
chart.fit_line()
chart.one_chart_per("subject")
chart.save("lesson7_per_subject.png")
```

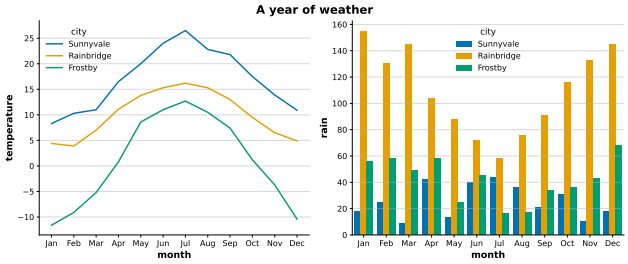


Lesson 7, task 1: `lesson7_per_subject.png`.

Answer One small chart for each subject (Maths, Science and English), each with its own fit line.

Task 2. Use `sdata.side_by_side(a, b)` instead of `sdata.one_above_other(a, b)` for the weather charts.

```
import simplydata as sdata
weather = sdata.example("weather")
a = sdata.chart(weather).lines(x="month", y="temperature", colour="city")
b = sdata.chart(weather).bars(x="month", y="rain", colour="city")
page = sdata.side_by_side(a, b)
page.title_top("A year of weather")
page.save("lesson7_side_by_side.png")
```

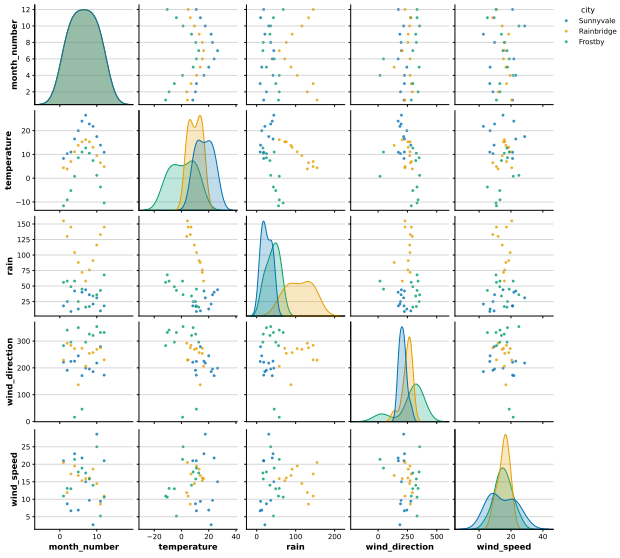


Lesson 7, task 2: `lesson7_side_by_side.png`.

Answer The same two charts, now next to each other instead of one above the other.

Task 3. Make `sdata.pairs` of `sdata.example("weather")` with `colour="city"`.

```
import simplydata as sdata
pairs = sdata.pairs(sdata.example("weather"), colour="city")
pairs.save("lesson7_weather_pairs.png")
```



Lesson 7, task 3: `lesson7_weather_pairs.png`.

Answer Every number column of the weather table against every other, one colour per city.

Back to lesson 7

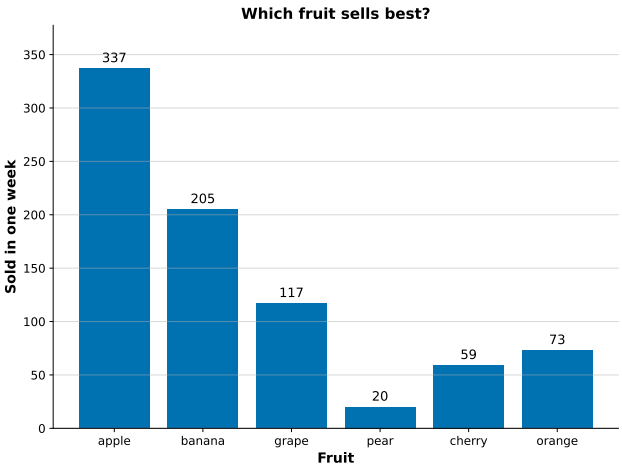
Lesson 8: Hard tricks, and your own program

Task 1. Write your own program, from an empty file, for a question you care about.

```
import simplydata as sdata
fruit = sdata.example("fruit")
print(fruit)
chart = sdata.chart(fruit)
chart.bars(x="fruit", y="count", show="total")
chart.numbers_on_bars()
chart.title_top("Which fruit sells best?")
chart.label_x("Fruit")
chart.label_y("Sold in one week")
chart.save("lesson8_fruit.png")
```

What it prints

```
    day          shop   fruit  count  price
0   Mon   Corner shop   apple     39   0.44
1   Mon  Market stall  banana     22   0.23
2   Mon  Market stall   grape     13   0.05
3   Mon  Market stall    pear      6   0.41
4   Tue   Corner shop   apple     39   0.44
5   Tue   Corner shop  cherry     15   0.11
6   Tue   Corner shop  orange     13   0.55
7   Tue  Market stall   apple     47   0.36
8   Tue  Market stall  banana     21   0.23
9   Wed   Corner shop   apple     19   0.44
10  Wed   Corner shop  banana     28   0.28
11  Wed   Corner shop  cherry     13   0.11
12  Wed   Corner shop  orange     23   0.55
13  Wed  Market stall   apple     28   0.36
14  Wed  Market stall  banana     18   0.23
15  Wed  Market stall   grape     29   0.05
16  Wed  Market stall  orange     19   0.45
17  Thu   Corner shop   apple     49   0.44
18  Thu   Corner shop  banana     39   0.28
19  Thu   Corner shop    pear      7   0.50
20  Thu  Market stall   apple     47   0.36
21  Thu  Market stall  banana     29   0.23
22  Thu  Market stall  cherry     15   0.09
23  Thu  Market stall   grape     28   0.05
24  Thu  Market stall  orange     18   0.45
25  Thu  Market stall    pear      7   0.41
26  Fri   Corner shop   apple     29   0.44
27  Fri   Corner shop  banana     23   0.28
28  Fri   Corner shop   grape     18   0.06
29  Fri  Market stall   apple     40   0.36
30  Fri  Market stall  banana     25   0.23
31  Fri  Market stall  cherry     16   0.09
32  Fri  Market stall   grape     29   0.05
```

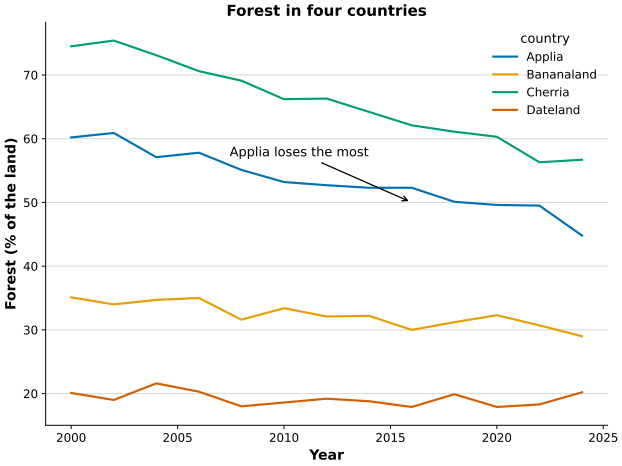


Lesson 8, task 1: `lesson8_fruit.png`.

Answer One possible question: which fruit sells best? Same plan as always: get the data, look, start, draw, words, save.

Task 2. Use any example table, or your own. Give it a title, axis names and a note.

```
import simplydata as sdata
growth = sdata.example("growth")
chart = sdata.chart(growth)
chart.lines(x="year", y="forest", colour="country")
chart.title_top("Forest in four countries")
chart.label_x("Year")
chart.label_y("Forest (% of the land)")
chart.note("Applia loses the most", x=2016, y=50, arrow=True)
chart.save("lesson8_forest.png")
```

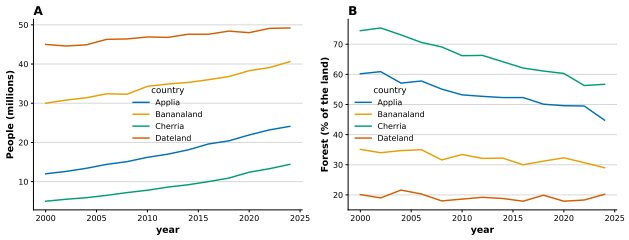


Lesson 8, task 2: `lesson8_forest.png`.

Answer Another table, with a title, both axis names and a note with an arrow.

Task 3. Make a second chart of the same table, put the two charts side by side and save the page as a PDF.

```
import simplydata as sdata
growth = sdata.example("growth")
people = sdata.chart(growth).lines(x="year", y="population", colour="country")
people.label_y("People (millions)")
forest = sdata.chart(growth).lines(x="year", y="forest", colour="country")
forest.label_y("Forest (% of the land)")
page = sdata.side_by_side(people, forest, letters=True)
page.save("lesson8_page.pdf")
```



Lesson 8, task 3: `lesson8_page.pdf`.

Answer Two charts of one table on one page, saved as a PDF because the name ends in `.pdf`.

Back to lesson 8

Lesson 9: Tables: open one and look at it

Task 1. Make a table of `sdata.example("weather")`. How many rows does it have?

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.count_rows()
```

What it prints

```
The table has 36 rows.
```

Answer 36 rows: 12 months for each of 3 cities.

Task 2. Describe its columns. Which column has the biggest numbers?

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.describe_columns()
```

What it prints

```
The table has 36 rows and 7 columns.

       column   kind  empty cells smallest biggest average different values
        month   text            0                                        12
 month_number number            0        1      12     6.5
         city   text            0                                         3
  temperature number            0    -11.6    26.5    9.35
```

```
          rain number            0        9     155   59.03
wind_direction number            0       16     354  244.42
    wind_speed number            0      2.7    28.6    15.1
```

Answer `wind_direction` has the biggest numbers (up to 354): it is an angle in degrees, from 0 to 360.

Task 3. Count each `"city"`. Does every city have the same number of rows?

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.count_each("city")
```

What it prints

```
How many rows have each "city" (3 different values): Frostby 12, Rainbridge 12, Sunnyvale 12.
```

Answer Yes: Frostby, Rainbridge and Sunnyvale have 12 rows each, one for every month.

Back to lesson 9

Lesson 10: Tables: choose rows and columns, and undo

Task 1. In the marks table, keep only class `"B"` and sort by `"hours"`. Who studied the longest?

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_rows("class", equal_to="B")
table.sort_by("hours", biggest_first=True)
table.show_first_rows(3)
```

What it prints

```
Kept 18 of 54 rows.
   name class  subject  hours  marks  height
0  Jade     B    Maths    9.4     81     146
1  Kofi     B  Science    8.5     74     146
2  Kofi     B    Maths    7.5     68     146
```

Answer Jade studied the longest: 9.4 hours, for Maths. `biggest_first=True` puts the longest on top.

Task 2. Undo the sorting, then sort by name instead.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_rows("class", equal_to="B")
table.sort_by("hours", biggest_first=True)
table.undo()
table.sort_by("name")
table.show_first_rows()
```

What it prints

```
Kept 18 of 54 rows.
Undone: sort by hours (biggest first)
   name class  subject  hours  marks  height
0   Gus     B    Maths    6.8     67     151
1   Gus     B  Science    4.9     69     151
2   Gus     B  English    2.2     58     151
3  Hana     B    Maths    6.1     91     144
4  Hana     B  Science    5.0     71     144
```

Answer `undo()` takes back only the sorting; class B stays. Text is sorted from A to Z.

Task 3. Keep 5 random rows with `table.keep_random_rows(5, seed=1)` and save them as `"five.csv"`.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_random_rows(5, seed=1)
print(table)
table.save("five.csv")
```

What it prints

```
Kept 5 of 54 rows.
simplydata table: 5 rows, 6 columns
   name class  subject  hours  marks  height
0  Kofi     B    Maths    7.5     68     146
1  Asha     A  English    7.6     77     144
2  Omar     C  English    6.0     49     151
3  Kofi     B  English    1.9     57     146
4  Kofi     B  Science    8.5     74     146
```

Answer `seed=1` picks the same 5 rows every time, so someone else can repeat it.

Back to lesson 10

Lesson 11: Tables: clean the data and add new columns

Task 1. In the first example, fill the empty marks with the median instead of the average.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.replace_values("marks", [61, 73], with_value=None)
table.count_empty_cells()
table.fill_empty_cells("marks", with_the="median")
table.count_empty_cells()
```

What it prints

```
Replaced 2 cells in marks.
Empty cells in each column:
  name     0
  class    0
  subject  0
  hours    0
  marks    2
  height   0
Total: 2 cells empty.
```

```
Filled 2 cells in marks.
Empty cells in each column:
  name     0
  class    0
  subject  0
  hours    0
  marks    0
  height   0
No empty cells.
```

Answer `with_the="median"` fills each empty mark with the middle mark of the others; afterwards no cell is empty.

Task 2. Add a column `"height_m"` with the height in metres: `from_column="height"`, `divide_by=100`.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.add_column("height_m", from_column="height", divide_by=100)
table.show_first_rows()
```

What it prints

```
   name class  subject  hours  marks  height  height_m
0  Asha     A    Maths    6.3     61     144      1.44
1  Asha     A  Science    3.8     66     144      1.44
2  Asha     A  English    7.6     77     144      1.44
3   Ben     A    Maths    6.4     73     152      1.52
4   Ben     A  Science    9.2     98     152      1.52
```

Answer The height in centimetres divided by 100: 144 becomes 1.44.

Task 3. In the last example, add a column `"best_two"` with the formula `"total - English"`.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.keep_columns("name", "class", "subject", "marks")
table.turn_rows_into_columns(names_from="subject", values_from="marks")
table.add_column("total", "Maths + Science + English")
table.add_column("best_two", "total - English")
table.show_first_rows()
```

What it prints

```
   name class  Maths  Science  English  total  best_two
0  Asha     A     61       66       77    204       127
1   Ben     A     73       98       71    242       171
2  Chen     A     81       59       46    186       140
3  Dina     A     74       79       57    210       153
4  Emil     A     47       86       76    209       133
```

Answer The formula can use a column made earlier (total); `best_two` is Maths + Science.

Back to lesson 11

Lesson 12: Tables: summaries, and from a table to a chart

Task 1. Make the fruit totals `for_each="shop"` instead. Which shop sold more?

```
import simplydata as sdata
table = sdata.table(sdata.example("fruit"))
totals = table.total_of("count", for_each="shop")
print(totals)
```

What it prints

```
The total of "count" for each "shop": Corner shop 354, Market stall 457.
simplydata table: 2 rows, 2 columns
           shop  total of count
0   Corner shop             354
1  Market stall             457
```

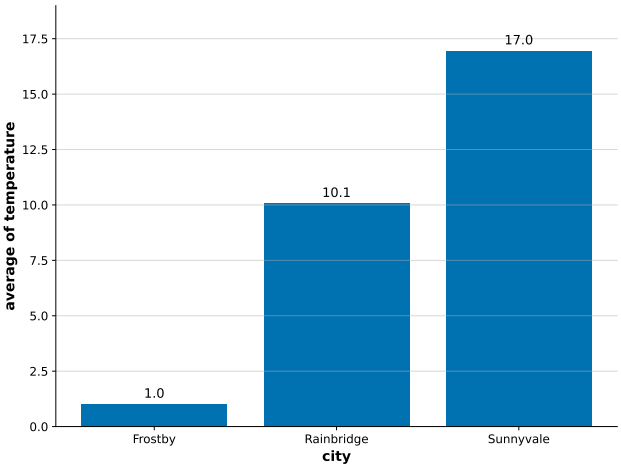
Answer The Market stall sold more: 457 pieces of fruit, against 354 at the Corner shop.

Task 2. Work out the average temperature `for_each="city"` in the weather table and draw it as bars.

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
averages = table.average_of("temperature", for_each="city")
chart = averages.chart()
chart.bars(x="city", y="average of temperature")
chart.numbers_on_bars(decimals=1)
chart.save("lesson12_temperature.png")
```

What it prints

```
The average of "temperature" for each "city": Frostby 1.02, Rainbridge 10.08, Sunnyvale 16.96.
```



Lesson 12, task 2: `lesson12_temperature.png`.

Answer Frostby 1.0, Rainbridge 10.1 and Sunnyvale 17.0 degrees. The summary table's column is called `"average of temperature"`.

Task 3. Is there a link between height and marks? Use `correlation_between` to find out.

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.correlation_between("height", "marks")
```

What it prints

```
The correlation between "height" and "marks" is r = 0.03 (Pearson, 54 rows): very weak, going up, and could
    ↪ easily be chance (p = 0.826).
```

Answer No: r = 0.03, very close to 0, and p = 0.83 says it could easily be chance.

Back to lesson 12

Lesson 13: Numbers: lists and grids of numbers

Task 1. Make the numbers 1 to 10, then square them in two ways: with ** 2 and with `power(2)`.

```
import simplydata as sdata
nums = sdata.numbers(1, to=10)
squares = nums ** 2
print(squares)
nums.power(2)
nums.show()
```

What it prints

```
simplydata numbers: 10 numbers
[  1,   4,   9,  16,  25,  36,  49,  64,  81, 100]
[  1,   4,   9,  16,  25,  36,  49,  64,  81, 100]
```

Answer Both give 1, 4, 9 … 100. ** 2 makes new numbers and leaves nums alone; `power(2)` changes nums itself (and `undo()` can take it back).

Task 2. Make 6 dice throws with `sdata.random_numbers(how_many=6, between=[1, 6], whole_numbers=True, seed=1)`. What is their total?

```
import simplydata as sdata
dice = sdata.random_numbers(how_many=6, between=[1, 6], whole_numbers=True, seed=1)
dice.show()
dice.total()
```

What it prints

```
[3, 4, 5, 6, 1, 1]
The total of the numbers is 20 (6 numbers).
```

Answer With `seed=1` the throws are 3, 4, 5, 6, 1 and 1, which add up to 20.

Task 3. Make a grid of 3 rows and 4 columns filled with 0: `sdata.number_grid(rows=3, columns=4, filled_with=0)`.

```
import simplydata as sdata
grid = sdata.number_grid(rows=3, columns=4, filled_with=0)
print(grid)
```

What it prints

```
simplydata number grid: 3 rows, 4 columns
[[0, 0, 0, 0],
 [0, 0, 0, 0],
 [0, 0, 0, 0]]
```

> **Answer** Three rows of four zeros: a grid to fill in later.

Back to lesson 13

Lesson 14: Statistical tests: real difference, or chance?

Task 1. Compare the average marks `between="subject"`. Could the differences be chance?

```
import simplydata as sdata
table = sdata.table(sdata.example("marks"))
table.compare_averages("marks", between="subject")
```

What it prints

```
The average "marks" is 62.3 in English, 62.8 in Maths and 69.7 in Science. Differences this big between the
    ↪ groups could easily be chance (p = 0.264); this does not show that there is no difference. Omega-
    ↪ squared = 0.0138: the groups explain about 1 % of the differences in "marks".
Test: Welch's one-way ANOVA (F = 1.39, df = 2 and 33.8), the usual test for three or more averages; it does
    ↪ not assume the groups have the same spread. To see which groups differ: table.compare_each_pair("
    ↪ marks", between="subject").
```

> **Answer** Yes, they could: p = 0.26 is well above 0.05. That does not prove the subjects are the same; these data simply cannot tell.

Task 2. Is the average temperature different between the cities of `sdata.example("weather")`?

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.compare_averages("temperature", between="city")
table.compare_each_pair("temperature", between="city")
```

What it prints

```
The average "temperature" is 1.02 in Frostby, 10.1 in Rainbridge and 17.0 in Sunnyvale. Differences this big
    ↪  between the groups are very unlikely to be chance (p = 0.0002). Omega-squared = 0.465: the groups
    ↪ explain about 46 % of the differences in "temperature".
Test: Welch's one-way ANOVA (F = 13.0, df = 2 and 20.8), the usual test for three or more averages; it does
    ↪ not assume the groups have the same spread. To see which groups differ: table.compare_each_pair("
    ↪ temperature", between="city").
Each pair of "city" groups compared on "temperature" with Welch's t-test; the p-values are adjusted for 3
    ↪ pairs with Holm's method, so the chance of any false alarm stays at 5 %.
  Frostby vs Rainbridge: the average of Frostby is 9.06 lower; unlikely to be chance (Holm-adjusted p =
    ↪ 0.011).
  Frostby vs Sunnyvale: the average of Frostby is 15.9 lower; very unlikely to be chance (Holm-adjusted p =
    ↪ 0.0002).
```

```
Rainbridge vs Sunnyvale: the average of Rainbridge is 6.88 lower; unlikely to be chance (Holm-adjusted p =
    ↪ 0.011).
```

Answer Yes: p = 0.0002, very unlikely to be chance. `compare_each_pair` then shows which cities differ from each other.

Task 3. Fit a straight line of `"forest"` against `"year"` in `sdata.example("growth")` for one country.

```
import simplydata as sdata
table = sdata.table(sdata.example("growth"))
table.keep_rows("country", equal_to="Applia")
table.fit_straight_line(x="year", y="forest")
```

What it prints

```
Kept 13 of 52 rows.
"forest" goes down by 0.574 for each 1 more "year" (95 % CI -0.672 to -0.477); this slope is very unlikely
    ↪ to be chance (p < 0.0001). R squared = 0.94: "year" explains about 94 % of the differences in "
    ↪ forest".
Line: forest = -0.574 * year + 1209, by least squares over 13 rows. A link does not show that one thing
    ↪ causes the other.
```

Answer In Applia the forest shrinks by about 0.57 percentage points a year (95 % CI 0.48 to 0.67), and the line explains 94 % of the differences.

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Lesson 15: A whole analysis: from a question to a figure

Task 1. Do a whole analysis of `sdata.example("weather")`: look at it, compare the average rain `between="city"`, draw the rain of each city as boxes, and save the chart and the table.

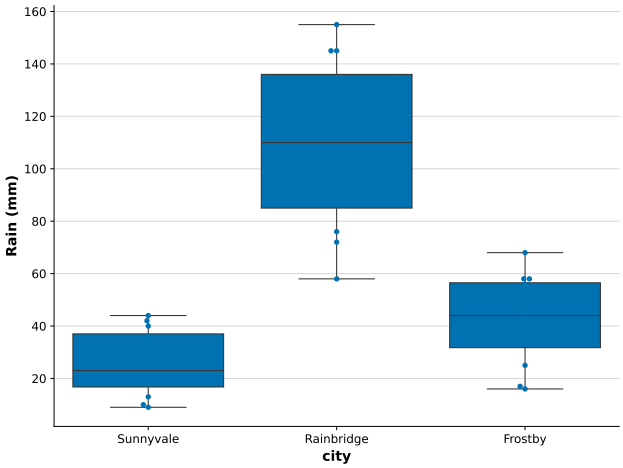
```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.describe_columns()
table.remove_columns("wind_direction", "wind_speed")
table.average_of("rain", for_each="city")
table.compare_averages("rain", between="city")
chart = table.chart()
chart.boxes(x="city", y="rain")
chart.all_dots(x="city", y="rain")
chart.label_y("Rain (mm)")
chart.save("lesson15_rain_boxes.png")
table.save("lesson15_weather.xlsx")
```

What it prints

```
The table has 36 rows and 7 columns.

       column   kind  empty cells smallest biggest average different values
        month   text            0                                        12
 month_number number            0        1      12     6.5
         city   text            0                                         3
  temperature number            0    -11.6    26.5    9.35
         rain number            0        9     155   59.03
wind_direction number           0       16     354  244.42
   wind_speed number            0      2.7    28.6    15.1
```

```
The average of "rain" for each "city": Frostby 42.08, Rainbridge 109.42, Sunnyvale 25.58.
The average "rain" is 42.1 in Frostby, 109 in Rainbridge and 25.6 in Sunnyvale. Differences this big between
    ↪  the groups are very unlikely to be chance (p < 0.0001). Omega-squared = 0.720: the groups explain
    ↪ about 72 % of the differences in "rain".
Test: Welch's one-way ANOVA (F = 33.9, df = 2 and 20.2), the usual test for three or more averages; it does
    ↪ not assume the groups have the same spread. To see which groups differ: table.compare_each_pair("
    ↪ rain", between="city").
```

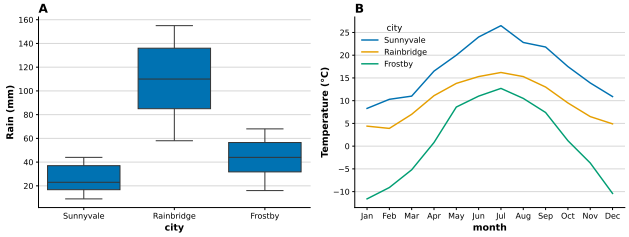


Lesson 15, task 1: `lesson15_rain_boxes.png`.

Answer Rainbridge is much wetter (109 mm a month on average) than Frostby (42) and Sunnyvale (26); $p < 0.0001$, so this is very unlikely to be chance.

Task 2. Make a second chart of the temperature, then put the two charts together with `sdata.side_by_side(..., letters=True)`.

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
rain = table.chart()
rain.boxes(x="city", y="rain")
rain.label_y("Rain (mm)")
warmth = table.chart()
warmth.lines(x="month", y="temperature", colour="city")
warmth.label_y("Temperature (°C)")
page = sdata.side_by_side(rain, warmth, letters=True)
page.save("lesson15_weather_page.png")
```



Lesson 15, task 2: `lesson15_weather_page.png`.

Answer Two charts of one table, side by side, with A and B so a report can point to each.

Task 3. Read the `steps()` of your table out loud. Would someone else understand what you did?

```
import simplydata as sdata
table = sdata.table(sdata.example("weather"))
table.remove_columns("wind_direction", "wind_speed")
table.keep_rows("month_number", between=[6, 8])
table.sort_by("rain", biggest_first=True)
table.steps()
```

What it prints

```
Kept 9 of 36 rows.
1. remove columns wind_direction, wind_speed
2. keep rows where month_number is between 6 and 8
3. sort by rain (biggest first)
```

Answer `steps()` lists each change in plain words, in order; those lines are a first draft of the methods part of a report.

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Index of words

Every word and option, A to Z, with the page of its dictionary entry. Other spellings of a word (such as `chart.bar()` for `chart.bars()`) point to the same entry.

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